KYUSHU-HET-354

Introduction to Generalized Symmetries

Justin Kaidi[1,2,3]

[1] *Institute for Advanced Study, Kyushu University, Fukuoka 819-0395, Japan*
[2] *Department of Physics, Kyushu University, Fukuoka 819-0395, Japan*
[3] *Quantum and Spacetime Research Institute (QuaSR), Kyushu University, Fukuoka 819-0395, Japan*

arXiv:2603.08798v3 [hep-th] 18 Sep 2026

Abstract

These notes were prepared for a series of intensive lectures delivered at Hokkaido University, Nagoya University, Kyoto University, and Kyushu University.

We begin with a brief review of higher-form symmetries, anomalies, and discrete gauge theories, before introducing non-invertible symmetries in $(1+1)$-dimensional systems. The basic structure of fusion categories is then discussed, including a discussion of categorical analogs of discrete gauging and representation theory. We subsequently turn to $(3+1)$-dimensional theories, where several physical applications of non-invertible symmetries are discussed. These notes are intended to be largely self-contained, and require no prior familiarity with subjects such as conformal field theory or lattice models.

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1 Introduction

These lectures give a broad overview of *generalized symmetries*, with a particular focus on the emerging notion of *non-invertible symmetries*. Symmetries have long been central to our understanding of quantum field theory (QFT): they constrain dynamics, classify phases of matter, and often dictate the structure of observables. Though they have traditionally been understood as transformations forming a group, developments over the last decade have revealed that this group-theoretic framework is too narrow, and that many quantum systems exhibit richer symmetry structures which fall outside the group paradigm. A hallmark of such structures is the lack of inverse for some transformations, giving rise to the name non-invertible symmetry.

The study of non-invertible symmetries has seen rapid progress in recent years, with connections to conformal field theory, lattice models, and topological quantum field theory, as well as implications for dualities and dynamics in strongly-coupled systems. In these lectures we will introduce the basic ideas and provide examples, aiming for a self-contained exposition. We will assume only a basic familiarity with QFT and undergraduate mathematics. The goal is to build intuition for how generalized and non-invertible symmetries appear, how they are characterized, and why they play an increasingly important role in modern theoretical physics.

We should emphasize that there is essentially nothing new in these lecture notes, and in various places we will borrow from previous reviews of generalized symmetries, which include [1–8]. The reader is encouraged to consult those references as well, which in some cases may be much clearer than the current reference. One potential "novelty" of the current notes is that, because some of the audiences to which these lectures were given were unfamiliar with conformal field theory (CFT), the presentation here makes no reference to CFT techniques—even the Ising model is treated only as a lattice model. This may make these notes more palatable to some readers, though less palatable to others.

Because these lectures were given over the course of three (long and arduous) days, the notes are correspondingly split into three sections. Each section ends with a short list of relevant references. Though we have tried to include a decent number of citations, it would be impossible to cite every paper on the subject, since the literature is vast and ever-growing. Apologies in advance to anyone whose work was not properly accounted for. I would be happy to add references to any relevant papers upon request.

Finally, an unfortunate fact about the literature is that the distinction between Euclidean and Lorentzian signature is often blurred. For example, one often refers to Hilbert spaces and operators acting on them, which are intrinsically Lorentzian notions, but at the same time discusses e.g. linkings of the operators in Euclidean space. The Euclidean conventions are also common when determining where to distribute factors of i throughout equations, though we will write most equations in a Lorentzian fashion here, in order to minimize the factors of i appearing in the action/Lagrangian/currents. For the most part we will be rather cavalier about this issue, commenting on it only when it seems relevant.

Acknowledgements

These notes were prepared for a series of intensive lectures given at Hokkaido University, Kyoto University, and Nagoya University. I would like to thank the organizers of those lectures—Tatsuo Kobayashi, Yu Nakayama, and Tadakatsu Sakai—as well as all of the participants for their many enlightening comments and questions. Almost all of the content of these lectures consists of things which I learned from my many excellent friends and collaborators, including Jan Albert, Vladimir Bashmakov, Yichul Choi, Michele Del Zotto, Thomas Dumitrescu, Wataru Harada, Azeem Hasan, Knight Hirasaki, Yamato Honda, Zohar Komargodski, Ying-Hsuan Lin, Yuefeng Liu, Julio Parra-Martinez, Emily Nardoni, Kantaro Ohmori, Sahand Seifnashri, Shu-Heng Shao, Xiaoyi Shi, Soichiro Shimamori, Zhengdi Sun, Yuji Tachikawa, Gabi Zafrir, and Yunqin Zheng. I look forward to learning more from them in the future. Finally, I thank Kota Miki for catching typos in the draft.

2 Topological Defects and Higher-Form Symmetries

We begin our discussion by reviewing various well-known properties of standard continuous symmetries, before generalizing to continuous higher-form global symmetries. This material can be found in any review on the topic, and we will follow [4] particularly closely. Having done so, we will then move on to a discussion of discrete symmetries (both zero- and higher-form), their gauging, their anomalies, and their mixing into higher-group symmetries. This material is again fairly standard (see e.g. [4–8]), but since all of the results discussed in this section will appear in some shape or form in the subsequent sections, we nevertheless include it for completeness.

2.1 Noether's theorem

We begin by considering the simplest case of a $U(1)$ symmetry. Familiar examples of theories with such a symmetry include complex scalar field theory,

$$S = \frac{1}{2}\int d^dx \, \left(\partial_\mu\phi\partial^\mu\phi^* + m^2|\phi|^2\right) \, , \tag{2.1}$$

which is invariant under

$$\phi(x) \to e^{i\alpha}\phi(x) \, , \qquad \phi^*(x) \to e^{-i\alpha}\phi^*(x) \, , \tag{2.2}$$

with α taking values in $\mathbb{R}/2\pi\mathbb{Z} \cong U(1)$, and the theory of a Dirac fermion,

$$S = \int d^dx \, \overline{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi \, , \tag{2.3}$$

which is invariant under

$$\Psi(x) \to e^{i\alpha}\Psi(x) \, , \qquad \overline{\Psi}(x) \to e^{-i\alpha}\overline{\Psi}(x) \, . \tag{2.4}$$

For reasons that will become clear later, we will refer to this as a $U(1)$ *zero-form* symmetry.

There is a famous theorem due to Noether stating the following,

Theorem 1 (Noether's theorem) *For every continuous symmetry there is an associated (equivalence class of) conserved current(s).*

The proof is as follows. Denote our fields (not necessarily scalars) by $\phi^a(x)$ and our Lagrangian density by $\mathcal{L}[\phi^a, \partial_\mu\phi^a]$. The action is given by

$$S = \int d^dx \, \mathcal{L}[\phi^a, \partial_\mu\phi^a] \, , \tag{2.5}$$

and the equations of motion are

$$\frac{\partial\mathcal{L}}{\partial\phi^a} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)}\right) = 0 \, . \tag{2.6}$$

We assume that our symmetry is continuous, so that we may consider transformations infinitesimally close to the identity, i.e. $\alpha \ll 1$ in the case of $U(1)$. Say that under such an infinitesimal transformation we have

$$\phi^a(x) \to \phi^a(x) + \alpha\, \delta\phi^a(x) \ . \tag{2.7}$$

In the case in which ϕ^a has charge q_a under a $U(1)$ symmetry, i.e. $\phi^a \to e^{iq_a\alpha}\phi^a$, we have simply $\delta\phi^a(x) = iq_a\phi^a(x)$.

For the above transformation to be a symmetry of the classical action, we must have

$$\delta_\alpha S = \int d^dx\, \delta_\alpha \mathcal{L} = 0 \ . \tag{2.8}$$

This requires that $\delta_\alpha\mathcal{L}$ be a total derivative, and we furthermore assume that it is exactly zero.[1] The variation can be written as

$$\begin{aligned} (\alpha)^{-1}\delta_\alpha\mathcal{L} &= \frac{\partial\mathcal{L}}{\partial\phi^a}\delta\phi^a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)}\delta(\partial_\mu\phi^a) = \frac{\partial\mathcal{L}}{\partial\phi^a}\delta\phi^a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)}\partial_\mu(\delta\phi^a) \\ &= \left[\frac{\partial\mathcal{L}}{\partial\phi^a} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)}\right)\right]\delta\phi^a + \partial_\mu\left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)}\delta\phi^a\right] \ , \end{aligned} \tag{2.9}$$

and on-shell, i.e. assuming that the equations of motion are satisfied, the first term vanishes, so we are left with

$$\partial_\mu\left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)}\delta\phi^a\right] = 0 \ . \tag{2.10}$$

Thus we find that

$$j_\mu := \frac{\partial\mathcal{L}}{\partial(\partial^\mu\phi^a)}\delta\phi^a \tag{2.11}$$

satisfies the divergence-free equation $\partial^\mu j_\mu = 0$. This j_μ is called a *conserved current*. In non-relativistic notation, the equation reads

$$\partial_0 j_0 - \partial_i j_i = 0 \ , \qquad i = 1, \ldots, d-1 \ , \tag{2.12}$$

where the minus sign appears because we have lowered the 0 index using $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, \ldots, 1)$. It is also useful to write this current in differential form notation, in which $j = j_\mu dx^\mu$ is a one-form. In this notation the conservation equation reads $d * j = 0$.[2]

[1] It is also possible to consider a Lagrangian for which the variation is a non-trivial total derivative, in which case we refer to the symmetry as a *quasi-symmetry* of the Lagrangian. In this case Noether's theorem still holds, but the expression for the current must be modified.

[2] Here $*$ is the Hodge star operation, which acts on differential k-forms in d dimensions via $* : \omega_{\mu_1,\ldots,\mu_k} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k} \mapsto \frac{1}{(d-k)!}\epsilon^{\mu_1\cdots\mu_k}{}_{\mu_{k+1}\ldots\mu_d}\omega_{\mu_1,\ldots,\mu_k} dx^{\mu_{k+1}} \wedge \cdots \wedge dx^{\mu_d}$ and satisfies $**\omega = \pm(-1)^{k(d-k)}\omega$, with $+$ for Euclidean signature and $-$ for Lorentzian signature.

From the conserved current we may define the *conserved charge*,

$$Q := -\int_{\text{space}} d^{d-1}x \; j_0(x) \; . \tag{2.13}$$

This is conserved since we have

$$\partial_0 Q = -\int d^{d-1}x \; \partial_0 j_0 = -\int d^{d-1}x \sum_{i=1}^{d-1} \partial_i j_i(x) = 0 \; , \tag{2.14}$$

where in the final step we have used that the integrand is a total derivative, and space is assumed to have no boundary.

Note that the current is subject to an ambiguity, allowing for shifts by so-called *improvement terms*,

$$j_0 \to j_0 + \partial_i X_i \; , \qquad\qquad j_i \to j_i + \partial_0 X_i + \partial_j Z^{[ij]} \; . \tag{2.15}$$

The shift in j_0 is allowed since it does not change the charge, but must be accompanied by the first shift in j_i so as to maintain the conservation equation. The second shift in j_i, namely the one involving $Z^{[ij]}$, is allowed since it changes neither the charge nor the conservation equation. These improvement terms are related to non-minimal couplings to background gauge fields, as we mention below.

2.2 Ward-Takahashi identities

At the quantum level, the conservation of the Noether current can be violated at the location of charged operators. Indeed, imagine that we have a charge q operator $\mathcal{O}_q(x)$, whose correlation functions are given by the following,

$$\langle \mathcal{O}_q(x) \dots \rangle = \frac{1}{Z} \int \mathcal{D}\phi^a \, \mathcal{O}_q(x) \dots e^{iS[\phi^a]} \; , \qquad\qquad Z = \int \mathcal{D}\phi^a e^{iS[\phi^a]} \; ; \tag{2.16}$$

the $\dots$ represent potential operator insertions at separated points. Assuming an invariant path-integral measure, a change of integration variables with a smooth, spacetime-dependent parameter α leaves the expectation value unchanged:

$$\delta_\alpha \langle \mathcal{O}_q(x) \dots \rangle = 0 \; . \tag{2.17}$$

On the other hand, note that

$$\begin{aligned} \delta_\alpha \langle \mathcal{O}_q(x) \dots \rangle \;\; &= \;\; \frac{1}{Z} \int \mathcal{D}\phi^a \left[(\delta_\alpha \mathcal{O}_q(x) + i\mathcal{O}_q(x)\delta_\alpha S[\phi^a]) \cdots + \mathcal{O}_q(x)\delta_\alpha(\dots) \right] e^{iS[\phi^a]} \\ &= \frac{1}{Z} \int \mathcal{D}\phi^a \left[\left(iq\alpha(x)\mathcal{O}_q(x) - i\mathcal{O}_q(x) \int d^d y \, \alpha(y) \, \partial^\mu j_\mu(y) \right) \cdots + \mathcal{O}_q(x)\delta_\alpha(\dots) \right] e^{iS[\phi^a]} \; . \end{aligned} \tag{2.18}$$

Choose α to vanish near the spectator insertions represented by $\ldots$. Since it is otherwise arbitrary, we obtain the following local *Ward-Takahashi identity*,

$$\langle \partial^\mu j_\mu(x) \mathcal{O}_q(y) \ldots \rangle = q\, \delta^{(d)}(x-y) \langle \mathcal{O}_q(y) \ldots \rangle \,. \tag{2.19}$$

Whereas the classical conservation law would have naively led us to believe that the left-hand side is zero, we see that we now get a non-zero value at coincident points. There are analogous contact terms at charged spectator insertions; in Lorentzian signature the correlators are time-ordered.

2.3 Conserved charges and topological defects

Now say, as above, that we have a classical current satisfying $\partial^\mu j_\mu = 0$. As we just said, the corresponding Noether charge at time t is given by the integral of $-j_0$ over space. We can also define a slightly more general charge, living on any codimension-1 (i.e. $(d-1)$-dimensional) submanifold M_{d-1} of spacetime,

$$Q(M_{d-1}) := \int_{M_{d-1}} ds^\mu\, j_\mu \,, \tag{2.20}$$

Here ds^μ is the oriented surface element, with $ds^\mu j_\mu = *j$ on M_{d-1}. On a constant-time slice oriented by $d^{d-1}x$, this gives $Q = -\int d^{d-1}x\, j_0$.

In this case, the analog of conservation in time is the fact that the charge $Q(M_{d-1})$ depends only on the homology class of M_{d-1} in spacetime with the charged insertions removed. Indeed, suppose these manifolds bound an oriented region N_d in spacetime that avoids charged insertions, such that $\partial N_d = M_{d-1} \sqcup \overline{M}'_{d-1}$. Then by Stokes' theorem,

$$Q(M_{d-1}) - Q(M'_{d-1}) = \int_{N_d} d^d x\, \partial^\mu j_\mu = 0 \,. \tag{2.21}$$

Thus homologous codimension-1 supports have the same charge, provided the interpolation avoids charged insertions.

We may now define the *symmetry generators*,

$$U_\alpha(M_{d-1}) := e^{i\alpha Q(M_{d-1})} \,. \tag{2.22}$$

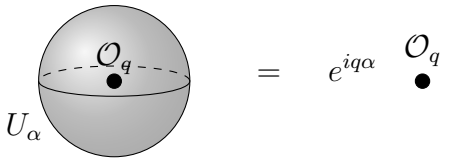
These again depend only topologically on M_{d-1}. In a quantum mechanical theory, how should we interpret the $U_\alpha(M_{d-1})$? When M_{d-1} is the whole space at a fixed time, then $U_\alpha(M_{d-1})$ should be interpreted as the conserved, unitary operator that acts on the Hilbert space $\mathcal{H}$. On the other hand, when M_{d-1} is localized in one of the spatial directions and extends along time, then the effect of $U_\alpha(M_{d-1})$ is to modify the boundary conditions along that spatial direction. Quantizing in the presence of these modified boundary conditions gives a new Hilbert space of states, which we denote by $\mathcal{H}_\alpha$. These are known as *twisted sectors*. For a more general M_{d-1} which is neither localized nor wrapping the entire time direction, the interpretation is less clear,

but what is always true is that moving an operator past these defects leads to the action of the $U(1)$ symmetry on the operator.

To see this, we may use the Ward-Takahashi identity (2.19) to compute

$$\begin{aligned} U_\alpha(M_{d-1})\mathcal{O}_q(x)U_{-\alpha}(M'_{d-1}) &= e^{i\alpha\int_{N_d} d^dy\,\partial^\mu j_\mu(y)}\mathcal{O}_q(x) \\ &= \sum_{n=0}^{\infty}\frac{(i\alpha)^n}{n!}\left(\int_{N_d} d^dy\,\partial^\mu j_\mu(y)\right)^n\mathcal{O}_q(x) \\ &= \sum_{n=0}^{\infty}\frac{(i\alpha q)^n}{n!}\left(\int_{N_d} d^dy\,\delta^{(d)}(x-y)\right)^n\mathcal{O}_q(x) \\ &= \sum_{n=0}^{\infty}\frac{(i\alpha q)^n}{n!}\mathcal{O}_q(x) \\ &= e^{iq\alpha}\,\mathcal{O}_q(x)\ , \end{aligned} \tag{2.23}$$

where M_{d-1} is a manifold enclosing x, and M'_{d-1} is a manifold which is continuously deformable to M_{d-1}, but which does not enclose x. Pictorially, this result can be drawn as



(2.24)

The symmetry generators $U_\alpha(M_{d-1})$ are our first examples of topological defects. Since everything is Abelian, it is easy to see that we have the following multiplication structure,

$$U_{\alpha_1}(M_{d-1})\times U_{\alpha_2}(M_{d-1}) = U_{\alpha_1+\alpha_2}(M_{d-1})\ , \tag{2.25}$$

which is the usual group structure for $U(1)$. Since each element has an inverse, these defects are referred to as *invertible*.

Example: As a concrete example, consider a massless complex scalar field in $(1+1)$-dimensions. As we mentioned before, there is a $U(1)$ global symmetry acting as $\phi\to e^{i\alpha}\phi$, and the Noether current is computed via (2.11) to be

$$j_\mu = \frac{i}{2}\left[\phi\partial_\mu\phi^* - \phi^*\partial_\mu\phi\right]\ . \tag{2.26}$$

If we take the space to be S^1 and quantize using the boundary conditions

$$\phi(t,x+2\pi) = \phi(t,x)\ , \tag{2.27}$$

then we get a Hilbert space $\mathcal{H}(S^1)$, and $U_\alpha(S^1) = e^{-i\alpha\int_{S^1} j_0}$ acts as a unitary operator on this Hilbert space. On the other hand, if we insert U_α along the time direction, then the boundary conditions for the scalar field are modified to

$$\phi(t,x+2\pi) = e^{i\alpha}\phi(t,x)\ , \tag{2.28}$$

and quantizing using these boundary conditions gives a new Hilbert space $\mathcal{H}_\alpha(S^1)$.

2.4 Coupling a current to a gauge field

Given a conserved current, we can couple it to a background gauge field by adding the following term, to first order in the background, to the action[3]

$$S[A] = \int d^d x \, A_\mu(x) j^\mu(x) = \int A \wedge *j \ . \tag{2.29}$$

At finite background field, further terms can be required for gauge invariance; for a charged scalar these include the term quadratic in A from the covariant kinetic term. Note that this is the *minimal* coupling of the gauge field to the current. We may also consider non-minimal couplings such as $F_{\mu\nu} Z^{[\mu\nu]}$, but these can be removed by changing the current via the improvement terms in (2.15).

The gauge field has the property that we may shift $A_\mu \to A_\mu + \partial_\mu \lambda(x)$, with $\lambda(x)$ a zero-form gauge field (i.e. a circle-valued field). For the linear coupling, current conservation gives

$$\delta_\lambda S[A] = \int d^d x \, \partial_\mu \lambda(x) \, j^\mu(x) = - \int d^d x \, \lambda(x) \partial_\mu j^\mu(x) = 0 \ . \tag{2.30}$$

Off-shell gauge invariance of the full action also requires transforming the charged fields, with phase $e^{-i q_a \lambda}$ for $A \to A + d\lambda$ in the present convention. Background gauge fields are an extremely useful tool in the study of symmetries, especially in the quantum setting. Indeed, the question of whether or not a symmetry is present (i.e. whether or not a current j_μ is conserved) can be translated into the question of whether or not the path-integral is invariant under gauge transformations of the background gauge field. This will be important for understanding anomalies, as will be discussed later.

Note that by "background" gauge field, we mean that the gauge field is non-dynamical, which means that it is not integrated over in the path integral. Coupling a theory to a background gauge field is *not* the same as gauging the symmetry.[4] To gauge a symmetry, we must couple to a background gauge field, and then make the gauge field *dynamical*. We will adopt the convention that background gauge fields are written with upper case letters $A, B, \ldots$, while dynamical gauge fields are written with lower case letters $a, b, \ldots$.

In fact, the background gauge field $A_\mu(x)$ is related to the symmetry defects discussed before in the following way. Note that inserting the defect $U_\alpha(M_{d-1}) = e^{i\alpha \int_{M_{d-1}} ds^\mu j_\mu}$ into the path integral is equivalent to coupling a background gauge field of the form

$$A = \alpha \, \omega_{M_{d-1}} \ , \tag{2.31}$$

where $\omega_{M_{d-1}}$ is the Poincaré dual of M_{d-1}, i.e. it is a 1-form such that $\int_{M_d} \omega_{M_{d-1}} \wedge \bullet = \int_{M_{d-1}} \bullet$. This is a singular gauge configuration, since $\omega_{M_{d-1}}$ is basically a delta-function localizing on M_{d-1}.

[3] Note that A is strictly speaking not a 1-form—it is a *1-form gauge field*, which means that it is locally a 1-form subject to a redundancy $A \sim A + d\lambda$ such that $\lambda \sim \lambda + 2\pi$, and that it must have properly quantized fluxes $\oint \frac{dA}{2\pi} \in \mathbb{Z}$.

[4] In condensed matter, it is sometimes referred to as *weakly gauging*, but this terminology is not common in high energy theory.

For continuous symmetries such as the $U(1)$ case that we are currently considering, one may in general consider smooth gauge fields, which can be thought of as some sort of smearing out of symmetry generators. Specifying the gauge field requires more than just the data of the symmetry generators though—one also needs the smearing function. In contrast, when we discuss discrete symmetries later on, we will see that there are no smooth gauge fields in the first place, and specification of the background gauge field is completely equivalent to specifying the symmetry generators inserted on the manifold.

Let us now use the background gauge field to reproduce the action of the symmetry generators on charged operators. We insert a symmetry generator $U_\alpha(M_{d-1})$ and consider moving the operator $\mathcal{O}_q$ from a point x_i on one side to a point x_f on the other along a path γ. Doing so leads to an Aharonov-Bohm phase,

$$\mathcal{O}_q(x_i) \;\to\; e^{iq\int_\gamma dx^\mu A_\mu}\mathcal{O}_q(x_f)\;. \tag{2.32}$$

Plugging in (2.31) and using the fact that γ intersects M_{d-1} at a single point, we then have

$$\mathcal{O}_q(x_i) \;\to\; e^{iq\alpha}\mathcal{O}_q(x_f)\;, \tag{2.33}$$

which is the expected symmetry action.

2.5 One-form symmetries

Thus far, our current j_μ had one tensor index. Next let us imagine that we had an anti-symmetric 2-tensor $j_{\mu\nu}$. It may not be obvious that such a thing is sensible, but we will see that it is, and will give various concrete examples later on. In this case the conservation law would look like

$$\partial^\mu j_{\mu\nu} = 0\;, \tag{2.34}$$

or in non-relativistic notation,[5]

$$\begin{aligned} \partial_0 j_{0i} - \partial_j j_{ji} &= 0\;, \qquad \nu = i \\ \partial_i j_{i0} &= 0\;, \qquad \nu = 0\;. \end{aligned} \tag{2.35}$$

In differential form notation, it is simply $d*j = 0$, where now j is a 2-form. As before, we are free to introduce improvement terms,

$$j_{0j} \to j_{0j} + \partial_i X_{[ij]}\;, \qquad j_{ij} \to j_{ij} + \partial_0 X_{[ij]} + \partial_k Z^{[kij]}\;. \tag{2.36}$$

In terms of these currents, we may now try to define two notions of charge,

[5]Note that from the non-relativistic point of view, the first of these looks like a conservation equation, while the second looks like an arbitrary constraint. If we are willing to get rid of Lorentz invariance, we could indeed relax the second constraint, which gives rise to so-called *vector symmetry* [9]. In these lectures we will restrict ourselves to relativistic systems. Let us also mention that we could relax the condition that the indices of $j_{\mu\nu}$ be anti-symmetric, which would give a more general notion of *higher-spin* symmetries. An example would be the energy-momentum tensor $T_{\mu\nu}$, whose corresponding charge is the 4-momentum p_μ.

1. **Vector charge:** We begin by defining charge as a vector,

$$Q_i := \int_{\text{space}} d^{d-1}x\, j_{0i}(x)\ . \tag{2.37}$$

This is conserved since

$$\partial_0 Q_i = \int_{\text{space}} d^{d-1}x\, \partial_0 j_{0i} = \int_{\text{space}} d^{d-1}x\, \partial_j j_{ji} = 0\ . \tag{2.38}$$

More generally, we can define $Q_i(M_{d-1})$, which depends topologically on M_{d-1}.

2. **Scalar charge:** Alternatively, we can define a scalar charge

$$Q(M_{d-2}) := \int_{M_{d-2}} ds^{\mu\nu} j_{\mu\nu}\ . \tag{2.39}$$

Here M_{d-2} is a codimension-2 manifold, and $ds^{\mu\nu} = \widehat{n}_1^\mu \widehat{n}_2^\nu\, d^{d-2}x$ is its normal element. The scalar charge is again topological, i.e.

$$Q(M_{d-2}) - Q(M'_{d-2}) = \int_{N_{d-1}} ds^\nu\, \partial_\mu j_{\mu\nu} = 0 \tag{2.40}$$

where M_{d-2} and M'_{d-2} are bordant within spacetime away from charged insertions and $\partial N_{d-1} = M_{d-2} \sqcup \overline{M}'_{d-2}$.

In these lectures, we will be concerned primarily with the scalar charges. Unlike the charges for $U(1)$ zero-form symmetries discussed in the previous sections, which were defined on codimension-1 submanifolds M_{d-1}, the scalar charges $Q(M_{d-2})$ appearing here are defined on codimension-2 submanifolds. The corresponding symmetry generators are defined as before,

$$U_\alpha(M_{d-2}) := e^{i\alpha Q(M_{d-2})}\ . \tag{2.41}$$

However, note that we can no longer define the action of $U_\alpha(M_{d-2})$ on local operators, since a codimension-2 operator cannot surround a point in a topologically non-trivial manner. For example, in three dimensions M_{d-2} is a line, but a line cannot non-trivially wrap a point. Instead, the things which can link non-trivially with codimension-2 surfaces are *lines*, and hence the objects which are charged under these symmetries are line operators.

Let us make this more explicit. By similar methods as before, one can obtain a Ward identity involving $j_{\mu\nu}$ and a charge q line operator $L_q(\gamma)$ supported on a line γ,

$$\langle \partial^\mu j_{\mu\nu}(x) L_q(\gamma) \dots \rangle = q\, \delta^{(d-1)}(x \in \gamma)\ \widehat{v}_\nu\, \langle L_q(\gamma) \dots \rangle\ , \tag{2.42}$$

where $\widehat{v}$ is a tangent vector to γ. The analog of (2.23) is then to consider two manifolds M_{d-2} and M'_{d-2}, the former of which links with γ and the second of which does not, and then compute

$$U_\alpha(M_{d-2}) L_q(\gamma) U_\alpha(M'_{d-2})^{-1} = e^{i\alpha q} L_q(\gamma)\ . \tag{2.43}$$

The computation is identical to what was done before. There is also a more direct path-integral derivation of this result, which we will give in an explicit example below.

Example: Consider $(3+1)$-dimensional Maxwell theory,[6]

$$S = \frac{1}{2e^2}\int f \wedge *f = \frac{1}{4e^2}\int d^4x\, f_{\mu\nu}f^{\mu\nu}\ , \tag{2.44}$$

where $f = da$ is the field strength. We write everything in lower case since they are dynamical here. We can define two currents

$$j^E_{\mu\nu} = \frac{1}{e^2}f_{\mu\nu}\ , \qquad j^M_{\mu\nu} = \frac{1}{4\pi}\epsilon_{\mu\nu\rho\sigma}f^{\rho\sigma}\ , \tag{2.45}$$

or in differential form notation

$$j^E = \frac{1}{e^2}f\ , \qquad j^M = \frac{1}{2\pi} * f\ . \tag{2.46}$$

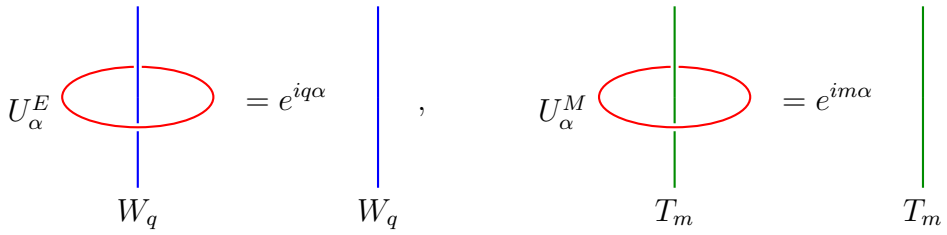
Both are conserved—indeed, we have

$$d * j^E = \frac{1}{e^2}d * f = 0\ , \qquad d * j^M = -\frac{1}{2\pi}df = 0\ , \tag{2.47}$$

where the first follows from the equations of motion and the second follows from the Bianchi identity. The symmetry operators live on codimension-2 surfaces M_2, with

$$U^E_\alpha(M_2) = e^{i\alpha\int_{M_2} *j^E}\ , \qquad U^M_\alpha(M_2) = e^{i\alpha\int_{M_2} *j^M}\ . \tag{2.48}$$

We denote the corresponding symmetries by $U(1)^{(1)}_E$ and $U(1)^{(1)}_M$ and refer to them as the "electric" and "magnetic" one-form symmetries. These symmetries act on Wilson and 't Hooft lines of the theory via linking, pictorially



(2.49)

Let us analyze $U(1)^{(1)}_E$ in more detail. Define the following Wilson line

$$W_q(\gamma) := e^{iq\oint_\gamma a}\ . \tag{2.50}$$

Its charge is measured by its linking with $U^E_\alpha(M_2)$, such that we have

$$U^E_\alpha(M_2)W_q(\gamma)U^E_\alpha(M_2')^{-1} = e^{i\alpha q\,\mathrm{Link}(M_2,\gamma)}W_q(\gamma)\ , \tag{2.51}$$

[6] As an aside, note that the coupling e is physically meaningful and cannot just be absorbed into redefinition of the integration variable a because a is constrained by the flux quantization condition $\oint \frac{f}{2\pi} \in \mathbb{Z}$; rescaling a would move e into the quantization condition, but will not get rid of it completely. In this sense, e is similar to the radius of the compact boson.

when M_2 links $\mathrm{Link}(M_2,\gamma)$ times with γ, and M_2' links none. We may verify this action from the path integral as follows. Inserting the Wilson line in the path integral gives

$$\int \mathcal{D}a\, W_q(\gamma)\, e^{\frac{i}{2e^2}\int_{M_4} f\wedge *f} = \int \mathcal{D}a\, e^{\int_{M_4} iq\omega_\gamma\wedge a + \frac{i}{2e^2} f\wedge *f}\ , \tag{2.52}$$

from which we see that the equations of motion are modified to

$$d*f = qe^2\omega_\gamma\ . \tag{2.53}$$

Then plugging this into

$$U^E_\alpha(M_2)W_q(\gamma)U^E_\alpha(M_2')^{-1} = e^{i\alpha\int_{N_3} d*j^E} W_q(\gamma)\ , \tag{2.54}$$

for $\partial N_3 = M_2 \sqcup \overline{M}_2'$, the right-hand side becomes

$$e^{\frac{i\alpha}{e^2}\int_{N_3} d*f} W_q(\gamma) = e^{iq\alpha\int_{N_3}\omega_\gamma} W_q(\gamma) = e^{iq\alpha\,\mathrm{Link}(M_2,\gamma)} W_q(\gamma)\ , \tag{2.55}$$

as expected.

On the other hand, the magnetic one-form symmetry $U(1)^{(1)}_M$ is more difficult to understand in the current presentation. This is because the 't Hooft lines T_m that are charged under it are not easily expressible in terms of a. Instead, here it is useful to pass to the electromagnetically dual description by defining the dual photon $\widehat{a}$ by $*da \propto d\widehat{a}$, in terms of which we can write the charge m 't Hooft line as

$$T_m(\gamma) = e^{im\oint_\gamma \widehat{a}}\ . \tag{2.56}$$

In this dual description, effectively the same results as for $U(1)^{(1)}_E$ hold for $U(1)^{(1)}_M$.

Exercise: Thus far, we have imposed the Bianchi identity $df = 0$ locally via $f = da$, while flux quantization follows from taking a to be a compact $U(1)$ gauge field. An alternative approach is to take f to be an unconstrained 2-form, and to impose the closedness and quantization via a Lagrange multiplier. In other words, we introduce a new dynamical field $\widehat{a}$ which couples as

$$S = \int \frac{1}{2e^2} f\wedge *f + \frac{1}{2\pi} f\wedge d\widehat{a}\ .$$

Integrating over local fluctuations of $\widehat{a}$ imposes $df = 0$, which gives the desired Bianchi identity. To furthermore get quantization of f, we take $\widehat{a}$ to be not a 1-form, but rather a *1-form gauge field*; see footnote 3. Explain why this leads to quantization of f. (*Hint: sum over fluxes of* $\widehat{a}$). Next, integrate out f to obtain a theory involving only $\widehat{a}$. What happens to the coupling constant when we do so? This is known as *electromagnetic-* or *S-duality*.

	degree of current	dimension of symmetry generator	dimension of charged object
0-form	1	$d-1$	0
1-form	2	$d-2$	1
p-form	$p+1$	$d-p-1$	p

Table 1: The degrees and dimensions of objects involved in higher-form symmetries.

2.6 Higher-form symmetries

The one-form symmetries discussed above are the simplest example of so-called *higher-form symmetries*. In particular, a continuous *p-form global symmetry* has a conserved current which is a $(p+1)$-index antisymmetric tensor $j_{\mu_1\dots\mu_{p+1}}$ satisfying a conservation equation

$$\partial^{\mu_1} j_{\mu_1\dots\mu_{p+1}} = 0 \qquad \Leftrightarrow \qquad d*j = 0\,. \tag{2.57}$$

In the same way as before, we define a $(d-p-1)$-dimensional topological symmetry generator

$$U_\alpha(M_{d-p-1}) := e^{i\alpha\int_{M_{d-p-1}} *j}\,, \tag{2.58}$$

and the fact that $*j$ is closed implies that the symmetry generator is topological. These symmetry generators can link with p-dimensional objects $W_q(\Gamma_p)$, and from the Ward identity

$$\langle d*j_{p+1}(x) W_q(\Gamma_p)\dots\rangle = q\,\delta^{(d-p)}(x\in\Gamma_p)\,\langle W_q(\Gamma_p)\dots\rangle \tag{2.59}$$

we obtain the following action

$$U_\alpha(M_{d-p-1}) W_q(\Gamma_p) U_{-\alpha}(M'_{d-p-1}) = e^{iq\alpha\,\mathrm{Link}(M,\Gamma)} W_q(\Gamma_p)\,. \tag{2.60}$$

The dimensions of the various ingredients are summarized in Table 1.

Note that, as for standard currents, we may also couple higher-form currents to background gauge fields. Indeed, a p-form global symmetry can be coupled to a $(p+1)$-form background gauge field A_{p+1} via

$$S_p = \int A_{p+1}\wedge *j_{p+1}\,. \tag{2.61}$$

In this case the gauge transformation parameter of A_{p+1} is a p-form gauge field,[7]

$$A_{p+1}\to A_{p+1} + d\Lambda_p\,. \tag{2.62}$$

Conservation of the current implies invariance of the path integral under these background gauge transformations.

[7] A p-form gauge field Λ_p is defined recursively as being locally a p-form, subject to the gauge redundancy $\Lambda_p \sim \Lambda_p + d\lambda_{p-1}$ with λ_{p-1} a $(p-1)$-form gauge field, and having properly quantized fluxes $\oint \frac{d\Lambda_p}{2\pi}\in\mathbb{Z}$.

Before moving on to an example, let us make an important comment about p-form symmetries for $p \geq 1$. Whereas 0-form symmetries can be either Abelian or non-Abelian, higher-form symmetries *must* be Abelian. The reason is that since the symmetry generators have codimension greater than one, there is no sense of ordering—the configurations $\langle U_{g_1}(M_{d-p-1})U_{g_2}(M'_{d-p-1})\dots\rangle$ and $\langle U_{g_2}(M'_{d-p-1})U_{g_1}(M_{d-p-1})\dots\rangle$ are topologically identical.

Example: Let us now give a concrete example of a p-form symmetry for $p > 1$. Consider a compact boson in $(3+1)$-dimensions,

$$S = \frac{R^2}{4\pi}\int d\phi \wedge *d\phi \,, \tag{2.63}$$

where $\phi \sim \phi + 2\pi$. In this theory, we may define the following currents

$$j_s = \frac{R^2}{2\pi} d\phi \,, \qquad j_w = \frac{1}{2\pi} * d\phi \,. \tag{2.64}$$

The first of these is a usual 1-form current, which is associated to the zero-form shift symmetry of ϕ, i.e. $\phi \to \phi + \alpha$ for any constant α. The fact that it is conserved follows from the equations of motion $d * d\phi = 0$. On the other hand, j_w is conserved by the Bianchi identity, and is a 3-form current. As such, the corresponding symmetry is a 2-form symmetry, whose interpretation is as the *winding* symmetry for the compact boson.

The symmetry generators corresponding to the two symmetries are

$$\begin{aligned} U(1)^{(0)}_s : \qquad U^s_\alpha(M_3) &:= e^{\frac{iR^2\alpha}{2\pi}\int_{M_3} *d\phi} \,, \\ U(1)^{(2)}_w : \qquad U^w_\alpha(M_1) &:= e^{\frac{i\alpha}{2\pi}\int_{M_1} d\phi} \,, \end{aligned} \tag{2.65}$$

and the charged objects with respect to these symmetries are the following

$$\begin{aligned} U(1)^{(0)}_s : \qquad & \mathcal{O}_q(x) = e^{iq\phi(x)} \,, \\ U(1)^{(2)}_w : \qquad & W_n(\Gamma_2) \,. \end{aligned} \tag{2.66}$$

Here $\mathcal{O}_q(x)$ is a local operator, while $W_n(\Gamma_2)$ is a vortex operator living on the 2-cycle Γ_2.[8]

[8]The vortex operator is defined by cutting out a tubular neighborhood of Γ_2 and imposing boundary conditions such that ϕ winds n times around the excised disc, i.e. $\phi(x + 2\pi) = \phi(x) + 2\pi n$.

Exercise: Unlike in $(3+1)$-dimensions, in $(1+1)$-dimensions the momentum and winding symmetries of the compact boson are both zero-form symmetries. Rewrite the action of the $(1+1)$-dimensional massless compact boson in terms of $f = d\phi$, and then treat f as an unconstrained 1-form, enforcing the closedness and quantization via a Lagrange multiplier field $\widehat{\phi}$. Show that upon integrating out f, one obtains a theory of only $\widehat{\phi}$, which is a compact boson at the inverse radius $1/R$. This is known as *T-duality*. It is the analog of electromagnetic duality mentioned above.

More generally, consider a theory of d massless compact bosons ϕ^μ for $\mu = 1, \ldots, d$ each of period 2π, with action

$$S = \frac{1}{4\pi} \int G_{\mu\nu} d\phi^\mu \wedge *d\phi^\nu - iB_{\mu\nu} d\phi^\mu \wedge d\phi^\nu \ . \tag{2.67}$$

Repeat the above steps to obtain a dual theory

$$S = \frac{1}{4\pi} \int \widehat{G}_{\mu\nu} d\widehat{\phi}^\mu \wedge *d\widehat{\phi}^\nu - i\widehat{B}_{\mu\nu} d\widehat{\phi}^\mu \wedge d\widehat{\phi}^\nu \ ,$$

where

$$\widehat{G} = (G - BG^{-1}B)^{-1} \ , \qquad \widehat{B} = -\widehat{G}BG^{-1} \ .$$

This is a more general version of T-duality.

2.7 Anomalies

We next briefly review the notion of an anomaly. Given a theory with a p-form global symmetry, we may couple the theory to a background gauge field A_{p+1}, in which case the partition function becomes a functional of the background gauge field,

$$Z[A_{p+1}] = \int \mathcal{D}\phi^a \, e^{iS[\phi^a, A_{p+1}]} \ . \tag{2.68}$$

If the current corresponding to the p-form symmetry is classically conserved, but the partition function is not invariant under gauge transformations of A_{p+1}, and this non-invariance cannot be removed by a local counterterm, then we say that the symmetry is *anomalous*. In particular, we have

$$Z[A_{p+1} + d\lambda_p] = e^{-i\int \mathcal{A}[A_{p+1}, \lambda_p]} \times Z[A_{p+1}] \ . \tag{2.69}$$

for some $\mathcal{A}[A_{p+1}, \lambda_p]$ such that $\mathcal{A}[A_{p+1}, 0] = 0$.

One way to phrase this phase ambiguity is to say that $Z[A_{p+1}]$ is a section of a complex line bundle, or a vector in a one-dimensional complex vector space without canonical basis. Recall

that for a one-dimensional complex vector space V, each element $v \in V$ is not just a number—to extract a number from v, we need to choose a basis vector b, at which point we can define $v/b \in \mathbb{C}$. But a different choice of basis vector will give a different choice of number, and the two are related by $\frac{v}{b} = \left(\frac{b'}{b}\right)\frac{v}{b'}$ where b'/b is the analog of our anomalous phase. As such, instead of thinking of the partition function as a number $Z[A_{p+1}] \in \mathbb{C}$, we should think of it as an element $Z[A_{p+1}] \in V(M_d)$, where $V(M_d)$ is a one-dimensional vector space associated with our spacetime M_d and its background fields.

This point of view gives rise to the following useful construction (see e.g. [10–13]). We would like to associate a one-dimensional vector space to any closed d-dimensional manifold. Recall that a d-dimensional QFT associates a vector space (i.e. the Hilbert space) to each $(d-1)$-dimensional manifold, and hence, when a compatible filling exists, we can consider M_d as the boundary of a $(d+1)$-dimensional QFT A on N_{d+1}, where $\partial N_{d+1} = M_d$. The $(d+1)$-dimensional QFT A must have the special property that its Hilbert space $\mathcal{H}_A$ is one-dimensional—such an A is called an *invertible* theory.[9] We then have that $Z[A_{p+1}] \in \mathcal{H}_A(M_d)$. The different choices of anomaly are captured by different choices of invertible field theory A in $(d+1)$ dimensions. This picture of anomalies as being captured by invertible theories in one higher dimension is known as *anomaly inflow* [15, 16]. Note that the coupled bulk-boundary system has no phase ambiguities, i.e. computing the partition function of the $(d+1)$-dimensional system on the manifold N_{d+1} with boundary leads to a well-defined number.

Slightly more explicitly, if we can write the total derivative of the anomalous phase $\mathcal{A}[A_{p+1}, \lambda_p]$ as the variation of some $(d+1)$-form depending on the background gauge field A_{p+1},

$$d\mathcal{A}[A_{p+1}, \lambda_p] = \delta_{\lambda_p}\widehat{\mathcal{A}}[A_{p+1}] \,, \tag{2.70}$$

then we can consider the following modified partition function

$$\widehat{Z}[A_{p+1}] = Z[A_{p+1}] \times e^{i\int_{N_{d+1}} \widehat{\mathcal{A}}[A_{p+1}]} \,, \tag{2.71}$$

and this is now gauge invariant. We can thus think of $\widehat{\mathcal{A}}[A_{p+1}]$ as the action for the invertible field theory A in $(d+1)$-dimensions.

We caution that there are two, very distinct notions of anomalies in the literature: ones that vanish when the background fields are turned off (so-called *'t Hooft anomalies*), and ones which do not (ABJ anomalies of global currents involving dynamical gauge fields). In the former case, the anomaly is a useful property of the symmetry, which has various nice features like anomaly matching and anomaly inflow. In the latter case, the anomaly means that the symmetry is simply broken.[10] In the modern parlance, when one refers to an anomaly without any qualifiers, one is almost always referring to the former, i.e. to an 't Hooft anomaly. However, both concepts are very important, and will play an important role in these lectures.

Note that it is also possible that we can have anomalies that appear only when multiple different background gauge fields are turned on. In this case we say we have a *mixed 't Hooft*

[9] If the Hilbert space is multi-dimensional, the theory on the boundary is not just anomalous, but *relative* [14].

[10] Though, as we will see later, the symmetry can sometimes be resurrected as a non-invertible symmetry.

anomaly between the corresponding symmetries. It is easiest to illustrate this by means of an example.

Example 1: In Section 2.5, we saw that $(3+1)$-dimensional Maxwell theory had two one-form symmetries, with currents as below,

$$j^E = \frac{1}{e^2} f \,, \qquad j^M = \frac{1}{2\pi} * f \,. \tag{2.72}$$

Let us now consider coupling these currents to background gauge fields by adding,

$$\begin{aligned} S_{\text{coupling}}[B_E, B_M] &= \int \frac{1}{e^2} B_E \wedge *f + \frac{1}{2\pi} B_M \wedge f \\ &= \int \frac{1}{2e^2} (B_E \wedge *f + f \wedge *B_E) + \frac{1}{2\pi} B_M \wedge f \,. \end{aligned} \tag{2.73}$$

We are also free to add any *local counterterms* (i.e. terms built out of only background fields) to the action—for example, we can add

$$S_{\text{l.c.}}[B_E] = \frac{1}{2e^2} \int B_E \wedge *B_E \,. \tag{2.74}$$

In total then, our action becomes

$$S[B_E, B_M] = \int \frac{1}{2e^2} (f + B_E) \wedge *(f + B_E) + \frac{1}{2\pi} f \wedge B_M \,. \tag{2.75}$$

This action is clearly invariant under background gauge transformations $B_M \to B_M + d\lambda_M$, since both a and λ_M have quantized fluxes, and hence $\frac{1}{2\pi} \int f \wedge d\lambda_M \in 2\pi\mathbb{Z}$. Indeed, $f/(2\pi)$ and $d\lambda_M/(2\pi)$ represent integral cohomology classes, whose cup product has an integer pairing with the closed oriented spacetime. On the other hand, consider background gauge transformations $B_E \to B_E + d\lambda_E$. In the first term, we can absorb this transformation by redefining $f \to f' = f + d\lambda_E$.[11] However, due to the second term, doing so gives rise to

$$\delta_{\lambda_E} S[B_E, B_M] = -\frac{1}{2\pi} \int d\lambda_E \wedge B_M \,. \tag{2.76}$$

This does not in general take values in $2\pi\mathbb{Z}$, since the periods of B_M need not be quantized. Thus we see that when B_M is turned on, we lose gauge invariance of B_E. Of course, we could always try to fix this non-invariance by adding in different local counterterms. For example, we could modify the action to

$$S'[B_E, B_M] = \int \frac{1}{2e^2} (f + B_E) \wedge *(f + B_E) + \frac{1}{2\pi} (f + B_E) \wedge B_M \,. \tag{2.77}$$

[11] Note that as long as λ_E is a proper 1-form gauge field, this does not spoil the quantization of f'.

Doing so leads to invariance under background gauge transformations of B_E, but now we lose invariance under $B_M \to B_M + d\lambda_M$,

$$\delta_{\lambda_M} S'[B_E, B_M] = \frac{1}{2\pi} \int B_E \wedge d\lambda_M \ . \tag{2.78}$$

This is the hallmark of a mixed 't Hooft anomaly—we can use local counterterms to shift the anomaly from one symmetry to another, but we can never get rid of it completely. The conclusion is that $U(1)_E^{(1)}$ and $U(1)_M^{(1)}$ have a mixed 't Hooft anomaly.

While we cannot use counterterms to get rid of the anomaly, note that we *can* get rid of it by introducing the 5d inflow term

$$\widehat{\mathcal{A}} = \frac{1}{2\pi} \int_{M_5} B_E \wedge dB_M \ , \qquad \partial M_5 = M_4 \ . \tag{2.79}$$

Indeed, under $B_E \to B_E + d\lambda_E$ we have

$$\delta_{\lambda_E} \widehat{\mathcal{A}} = \frac{1}{2\pi} \int_{M_5} d\lambda_E \wedge dB_M = \frac{1}{2\pi} \int_{M_5} d(d\lambda_E \wedge B_M) = \frac{1}{2\pi} \int_{M_4} d\lambda_E \wedge B_M \ , \tag{2.80}$$

which precisely cancels the variation in (2.76), while the gauge invariance under gauge transformations of B_M is also preserved. The catch is that the coupled theory now depends on the auxiliary manifold M_5.

Example 2: We next give the most familiar example of an anomaly of a global current, namely the *ABJ anomaly* of massless QED. By massless QED, we mean Maxwell theory coupled to a charge-1 massless Dirac fermion,

$$\mathcal{L} = \frac{1}{4e^2} f_{\mu\nu} f^{\mu\nu} + i\overline{\Psi}(\partial_\mu - i a_\mu)\gamma^\mu \Psi \ . \tag{2.81}$$

Classically, there is a $U(1)_A$ axial global symmetry that acts on the fermion as[12]

$$\Psi \to e^{i\alpha\gamma_5/2}\Psi \ , \tag{2.82}$$

with the conventional axial current, twice the current normalized to generate the parameter α above,

$$j^A_\mu = \overline{\Psi}\gamma_5\gamma_\mu\Psi \ . \tag{2.83}$$

By using e.g. the Fujikawa method described in any standard textbook on QFT, one can show that this current is not conserved, and we instead have

$$d * j^A = \frac{1}{4\pi^2} f \wedge f \ . \tag{2.84}$$

[12]The normalization in the exponent is chosen such that $\alpha \sim \alpha + 2\pi$. Indeed, for $\alpha = 2\pi$ we have $\Psi \to e^{i\pi\gamma_5}\Psi = -\Psi$, which is already part of the $U(1)$ gauge group.

Clearly the right-hand side does not vanish even when background fields are set to zero—it depends on the dynamical field f, and hence indicates that the $U(1)_A$ axial symmetry is broken by a local nonconservation equation. In Section 4.3, we will see that part of this broken symmetry can actually be resurrected as a non-invertible symmetry.

Note that ABJ anomalies can be thought of as originating from a mixed 't Hooft anomaly in the ungauged theory (i.e. the theory of just a massless Dirac fermion). Gauging of $U(1)_V$ turns the corresponding background gauge field into a dynamical gauge field, and leads to breaking of the $U(1)_A$ symmetry. In general, when we have a mixed 't Hooft anomaly, we are allowed to gauge one of the symmetries (as long as there are no self-anomalies), and this will lead to breaking of the other symmetry.

2.8 Discrete gauge theories

Thus far we have been focusing on continuous symmetries, but discrete symmetries also play an important role in the study of generalized symmetries. In particular, it will be important for us to understand how to couple such symmetries to gauge fields.

To do so, let us first recall some facts about the description of continuous gauge fields. We proceed as follows,

1. Choose an open cover $\{U_i\}$ of the spacetime M_d.
2. Choose a trivialization (i.e. a local section) in each patch.
3. On overlaps, the trivializations are related by transition functions $g_{ij} : U_i \cap U_j \to G$.
4. On triple overlaps $U_i \cap U_j \cap U_k$, the g_{ij} are required to satisfy $g_{ij}g_{jk}g_{ki} = 1$.

To specify a gauge field, on each patch we choose a Lie-algebra-valued connection $A^{(i)}$, and the connections on adjacent patches are related via

$$A^{(j)} = g_{ij}^{-1} A^{(i)} g_{ij} + g_{ij}^{-1} dg_{ij} \ . \tag{2.85}$$

The gauge field contains important geometric data about the bundle, captured by the curvature $F = dA + A \wedge A$. It also determines holonomies around loops, which depend only on their homotopy classes when the connection is flat.

For a discrete group G on the other hand, there is no nonzero Lie algebra, and hence no local connection degrees of freedom or curvature. Indeed, discrete gauge fields must always be (locally) flat. One way to understand this is to note that the holonomy should depend continuously on the path γ, but since the holonomy is valued in a discrete group it cannot change continuously, and hence is expected to be constant upon continuous deformation of γ. This means that the holonomy of any contractible cycle is trivial, which expresses local flatness.

Thus in the discrete case, all of the data of the gauge field is contained in the holonomies around non-contractible cycles, and this data is in turn contained in the transition functions

described above. As such, specifying the gauge field basically just amounts to specifying the transition functions, each of which we may imagine as a symmetry generator $U_{g_{ij}}(M_{d-1})$ implementing the corresponding g_{ij}.[13] In other words, in the discrete case, a background gauge field is equivalent to a fine mesh of topological defects. By "fine mesh," we mean that we first choose a triangulation of the manifold, and then insert codimension-1 defects on the corresponding cells of the dual decomposition. Different triangulations give rise to different meshes, each of which are related by gauge transformations.

2.8.1 BF theory description

In the case of an Abelian discrete group such as $\mathbb{Z}_n$, we may write the corresponding discrete gauge theory as follows,

$$S_{BF} = \frac{n}{2\pi} \int_{M_d} a_{p+1} \wedge db_{d-p-2} \tag{2.86}$$

with a and b respectively the compact $U(1)$ gauge fields for p- and $(d-p-3)$-form symmetries, whose field strengths have periods in $2\pi\mathbb{Z}$. If b_{d-p-2} is dynamical, we claim that the above action makes a_{p+1} into a $\mathbb{Z}_n$-valued $(p+1)$-form gauge field.[14]

First, let us check that S_{BF} is invariant under the following gauge transformations,

$$a_{p+1} \to a_{p+1} + d\lambda_p \ , \qquad b_{d-p-2} \to b_{d-p-2} + d\widehat{\lambda}_{d-p-3} \ . \tag{2.87}$$

For example, the first gives

$$\delta_\lambda S_{BF} = \frac{n}{2\pi} \int_{M_d} d\lambda_p \wedge db_{d-p-2} \ , \tag{2.88}$$

and on closed, oriented M_d, the integral below is the pairing of integral cohomology classes,

$$\delta_\lambda S_{BF} = 2\pi n \int_{M_d} \frac{d\lambda_p}{2\pi} \wedge \frac{db_{d-p-2}}{2\pi} \in 2\pi n\mathbb{Z} \ . \tag{2.89}$$

Hence as long as $n \in \mathbb{Z}$, the theory is gauge invariant.

It is clear that the equation of motion of b_{d-p-2} imposes that a_{p+1} is locally flat, i.e. $da_{p+1} = 0$, as expected for a discrete gauge field. But we can actually say more. To illustrate the global constraint, consider one independent flux through Σ_{d-p-1} and the holonomy on its dual cycle Σ_{p+1}. Note that the gauge field b_{d-p-2} may be split into different topological sectors labelled by $k := \oint_{\Sigma_{d-p-1}} \frac{db_{d-p-2}}{2\pi} \in \mathbb{Z}$. In each topological sector, we may then write the gauge field as a sum over two pieces—a certain representative in that sector, together with an element in the trivial topological sector. Hence when we perform the path integral in this theory, we may split

[13] Here we are considering a zero-form symmetry, so that the symmetry generators live on codimension-1 loci. For higher-form symmetries, analogous statements hold, see e.g. [17].

[14] We can take a_{p+1} to be either background or dynamical. When it is dynamical, this action also turns b_{d-p-2} into a $\mathbb{Z}_n$-valued gauge field.

the path integral over the gauge field b_{d-p-2} into a sum over k, as well as a path integral over the fields in the zero-flux sector. Integrating over the zero-flux fluctuations imposes $da_{p+1} = 0$. The sum over the remaining integer flux then gives

$$\sum_{k\in\mathbb{Z}} e^{ink\int_{\Sigma_{p+1}} a_{p+1}} \;=\; 2\pi \sum_{k\in\mathbb{Z}} \delta\left(n\int_{\Sigma_{p+1}} a_{p+1} - 2\pi k\right) , \tag{2.90}$$

with an analogous constraint for each independent holonomy. The periodic delta function means that Wilson $(p+1)$-surfaces are forced to satisfy $W^n = 1$, which is exactly as expected for a $\mathbb{Z}_n$ gauge theory.

Note that in the special case of $p = 1$ in $d = 4$, it is conventional to write the 2-form gauge field a_2 as b and the 2-form field strength db_1 as f, in which case the action looks like

$$S_{BF} = \frac{n}{2\pi}\int_{M_4} b\wedge f \; . \tag{2.91}$$

For this reason, discrete gauge theories written in this way are known as *BF theories*. They will play a role in several places below.

2.8.2 Topological boundary conditions for BF theory

It will also be useful to understand the topological boundary conditions of BF theory. There are two obvious boundary conditions that one can consider,[15]

$$\begin{aligned} \text{Dirichlet}: &\quad a_{p+1}|_\partial = 0 \, , \\ \text{Neumann}: &\quad b_{d-p-2}|_\partial = 0 \, , \end{aligned} \tag{2.92}$$

where the notions of "Dirichlet" and "Neumann" depend on whether we are considering the theory as an a_{p+1} gauge theory or a b_{d-p-2} gauge theory—here we are taking the former perspective. That these are legitimate boundary conditions follows from the fact that

$$\begin{aligned} \delta S \;&=\; \frac{n}{2\pi}\int_{M_d^{\geq 0}} \left[\delta a_{p+1}\wedge db_{d-p-2} + a_{p+1}\wedge d(\delta b_{d-p-2})\right] \\ &=\; \frac{n}{2\pi}\int_{M_d^{>0}} \left[\delta a_{p+1}\wedge db_{d-p-2} - (-1)^{p+1} da_{p+1}\wedge\delta b_{d-p-2}\right] \\ &\qquad\qquad +(-1)^{p+1}\frac{n}{2\pi}\int_{M_{d-1}|_0} a_{p+1}\wedge\delta b_{d-p-2} \, , \end{aligned} \tag{2.93}$$

where the boundary term vanishes in both cases.

An important property of these boundary conditions is that they are *topological*, in the sense that any infinitesimal change in the location of the boundary does not affect correlation

[15] More generally, one could fix the right-hand side to a non-trivial flat background field, but for simplicity we just set it to be zero.

functions. This may be shown by an argument similar to the proof of topologicalness of the symmetry generators in Section 2.3. For example, for the Dirichlet boundary condition we imagine changing from boundary M_{d-1} to some infinitesimally different M'_{d-1} continuously along a direction r. Choosing a gauge in which the normal components of a_{p+1} vanish near the boundary gives

$$a_{p+1}|_{M_{d-1}} - a_{p+1}|_{M'_{d-1}} = \int_r da_{p+1} = 0 \,, \tag{2.94}$$

where we have used that $da_{p+1} = 0$ locally by the equations of motion.

In higher dimensions there are many additional topological boundary conditions.[16] Later, we will use the topological nature of the Dirichlet boundary condition to construct non-invertible defects by gauging a discrete symmetry in half of space.

2.8.3 Discrete cocycle description

An alternative description of discrete gauge theory is to work directly with discrete p-cochains $a \in C^p(M_d, \mathbb{Z}_n)$, which satisfy $\oint a \in \mathbb{Z}_n$ by definition. As opposed to the case of continuous gauge fields, for which the Wilson p-surfaces were written as $e^{i\oint a}$, in this case it is more natural to normalize the Wilson p-surfaces as

$$W(\Gamma_p) = e^{i\frac{2\pi}{n}\oint_{\Gamma_p} a} \,; \tag{2.95}$$

this ensures that the $a \sim a + n$ redundancy in $\mathbb{Z}_n$ p-cochains does not lead to an ambiguity in W, i.e. $W(\Gamma_p) = e^{i\frac{2\pi}{n}\oint_{\Gamma_p} a} \cong e^{i\frac{2\pi}{n}\oint_{\Gamma_p}(a+n)} = e^{2\pi i} e^{i\frac{2\pi}{n}\oint_{\Gamma_p} a} = W(\Gamma_p)$. More generally, given an expression written in terms of continuous gauge fields, we may obtain one in terms of discrete gauge fields by making the replacement

$$a^{\text{cont.}} \longrightarrow \frac{2\pi}{n} a^{\text{disc.}} \,. \tag{2.96}$$

Additionally, in the discrete notation the wedge product and exterior derivative are replaced by analogous operations,

$$\wedge \longrightarrow \cup \,, \qquad\qquad d \longrightarrow \delta \,. \tag{2.97}$$

Thus, for example, in discrete notation the BF action in (2.86) would take the following form,

$$S_{BF} = \frac{2\pi}{n} \int_{M_d} a_{p+1} \cup \delta b_{d-p-2} \,. \tag{2.98}$$

One might think that the BF term is somewhat meaningless now that a and b are already assumed to be $\mathbb{Z}_n$-valued, but it still tells us that $\delta a_{p+1} = 0$, so that a_{p+1} is not just a $\mathbb{Z}_n$-cochain, but a $\mathbb{Z}_n$-cocycle.

[16] In $d \geq 3+1$, there are an infinite number of simple topological boundary conditions that can be obtained by stacking with decoupled $(d-1)$-dimensional TQFTs. In $d = 2+1$ the situation is better, since stacking with non-trivial $(1+1)$-dimensional TQFTs gives non-simple boundary conditions.

Finally, let us mention that because the gauge fields are now discrete, all path integrals involving $a^{\text{cont.}}$ are to be replaced by sums over $a^{\text{disc.}}$. In particular, we would like to replace the integral over $a^{\text{cont.}}$ with a sum over flat $a^{\text{disc.}} \in Z^{p+1}(M_d, \mathbb{Z}_n)$, with Z^{p+1} being the set of $(p+1)$-cocycles. However, note that a is a cocycle subject to a gauge redundancy, $a_{p+1} \sim a_{p+1} + \delta\lambda_p$, and we must divide out by this redundancy. At first glance, λ_p is an element of $C^p(M_d, \mathbb{Z}_n)$, i.e. it is a p-cochain, but note that λ_p is itself subject to gauge transformations $\lambda_p \to \lambda_p + \delta\lambda_{p-1}$, with λ_{p-1} an element of $C^{p-1}(M_d, \mathbb{Z}_n)$. Likewise, λ_{p-1} is subject to a gauge transformation in terms of an element of $C^{p-2}(M_d, \mathbb{Z}_n)$, and so on and so forth. The correct replacement is thus as follows,

$$\int \mathcal{D}a \quad \longrightarrow \quad \prod_{i=1}^{p+1} |C^{p+1-i}(M_d, \mathbb{Z}_n)|^{(-)^i} \sum_{a \in Z^{p+1}(M_d, \mathbb{Z}_n)} . \tag{2.99}$$

In fact, it is more common to rewrite the sum in a slightly different way. To do so, recall that we may define cohomology groups as

$$H^p(M_d, \mathbb{Z}_n) = \frac{Z^p(M_d, \mathbb{Z}_n)}{B^p(M_d, \mathbb{Z}_n)} , \tag{2.100}$$

where B^p is the group of exact p-cochains. In particular, we have $|B^p| = |C^{p-1}|/|Z^{p-1}|$ and $|B^0| = 1$, where for simplicity we dropped the arguments of the various groups involved. These then give

$$|B^p| = \frac{|C^{p-1}|}{|Z^{p-1}|} = \frac{|C^{p-1}|}{|H^{p-1}||B^{p-1}|} = \frac{|C^{p-1}||Z^{p-2}|}{|H^{p-1}||C^{p-2}|} = \frac{|C^{p-1}||H^{p-2}|}{|H^{p-1}||C^{p-2}|}|B^{p-2}| , \tag{2.101}$$

and repeating this sort of manipulation eventually gives

$$|B^p| = \prod_{i=1}^{p} \left(\frac{|H^{p-i}|}{|C^{p-i}|} \right)^{(-)^i} . \tag{2.102}$$

Using this, we conclude that as long as the summand is gauge invariant, we may rewrite

$$\begin{aligned} \prod_{i=1}^{p+1} |C^{p+1-i}|^{(-)^i} \sum_{a \in Z^{p+1}} &= |B^{p+1}| \prod_{i=1}^{p+1} |C^{p+1-i}|^{(-)^i} \sum_{a \in H^{p+1}} \\ &= \prod_{i=1}^{p+1} |H^{p+1-i}|^{(-)^i} \sum_{a \in H^{p+1}} . \end{aligned} \tag{2.103}$$

Thus, for example, for a zero-form symmetry we replace

$$\int \mathcal{D}a \quad \longrightarrow \quad \frac{1}{|H^0(M_d, \mathbb{Z}_n)|} \sum_{a \in H^1(M_d, \mathbb{Z}_n)} , \tag{2.104}$$

while for a one-form symmetry we replace

$$\int \mathcal{D}a \quad \longrightarrow \quad \frac{|H^0(M_d, \mathbb{Z}_n)|}{|H^1(M_d, \mathbb{Z}_n)|} \sum_{a \in H^2(M_d, \mathbb{Z}_n)} . \tag{2.105}$$

Bocksteins: Though it will not play much of a role below, let us briefly mention a bit of math that appears throughout the literature on this subject; this material can be skipped on a first reading. Associated to any short exact sequence of Abelian groups

$$0 \to K \to G \to H \to 0 \tag{2.106}$$

is a long exact sequence of cohomology groups

$$\cdots \to H^p(M_d, G) \to H^p(M_d, H) \xrightarrow{\beta} H^{p+1}(M_d, K) \to \dots \tag{2.107}$$

We will be interested in the map $\beta : H^p(M_d, H) \to H^{p+1}(M_d, K)$, which can be described as follows. Begin with a cocycle $a \in C^p(M_d, H)$ with $\delta a = 0$, and then lift it to a cochain $\tilde{a} \in C^p(M_d, G)$, such that the projection to H reproduces a. Note that $\delta\tilde{a} \in C^{p+1}(M_d, G)$ is not necessarily 0, but since a was closed it must be in the image of the map from K to G. We may then use this to define a $[\delta\tilde{a}] \in H^{p+1}(M_d, K)$, identifying K with its image in G, which we denote by $\beta([a]) := [\delta\tilde{a}]$, referred to as the *Bockstein* of a. Morally, it tells us the obstruction to a being flat when lifted to G.

As a warm-up example, consider the short-exact sequence[17]

$$0 \to \mathbb{Z}_2 \xrightarrow{\imath} \mathbb{Z}_4 \xrightarrow{\pi} \mathbb{Z}_2 \to 0 \tag{2.108}$$

with $\imath$ the inclusion map and π a projection. Given a cocycle $a \in Z^p(M_d, \mathbb{Z}_2)$, we choose a lift $\tilde{a} \in C^p(M_d, \mathbb{Z}_4)$ such that $\pi(\tilde{a}) = a$, and in particular $\pi(\delta\tilde{a}) = \delta a = 0$. The latter means that $\delta\tilde{a}$ lies in the image of $\imath : \mathbb{Z}_2 \to \mathbb{Z}_4$, so there exists a unique $x \in C^{p+1}(M_d, \mathbb{Z}_2)$ such that $\delta\tilde{a} = \imath(x)$, and from our above general discussion $\beta(a) = x$. Concretely, the inclusion map $\imath$ is given simply by multiplying by 2, so we have $\delta\tilde{a} = 2\beta(a)$, or

$$\beta(a) = \frac{1}{2}\delta\tilde{a} \mod 2 \,, \tag{2.109}$$

where the division is done in the even subgroup of $\mathbb{Z}_4$. Note that we can change the choice of $\mathbb{Z}_4$ lift $\tilde{a}$ by an element of the kernel of π, i.e. $\tilde{a} \to \tilde{a} + 2\lambda$ for $\lambda \in C^p(M_d, \mathbb{Z}_2)$, but this only gives rise to a change $\beta(a) \to \beta(a) + \delta\lambda$, so that the cohomology class of $\beta(a)$ is independent of the particular lift.

We now move on to the example of particular interest to us, namely the short-exact sequence

$$0 \to \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \to \mathbb{Z}_n \to 0 \,. \tag{2.110}$$

If $a \in Z^p(M_d, \mathbb{Z}_n)$ is a discrete cocycle, let $\tilde{a} \in C^p(M_d, \mathbb{Z})$ be an integer lift. Because $\delta a = 0 \mod n$, we have that $\delta\tilde{a} = nx$ for some integer $(p+1)$-cochain x, and the Bockstein is given by

$$\beta(a) = \frac{1}{n}\delta\tilde{a} \,. \tag{2.111}$$

[17] In this context, the Bockstein operation is equivalent to the first Steenrod square.

We can also write the discrete BF action using integer lifts. First, starting with $a, b \in C^*(M_d, \mathbb{Z}_n)$, we choose integer lifts $\tilde{a}, \tilde{b} \in C^*(M_d, \mathbb{Z})$, and then write

$$S = \frac{2\pi}{n} \int a \cup \delta b = \frac{2\pi}{n} \int \tilde{a} \cup \delta \tilde{b} \tag{2.112}$$

where the choice of integer lift does not matter since the action is only defined modulo $2\pi\mathbb{Z}$. Now if we further take b to be flat, then we may make use of the Bockstein from above, which gives its value on flat b

$$S = 2\pi \int \tilde{a} \cup \beta(b) \ = 0 \mod 2\pi \ . \tag{2.113}$$

This vanishing on flat b does not replace the off-shell BF action, whose sum over b imposes flatness of a.

2.9 Dual Symmetries and Twisted Sectors

We next discuss the gauging of non-anomalous $\mathbb{Z}_n$ p-form symmetries at the level of the partition function/spectrum of the theory. To do this, we couple the theory $\mathcal{X}$ to a background gauge field A_{p+1} for $\mathbb{Z}_n$, and then make it dynamical by performing a sum to get

$$Z_{\mathcal{X}/\mathbb{Z}_n} \propto \sum_{a_{p+1}} Z_{\mathcal{X}}[a_{p+1}] \ . \tag{2.114}$$

When we perform the sum, we are also free to add a coupling to another background field $\widehat{A}_{d-p-1}$ to give

$$Z_{\mathcal{X}/\mathbb{Z}_n}[\widehat{A}_{d-p-1}] \propto \sum_{a_{p+1}} e^{\frac{2\pi i}{n} \int a_{p+1} \cup \widehat{A}_{d-p-1}} Z_{\mathcal{X}}[a_{p+1}] \ . \tag{2.115}$$

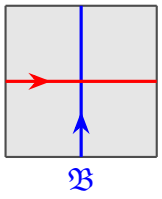
This field $\widehat{A}_{d-p-1}$ can be thought of as the background field for a $\mathbb{Z}_n$ $(d-p-2)$-form symmetry, which is known as the *dual* or *quantum symmetry* [18].

A special class of examples are those for which $p = d - p - 2$, i.e. $p = \frac{d-2}{2}$. This includes zero-form symmetries in $d = 2$ or one-form symmetries in $d = 4$. In this case the original symmetry and the dual symmetry are the same, and thus it is sometimes possible that the theory after gauging is the *same* as the theory before gauging. As will be discussed in Sections 3 and 4, this leads to one of the simplest constructions of non-invertible defects, known as "half-space gauging."

Let us now focus for simplicity on the case of zero-form symmetries in two-dimensions. In this case, we will write simply $a = a_1$, and we have

$$Z_{\mathcal{X}/\mathbb{Z}_n}[\widehat{A}] = \frac{1}{n} \sum_{a} e^{\frac{2\pi i}{n} \int a \cup \widehat{A}} Z_{\mathcal{X}}[a] \ , \tag{2.116}$$

with overall coefficient as given in (2.104); note that $|H^0(M_2, \mathbb{Z}_n)| = n$ for connected M_2. Now let us translate this into the notation of topological defects. For simplicity, take the space-time to be a torus, with $\mathfrak{A}$- and $\mathfrak{B}$-cycles as follows,



, (2.117)

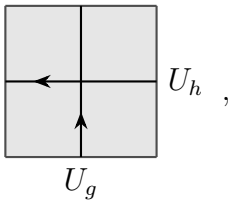
and with intersection pairing $\langle \mathfrak{A}, \mathfrak{B} \rangle = -\langle \mathfrak{B}, \mathfrak{A} \rangle = 1$. A generic 1-cocycle can be written, up to a coboundary, as a linear combination of the Poincaré duals of $\mathfrak{A}$ and $\mathfrak{B}$, and we choose to write

$$a = -h\,\omega_{\mathfrak{A}} + g\,\omega_{\mathfrak{B}}\,, \qquad \widehat{A} = -\widehat{h}\,\omega_{\mathfrak{A}} + \widehat{g}\,\omega_{\mathfrak{B}} \tag{2.118}$$

such that

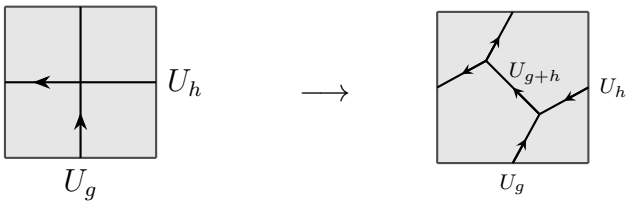
$$\int_{\mathfrak{A}} a = g\,, \qquad \int_{\mathfrak{B}} a = h\,, \qquad \int_{\mathfrak{A}} \widehat{A} = \widehat{g}\,, \qquad \int_{\mathfrak{B}} \widehat{A} = \widehat{h}\,, \tag{2.119}$$

with $g, h \in \mathbb{Z}_n$. In other words, the gauge field a on the torus corresponds to the following mesh of defects



, (2.120)

or, more precisely, since only trivalent junctions of defects are defined,[18]



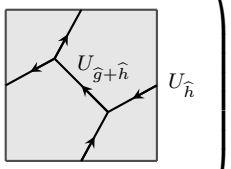
. (2.121)

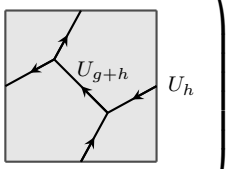
The sum over a then becomes simply a sum over g, h. As for the coupling to the dual gauge field, we have

$$\begin{aligned} \int a \cup \widehat{A} &= \int (-h\,\omega_{\mathfrak{A}} + g\,\omega_{\mathfrak{B}}) \cup (-\widehat{h}\,\omega_{\mathfrak{A}} + \widehat{g}\,\omega_{\mathfrak{B}}) \\ &= -h\widehat{g} \int \omega_{\mathfrak{A}} \cup \omega_{\mathfrak{B}} - g\widehat{h} \int \omega_{\mathfrak{B}} \cup \omega_{\mathfrak{A}} \\ &= g\widehat{h} - h\widehat{g}\,. \end{aligned} \tag{2.122}$$

[18] There is an obvious ambiguity in how we resolve the 4-valent junction into trivalent vertices, but the two possibilities should give the same result as long as the symmetry is non-anomalous. This will be discussed more in Section 3.1.

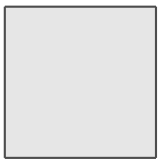
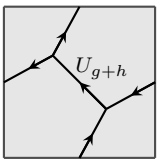
We thus conclude that the expression for the gauged partition function (2.116) can be written in terms of defects as follows,

$$Z_{\mathcal{X}/\mathbb{Z}_n}\left(\right.$$ 

$$\left.\right) = \frac{1}{n}\sum_{g,h\in\mathbb{Z}_n} e^{\frac{2\pi i}{n}(g\widehat{h}-h\widehat{g})}\; Z_{\mathcal{X}}\left(\right.$$ 

$$\left.\right) , \qquad (2.123)$$

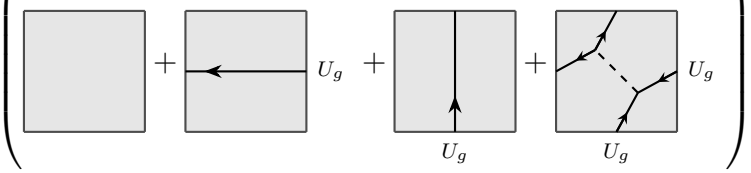
where g, h capture the dynamical gauge field a and are thus summed over, whereas $\widehat{g}, \widehat{h}$ specify the background gauge field $\widehat{A}$. In particular, if we are only interested in the case in which $\widehat{A}$ is turned off, then we have the simpler expression

$$Z_{\mathcal{X}/\mathbb{Z}_n}\left(\right.$$ 
$$\left.\right) = \frac{1}{n}\sum_{g,h\in\mathbb{Z}_n} Z_{\mathcal{X}}\left(\right.$$ 

$$\left.\right) . \qquad (2.124)$$

Physically, this may be understood as follows. First, if we focus on the terms with $g = 0$ on the right-hand side, we see that we are simply inserting $\frac{1}{n}\sum_{h\in\mathbb{Z}_n} U_h$ on the horizontal cycle. If we take our foliation to be in the vertical direction, this corresponds to inserting the operator $\frac{1}{n}\sum_{h\in\mathbb{Z}_n} U_h$ into the path integral, which projects onto $\mathbb{Z}_n$-invariant states. Thus the terms with $g = 0$ just capture the states in the original theory which are invariant under $\mathbb{Z}_n$, and hence which survive the gauging. On the other hand, the states with g inserted vertically correspond to *twisted sector* states, as we have already discussed below (2.22). For each fixed g, we again have the projector $\frac{1}{n}\sum_{h\in\mathbb{Z}_n} U_h$ inserted, so we are projecting onto invariant states in the twisted sectors as well.

To be even more explicit, let us focus on the case of $n = 2$, in which case we have

$$Z_{\mathcal{X}/\mathbb{Z}_2} = \frac{1}{2}\left(\right.$$ 

$$\left.\right) \qquad (2.125)$$

in terms of the single generator g. Note that we have dropped the $Z_{\mathcal{X}}$ on the right-hand side and represented the partition function simply by the picture of the torus itself, as is standard in the literature. The first two terms give the $\mathbb{Z}_2$-invariant states of the original theory, while the second two terms correspond to the even twisted sector states of the original theory.

An extremely confusing point is that in the older literature, the first two terms are referred to as the untwisted sector of the gauged theory, while the second two terms are referred to as the

twisted sector of the gauged theory. On the other hand, in the modern terminology, one would call both of these sectors untwisted in the gauged theory. Indeed, in modern parlance, any state which can be prepared using a local operator (and hence is counted by the partition function) is untwisted. In the modern language, the original theory contains both an untwisted sector and twisted sector, and only the former contributes to the original partition function—states in the latter are created by twist operators attached to a $\mathbb{Z}_2$ line, and hence do not contribute to the partition function. Then when we gauge the $\mathbb{Z}_2$ symmetry, part of the untwisted sector is projected out into the twisted sector, and conversely part of the twisted sector becomes untwisted. The gauged partition function ultimately only counts untwisted sector states in the gauged theory, but the origin of these untwisted sectors can be either the untwisted or twisted sectors in the ungauged theory.

More explicitly, let us separate states of the ungauged theory $\mathcal{X}$ into four sectors, depending on if they are $\mathbb{Z}_2$ even/odd, and if they are untwisted/twisted (i.e. if the operator creating them is a local operator, or an operator attached to a $\mathbb{Z}_2$ line),

$\mathcal{X}$	untwisted	twisted
even	S	U
odd	T	V

(2.126)

Upon gauging, we lose the $\mathbb{Z}_2$ symmetry (all surviving states are invariant under it) but we gain a dual $\widehat{\mathbb{Z}}_2$ symmetry, and can again separate states of the gauged theory $\mathcal{X}/\mathbb{Z}_2$ into four sectors, depending on if they are $\widehat{\mathbb{Z}}_2$ even/odd, and if they are untwisted/twisted (i.e. if the operator creating them is a local operator, or an operator attached to a $\widehat{\mathbb{Z}}_2$ line),

$\mathcal{X}/\mathbb{Z}_2$	untwisted	twisted
even	S	T
odd	U	V

(2.127)

We see that upon discrete gauging, we never really lose or gain states—they just get reorganized. In the current case of $\mathbb{Z}_2$ gauging, the $\mathbb{Z}_2$-even twisted sector states in $\mathcal{X}$ become $\widehat{\mathbb{Z}}_2$-odd untwisted sector states in $\mathcal{X}/\mathbb{Z}_2$ and vice versa. The partition function only ever contains contributions from the first column.

2.10 Higher-group symmetries

Above we have considered theories with both zero-form and higher-form symmetries—for example, the compact boson in $(3+1)$ dimensions. An interesting phenomenon is that sometimes the zero- and higher-form symmetries can combine in a non-trivial extension to form a *higher-group symmetry*, see e.g. [19–21]. Concretely, consider a theory with a collection of p_i-form symmetries with $p_1 < p_2 < \cdots < p_n$. If we had a simple direct product of these symmetries, then each would have independent gauge transformations,

$$\delta_i A_{p_i+1} = d\lambda_{p_i} \ . \tag{2.128}$$

But we can also consider the case in which the gauge transformations are given by

$$\delta A_{p_i+1} = d\lambda_{p_i} + \sum_{j\leq i} \lambda_{p_j} \wedge \alpha_j^{(i)}(\{A_{p_j+1}\}) + \dots \tag{2.129}$$

where $\alpha_j^{(i)}$ are $(p_i - p_j + 1)$-forms that depend on A_{p_j+1} for $j < i$, and the dots represent terms which are non-linear in gauge transformation parameters. This corresponds to a non-trivial extension of the p_1-form symmetry by the higher-form symmetries, and we call the result a $(p_n + 1)$-*group* symmetry.

It is useful to jump directly into an example here. Say that we have a $(3 + 1)$-dimensional theory on M_4 with a zero-form symmetry $\mathbb{Z}_2^{(0)}$ and a one-form symmetry $\mathbb{Z}_2^{(1)}$. Let us denote the background gauge fields for these two symmetries by A_1 and B_2, and further assume that they have a mixed 't Hooft anomaly given by the inflow action

$$\widehat{\mathcal{A}}[A_1, B_2] = \pi \int_{M_5} A_1 \cup B_2 \cup B_2 \,, \qquad \partial M_5 = M_4 \,. \tag{2.130}$$

We now gauge $\mathbb{Z}_2^{(0)}$ by promoting the background field A_1 to a dynamical field a_1,

$$Z_{\mathcal{X}/\mathbb{Z}_2}[\widehat{A}_3, B_2] = \frac{1}{2} \sum_{a_1 \in H^1(M_4, \mathbb{Z}_2)} (-1)^{\int a_1 \cup \widehat{A}_3} Z_{\mathcal{X}}[a_1, B_2] \tag{2.131}$$

where $\widehat{A}_3$ is the background gauge field for the dual $\mathbb{Z}_2^{(2)}$ symmetry.

When we perform this gauging, the anomaly (2.130) naively gives us two choices: we either remain in $(3 + 1)$ dimensions and break the $\mathbb{Z}_2^{(1)}$ symmetry, or we couple the $(3 + 1)$-dimensional theory to a $(4+1)$-dimensional bulk theory cancelling the anomaly, in which case we preserve the $\mathbb{Z}_2^{(1)}$ symmetry. However, there is actually another option. Let us begin by adding in the inflow term, which gives the gauged partition function,

$$Z_{\mathcal{X}/\mathbb{Z}_2}[\widehat{A}_3, B_2] = \frac{1}{2} \sum_{a_1} Z_{\mathcal{X}}[a_1, B_2]\, e^{i\pi \int_{M_5} a_1 \cup (B_2)^2} e^{i\pi \int_{M_4} a_1 \cup \widehat{A}_3} \,. \tag{2.132}$$

From this expression, we see that we can actually remove the M_5 dependence by modifying the standard cocycle condition $\delta \widehat{A}_3 = 0$ of the dual background gauge field to the twisted cocycle condition $\delta \widehat{A}_3 = -B_2 \cup B_2$. One effect of this is that the gauge transformations of B_2 and $\widehat{A}_3$ can no longer be separated—indeed, transformations of B_2 must be accompanied by transformations of $\widehat{A}_3$ in order to maintain the twisted cocycle condition. The fact that the $\mathbb{Z}_2^{(1)}$ and $\widehat{\mathbb{Z}}_2^{(2)}$ gauge transformations are related in this way means that the full group is a non-trivial extension of the former by the latter,

$$0 \to \widehat{\mathbb{Z}}_2^{(2)} \to G^{(3)} \to \mathbb{Z}_2^{(1)} \to 0 \,. \tag{2.133}$$

This is a concrete example of a *3-group*.

Further References

The realization of symmetries as topological defects and the subsequent generalization to higher-form symmetries is due to [17, 22, 23], with some earlier results pointing to the existence of higher-form symmetries in [24–26]. Perhaps the earliest "killer app" of higher-form symmetry was that of [27], and by now the techniques have become absolutely ubiquitous, see e.g. [28–62]. The generalization to higher-group symmetries and the exploration of their applications can be found in [19–21, 63–72]. The anomaly inflow formalism originated in the work of [15, 16], with the modern formulation being given in [11–13]. A useful summary of many of the techniques involved in the study of discrete gauge theories can be found in [17].

3 Topological Defect Lines in Two Dimensions

Thus far, all of the symmetries that we have discussed have had the structure of a (higher-)group. In particular, the fusion of two symmetry generators always gave a unique third element, labelled by the composition of group elements,

$$U_{g_1} \times U_{g_2} = U_{g_1 g_2} \ , \tag{3.1}$$

and every symmetry defect U_g had an inverse $U_{g^{-1}}$ labelled by the inverse element.

More generally, it can be possible for the fusion of two topological defects to produce a *sum* of topological defects,

$$U_{g_1} \times U_{g_2} = U_{g_3} + U_{g_4} + \dots \ . \tag{3.2}$$

The meaning of the sum here is that, when inserted into a correlation function, we have

$$\langle U_{g_1} \times U_{g_2} \dots \rangle = \langle U_{g_3} \dots \rangle + \langle U_{g_4} \dots \rangle + \dots \ . \tag{3.3}$$

One can likewise define the Hilbert space associated with $U_{g_3} + U_{g_4} + \dots$ to be $\mathcal{H}_{g_3} \oplus \mathcal{H}_{g_4} \oplus \dots$. Requiring that there be a Hilbert space interpretation tells us that the coefficients in these linear combinations must be non-negative integers.

Because topological defects with the above fusion rules do not have an inverse, we say that they generate *non-invertible symmetries*. The reason that they are still referred to as symmetries in the first place is that, as we will see later, just like standard symmetries they are invariants of symmetry-preserving RG flows, and they can be used to obtain a variety of selection rules. It is also possible to label operators in the theory by representations of these non-invertible symmetries.

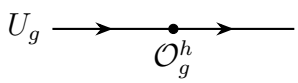
By far the most well-understood examples of non-invertible symmetries are in two-dimensions, where the mathematical structure takes the form of a *fusion category*. Standard physics references on this topic include [73–75]. Below, we will begin by rephrasing the structure of familiar group symmetry in the categorical language, before moving on to the more general case. Our treatment will be far from mathematically rigorous, and will only highlight the salient features needed for practicing physicists. Readers looking for a more rigorous treatment should consult one of the works listed above.

3.1 Group symmetry in categorical language

Let us begin by focusing on the case of an invertible finite zero-form symmetry G in $(1+1)$-dimensions. We may describe this symmetry with the following data,

- **Objects:** topological lines U_g for each $g \in G$. Each line is endowed with an orientation, and every line U_g has a unique orientation reversal $\overline{U}_g$ given by the inverse element $\overline{U}_g = U_{g^{-1}}$.

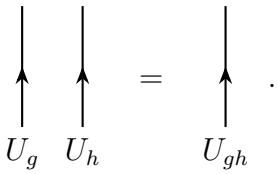
- **Morphisms:** topological local operators $\mathcal{O}_g^h$ that act as homomorphisms between two topological lines, i.e. $\mathcal{O}_g^h \in \mathrm{Hom}(U_g, U_h)$. Pictorially,



(3.4)

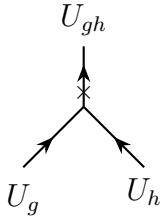
  By definition, for *simple* lines $g \neq h$, the morphism space $\mathrm{Hom}(U_g, U_h)$ is trivial, i.e. there does not exist any topological local operator between the two lines, whereas if $g = h$ the morphism space is one-dimensional, corresponding to the trivial operator on the line U_g. For non-simple lines, the morphism space contains projection operators onto simple constituents.

The data of objects and morphisms furnishes the structure of a category, but for categories describing symmetries, there is an extra necessary ingredient known as the *fusion* of line operators. Globally, this fusion can be thought of as follows,

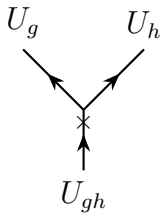


(3.5)

Locally, two lines can be fused to obtain a third line by using an operator in $\mathrm{Hom}(U_g \otimes U_h, U_{gh})$. A similar thing is done for the splitting junction, and each of these can be represented by a trivalent vertex,

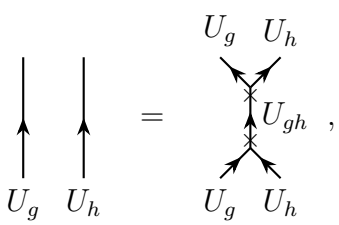


$\in \mathrm{Hom}(U_g \otimes U_h, U_{gh})$,

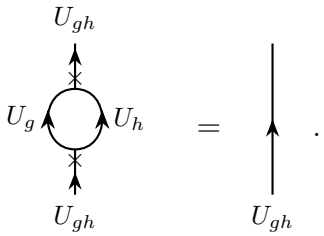


$\in \mathrm{Hom}(U_{gh}, U_g \otimes U_h)$. (3.6)

We mark one of the lines by an $\times$ to make it explicit which hom-space the junction belongs to, although it is already implicit in the direction of the arrows. As we mentioned above, in the group case these hom-spaces are one-dimensional. As such, we have the following completeness and orthogonality relations,



,



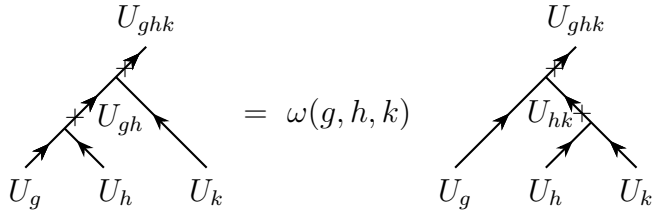
. (3.7)

In the dual picture of open patches on a manifold (see the discussion in Section 2.8), each line represents a transition function between two open patches, and the above relations correspond to the requirement that changing our open cover does not change any physical results.

An important property of the fusion is that it is not necessarily strictly associative (even though the group multiplication is always associative). Indeed, there can exist a non-trivial *associator homomorphism*,

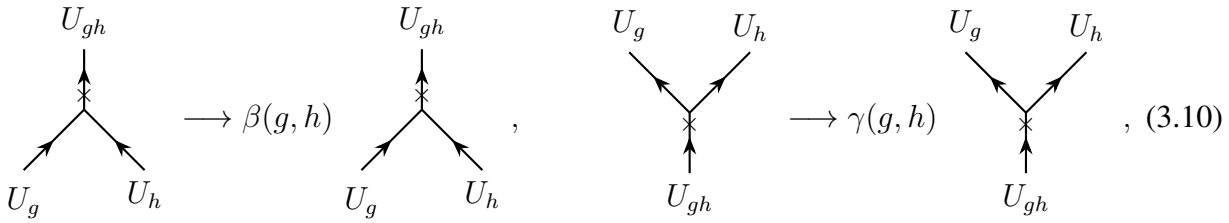
$$\omega \in \mathrm{Hom}\left((U_g \otimes U_h) \otimes U_k, U_g \otimes (U_h \otimes U_k)\right) \ . \tag{3.8}$$

Pictorially, this is captured by the following,



(3.9)

Note that one is always free to make a redefinition of the hom-space basis vectors via factors β, γ,



(3.10)

which is referred to as a *gauge transformation*.[19] We take $\gamma(g,h) = \beta(g,h)^{-1}$ such that the conventions in (3.7) remain unchanged. Upon such a gauge transformation, the associator is modified via

$$\omega(g,h,k) \to \frac{\beta(g,hk)\beta(h,k)}{\beta(g,h)\beta(gh,k)}\omega(g,h,k) \ , \tag{3.11}$$

and hence the associators themselves are not, in general, gauge invariant. At the same time, the associators are subject to the so-called *pentagon identity*, illustrated in Figure 1, which in equation form is

$$\omega(g,h,k)\omega(g,hk,\ell)\omega(h,k,\ell) = \omega(gh,k,\ell)\omega(g,h,k\ell) \ . \tag{3.12}$$

Now let us define an operation

$$\begin{aligned} \delta f(g_1,\ldots,g_{n+1}) \ &:= \ f(g_2,\ldots,g_{n+1}) f(g_1,g_2,\ldots,g_n)^{(-1)^{n+1}} \\ &\quad \times \prod_{i=1}^{n} f(g_1,g_2,\ldots,g_{i-1},g_i g_{i+1},g_{i+2},\ldots,g_{n+1})^{(-1)^i} \ . \end{aligned} \tag{3.13}$$

[19] This should not be confused with the background gauge transformations of G, which correspond to changing our mesh of topological defects.

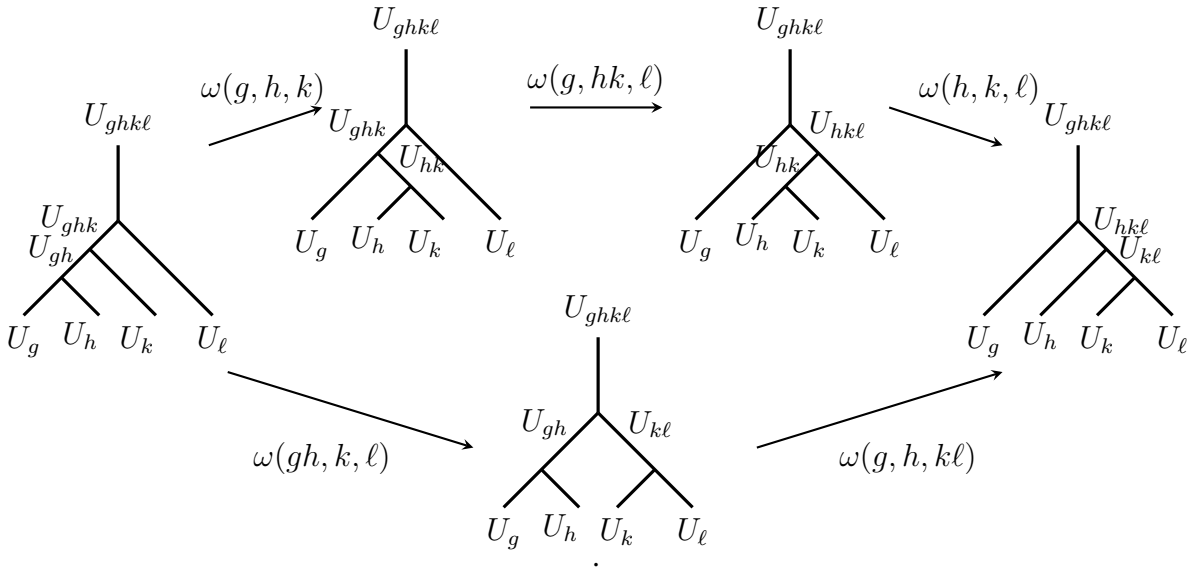


Figure 1: The pentagon identity for invertible defects. This condition translates to the closedness of ω in group cohomology.

In terms of this operation, the condition in (3.12) becomes

$$\delta\omega(g,h,k,\ell) = 1 \ , \tag{3.14}$$

while the redundancy in (3.11) becomes

$$\omega(g,h,k) \sim \delta\beta(g,h,k)\,\omega(g,h,k) \ . \tag{3.15}$$

Exercise: Verify equations (3.14) and (3.15).

This should be reminiscent of the definition of a cohomology class—namely closed forms modulo exact forms. Indeed, an element $\omega(g,h,k)$ which satisfies (3.12) is known as a *3-cocycle*; its equivalence class under the redundancy (3.15) is an element of *group cohomology* $H^3(G,U(1))$. This is the same space in which potential anomalies of G take values, so the physical interpretation of the associator $\omega(g,h,k)$ is as the anomaly of G. Physically, changing the mesh from the left-hand side of (3.9) to the right-hand side is like performing a background gauge transformation for G, and $\omega(g,h,k)$ captures non-invariance of the partition function under this transformation. The fusion category describing a discrete symmetry G with anomaly ω is denoted by Vec_G^ω.

3.2 General fusion categories

We now generalize the discussion above to more general fusion categories $\mathcal{C}$. As before, the basic data of the category is

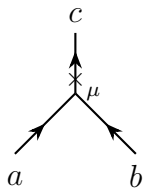
- **Objects:** topological lines $a, b, c, \cdots \in \mathcal{C}$.[20] Every element a has a unique orientation-reversal, or *dual*, element $\bar{a}$. Note that $\bar{a}$ is not necessarily the inverse of a; indeed, a will not in general admit an inverse.

- **Morphisms:** topological local operators $\mathcal{O}_a^b$ that act as homomorphisms between two topological lines, i.e. $\mathcal{O}_a^b \in \mathrm{Hom}(a, b)$. A *simple object* is one such that $\dim \mathrm{Hom}(a, a) = 1$.

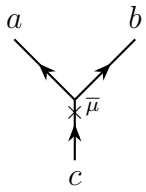
We also need a notion of fusion, with a, b, c now labelling simple objects,

$$a \otimes b = \bigoplus_c N_{ab}^c \, c \,, \tag{3.16}$$

where the fusion coefficients N_{ab}^c are non-negative integers. We assume that the fusion rules are commutative, so that $N_{ab}^c = N_{ba}^c$. We also have $N_{a\bar{a}}^1 = N_{\bar{a}a}^1 = 1$ for any simple a. Corresponding to these are trivalent junctions representing elements of the hom-spaces,

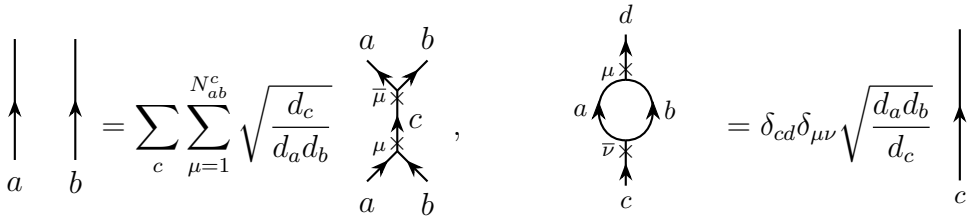


$\in \mathrm{Hom}(a \otimes b, c)$,



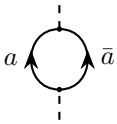
$\in \mathrm{Hom}(c, a \otimes b)$, (3.17)

where $\mu = 1, \ldots, N_{ab}^c$. Similar to before, the basis vectors of the hom-spaces are chosen such that they satisfy completeness and orthogonality relations, which in the current case take the following form,



. (3.18)

Here d_a is the *quantum dimension* of the line a, which is additive under direct sums and multiplicative under fusion, $d_a d_b = \sum_c N_{ab}^c d_c$, satisfying $d_a = d_{\bar{a}}$ and $d_a > 0$ in a unitary theory. The quantum dimension can be defined in a number of ways, one of which is as the expectation value of a loop of a and $\bar{a}$,

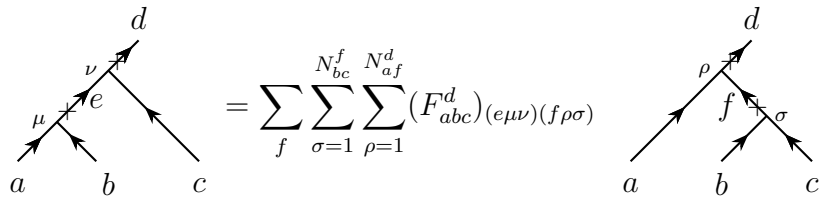


$= \; d_a \,;$ (3.19)

[20] These are not to be confused with dynamical gauge fields $a, b, c, \ldots$ appearing in other sections. In the current subsection we will not discuss dynamical gauge fields, so this should not lead to any confusion.

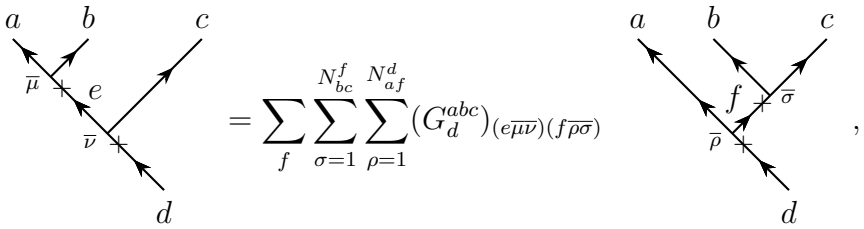
in other words, we just define it in terms of the second equation of (3.18). Another definition is as the normalized vacuum expectation value of a loop of a wrapping the non-trivial cycle on a cylinder. Note that for invertible elements we have $d_a = 1$ in a unitary theory, as follows from multiplicativity and $d_a > 0$.

As before, the fusion rules can have a non-trivial associator, whose matrix elements in the current context are known as the *F-symbols* of the category. Pictorially, they may be represented as follows,



. (3.20)

The F-symbols play an analogous role as the anomaly $\omega(g,h,k)$ in (3.9), but are more complicated due to the various sums and non-trivial junction vectors that appear in the non-invertible case. There are also so-called G-symbols, defined by

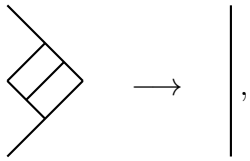


, (3.21)

which are related to the F-symbols by

$$(G_d^{abc})_{(e\overline{\mu\nu})(f\overline{\rho\sigma})} = (F_{abc}^d)^{-1}_{(f\rho\sigma)(e\mu\nu)} \; ; \tag{3.22}$$

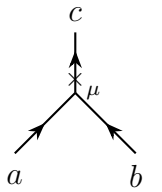
this relation is derived by considering two ways of going between the following two diagrams,

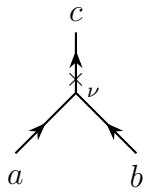

$\longrightarrow$

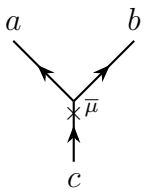
, (3.23)

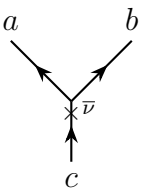
namely using an F-move together with two applications of the orthogonality relation of (3.18), or a G-move with two applications of the orthogonality relation.

As before, one is always free to make a redefinition of the hom-space basis vectors via unitary matrices,



$\longrightarrow \sum_{\nu=1}^{N_{ab}^c} (U_{ab}^c)_{\mu\nu}$


,


$\longrightarrow \sum_{\nu=1}^{N_{ab}^c} (V_c^{ab})_{\overline{\mu\nu}}$


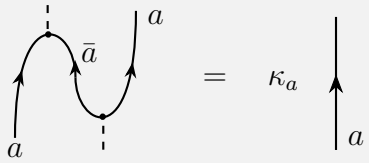
. (3.24)

We take $(V_c^{ab})_{\overline{\mu\nu}} = (U_{ab}^c)^{-1}_{\nu\mu}$ such that the conventions in (3.18) remain unchanged. Upon such a gauge transformation, the F-symbols are modified via

$$(F^d_{abc})_{(e\mu\nu)(f\rho\sigma)} \to \sum_{\alpha=1}^{N_{ab}^e}\sum_{\beta=1}^{N_{ec}^d}\sum_{\gamma=1}^{N_{af}^d}\sum_{\delta=1}^{N_{bc}^f}(U^e_{ab})^{-1}_{\mu\alpha}(U^d_{ec})^{-1}_{\nu\beta}(F^d_{abc})_{(e\alpha\beta)(f\gamma\delta)}(U^d_{af})_{\gamma\rho}(U^f_{bc})_{\delta\sigma}\ , \tag{3.25}$$

and hence the F-symbols themselves are not, in general, gauge invariant. At the same time, the F-symbols are subject to a pentagon identity of exactly the same form as in Figure 1.

Exercise: For a self-dual simple object $a = \bar{a}$, using the definition of the quantum dimension and F-symbols, show that we have

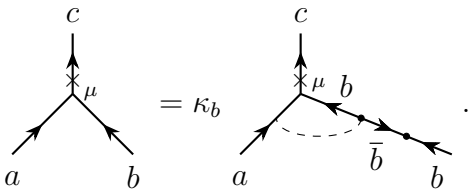


where

$$\kappa_a = d_a\,(F^a_{a\bar{a}a})_{11}\ .$$

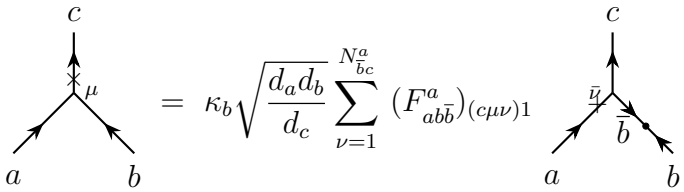
(*Hint: Take the upper identity line to end on the upwards-going a line, and then perform an F-move with this as the internal line*.) Then check that this quantity is gauge invariant with the unit vertices held fixed. In fact, this quantity can be shown to take values ± 1 and is known as the *Frobenius-Schur (FS) indicator* of a. A non-trivial FS indicator is a sign of a mixed anomaly between isotopy and orientation-reversal invariances [73, 75, 76].

For $a \neq \bar{a}$, the bending phase κ_a is gauge dependent; below we choose it to be $+1$. Note that using the phase κ_b, we may re-express the fusion trivalent junctions in (3.17) as



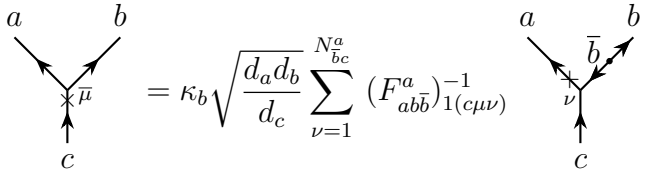
. (3.26)

Then performing a G-move with internal line being the identity line shown above, and shrinking the resulting bubble using (3.18), we obtain



(3.27)

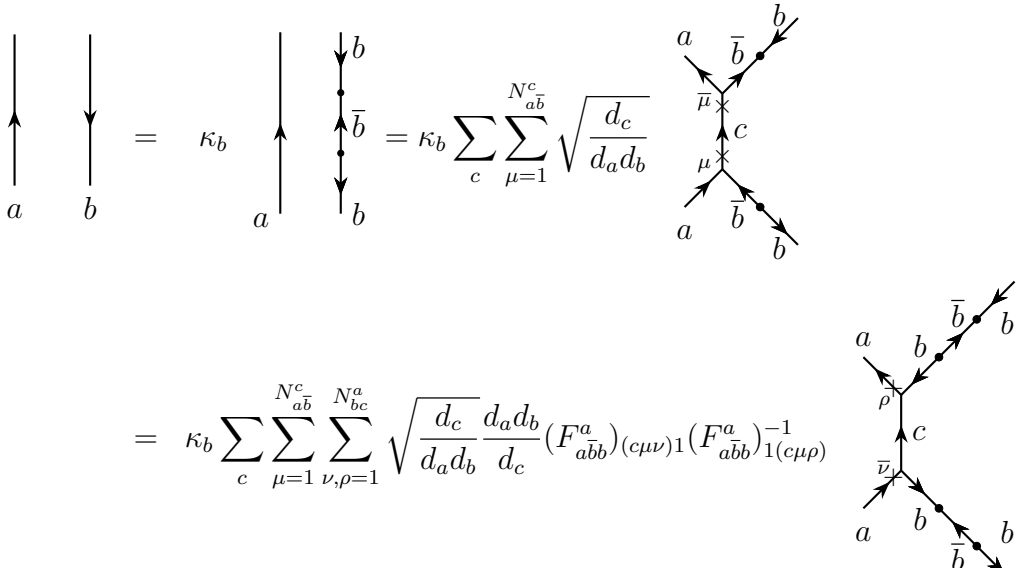
and likewise for the splitting junctions,



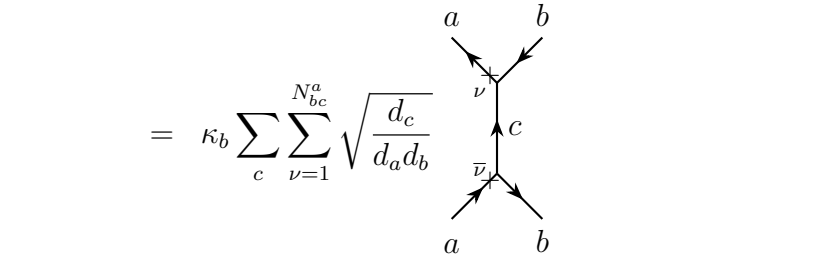
(3.28)

The coefficients appearing here are referred to as B-*symbols* in some of the literature [73, 77].

To give just one example of the use of these definitions in a non-trivial computation, let us derive the opposite orientation version of the completeness relation in (3.18). In particular, we have



(3.29)



(3.30)

where in the first step we have used the definition of the FS indicator, in the second step we have used the usual completeness relation, in the third step we have used the results of (3.27) and (3.28), and in the final step we have again used the definition of the FS indicator.

Returning to the discussion of F-symbols and the pentagon identity, a result known as *Ocneanu rigidity* in the math literature [78, 79] states that the solutions to the pentagon identity are discrete up to gauge equivalence. Hence given a set of fusion rules, there do not exist continuous families of categories with different F-symbols—only a discrete set. Because there are only discrete sets of possibilities, we do not expect the category to change under continu-

ous processes like symmetry-preserving RG flows, which is why it is natural to think of this structure as describing some generalization of symmetry.[21]

3.3 Low-rank fusion categories

The number of simple objects in a fusion category is known as its *rank*. For low-rank pivotal fusion categories, one can actually classify the full set of solutions to the pentagon identities, as was done for ranks 2 and 3 in [80] and [81], respectively.[22] We will now review this classification.

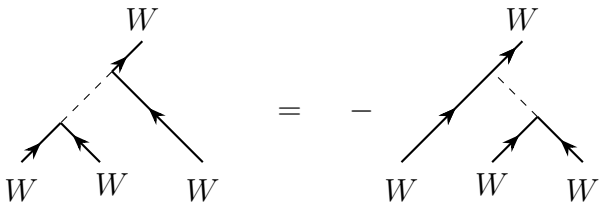
3.3.1 Rank 2

We begin with rank 2, for which there are two simple topological lines 1 and W. The fusion algebra takes the general form $W^2 = 1 + nW$, where n is *a priori* any non-negative integer. However, it turns out that only $n = 0, 1$ allow for solutions to the pentagon identity.

- $n = 0$: In this case the fusion rules are $W^2 = 1$, so W is just the generator of a $\mathbb{Z}_2$ global symmetry. Thus we reduce to the group-like case studied in Section 3.1. The pentagon identities simplify in this case to a single non-trivial identity $\omega(W, W, W)^2 = 1$, from which we conclude that there are two solutions

$$\omega(W, W, W) = \pm 1 \,. \tag{3.31}$$

  These solutions correspond to a $\mathbb{Z}_2$ global symmetry with and without 't Hooft anomaly. In particular, the anomalous case has the following crossing relation,



(3.32)

- $n = 1$: In this case the fusion algebra is $W^2 = 1 + W$, which is known as the *Fibonacci* or *Lee-Yang algebra*. Since the quantum dimension is additive and multiplicative, we must have $d_W^2 = 1 + d_W$, which has solutions $d_W = \frac{1}{2}(1 \pm \sqrt{5})$. The choice of $d_W = \frac{1}{2}(1 + \sqrt{5}) = \varphi$, i.e. the golden ratio, gives the so-called *Fibonacci category*. The non-trivial F-symbols are uniquely determined up to gauge equivalence; in a convenient gauge they are

$$F^W_{WWW} = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix} \,. \tag{3.33}$$

[21] Of course, it is still conceivable that there are discontinuous jumps, but this is also the case for standard symmetry.

[22] Even stronger results exist in the literature—for example, modular categories have been classified up to at least rank 5 [82], while multiplicity-free pivotal fusion categories have been classified up to rank 7 [83].

On the other hand, the choice of $d_W = \frac{1}{2}(1-\sqrt{5})$ gives the *Lee-Yang category*. It cannot be realized in a unitary model, but underlies e.g. the non-unitary minimal model $\mathcal{M}(2,5)$.

3.3.2 Rank 3

We next consider rank 3. In this case there are 3 simple topological lines $1, X, Y$. There are now five allowed fusion algebras, and various ways to complete each of these algebras into a fusion category,

1. $X^2 = Y, Y^2 = X, XY = 1$: three solutions to the pentagon identity
2. $X^2 = 1, Y^2 = 1 + X, XY = Y$: two solutions to the pentagon identity
3. $X^2 = 1, Y^2 = 1 + X + Y, XY = Y$: three solutions to the pentagon identity
4. $X^2 = 1 + Y, Y^2 = 1 + X + Y, XY = X + Y$: three solutions to the pentagon identity
5. $X^2 = 1, Y^2 = 1 + X + 2Y, XY = Y$: four solutions to the pentagon identity

In the first case, we recognize the fusion rules as those for a $\mathbb{Z}_3$ global symmetry. The three solutions to the pentagon identity correspond to the three different values of the mod 3 't Hooft anomaly. On the other hand, the remaining four cases are all non-invertible.

Let us now focus on the second case. In this case, one often uses the notation $X = \eta$ and $Y = \mathcal{N}$, so that the fusion rules read

$$\eta^2 = 1\,, \qquad \eta \times \mathcal{N} = \mathcal{N}\,, \qquad \mathcal{N} \times \mathcal{N} = 1 + \eta\,. \tag{3.34}$$

These are known as the *Ising fusion rules*, since they are realized in the critical Ising model, as we will discuss below. We see that η generates a standard $\mathbb{Z}_2$ symmetry, but $\mathcal{N}$ is a non-invertible object. One can check that the two solutions to the pentagon identities are given by

$$F^{\mathcal{N}}_{\eta\mathcal{N}\eta} = F^{\eta}_{\mathcal{N}\eta\mathcal{N}} = -1\,, \qquad F^{\mathcal{N}}_{\mathcal{N}\mathcal{N}\mathcal{N}} = \pm\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\,, \tag{3.35}$$

with all other F-symbols being trivial. The choice with the $+$ sign is often referred to as the *Ising fusion category*.

The Ising category is in some sense the simplest and most famous of all fusion categories. One of the reasons why it is so famous is that it is realized in the simplest non-trivial unitary conformal field theory (CFT), the critical Ising model, as we will now discuss.

3.4 The Ising model

The Ising model can be described as a lattice model in the following way. We begin with a two-dimensional Euclidean square lattice, which for simplicity we take to be periodic in both directions. We assign a spin $0, 1$ to each lattice site, and then couple the sites together via nearest neighbor interactions such that the total energy is

$$E[s] = -J \sum_{\langle ij \rangle} (-1)^{s_i + s_j} \ , \tag{3.36}$$

where $J > 0$ is a coupling constant, s_i is the spin associated with site i, and $\langle ij \rangle$ labels nearest neighbor pairs. We choose J to be positive so that the energy decreases when the spins are aligned—in this case the system acts as a ferromagnet. The partition function of the Ising model at inverse temperature β is then given by

$$Z_{\text{Ising}}(\beta, J) = \sum_{\{s\}} e^{-\beta E[s]} = \sum_{\{s\}} e^{\beta J \sum_{\langle ij \rangle} (-1)^{s_i + s_j}} \ . \tag{3.37}$$

3.4.1 Kramers-Wannier duality

An important property of the Ising model is that it has a $\mathbb{Z}_2$ spin flip symmetry which flips all s_i simultaneously, i.e. $s_i \to s_i + 1 \bmod 2$. This symmetry is non-anomalous and can be gauged. Doing so gives the gauged theory $\text{Ising}/\mathbb{Z}_2$ with partition function

$$Z_{\text{Ising}/\mathbb{Z}_2}(\beta, J) = \frac{1}{2} \sum_{\text{b.c.}} \sum_{\{s\}} e^{\beta J \sum_{\langle ij \rangle} (-1)^{s_i + s_j}} \ , \tag{3.38}$$

where the sum $\sum_{\text{b.c.}}$ is a sum over boundary conditions along the two cycles of the lattice, which is analogous to (2.125) on the continuum torus.

Since the dependence on β and J above comes only in the combination $K := \beta J$, we will denote the partition functions as $Z_{\text{Ising}}(K)$ and $Z_{\text{Ising}/\mathbb{Z}_2}(K)$, and will refer to K as the temperature. One of the most important features of the Ising model is so-called *Kramers-Wannier (KW) duality* [84, 85], which states that[23]

$$Z_{\text{Ising}/\mathbb{Z}_2}(K) = (\sinh 2K)^N Z_{\text{Ising}}(\widetilde{K}) \ , \tag{3.39}$$

where N is the number of lattice sites, and $\widetilde{K}$ is related to K via

$$\sinh 2K \sinh 2\widetilde{K} = 1 \ . \tag{3.40}$$

Note that when K is very large, $\widetilde{K}$ is very small, and vice versa, so this is roughly speaking a duality between the Ising model at high temperature and the $\mathbb{Z}_2$-gauged Ising model at

[23] The overall factor of $\sinh$ can be removed by addition of appropriate local counterterms, and hence is not particularly important for the statement of the duality.

low temperature, and vice versa. There is a special, *critical temperature* $K_c = \widetilde{K}_c$ at which Kramers-Wannier is not just a duality between distinct theories, but a self-duality of a single theory. At this temperature, the continuum limit is a CFT and has a new non-invertible symmetry, as we will discuss momentarily.

Before discussing the critical point, let us try to understand explicitly how KW duality arises. Here we follow closely the discussion in [86]. Our starting point will be the following function of K,

$$\mathcal{Z}(K) = \frac{1}{2^N} \sum_{\{a\}} \sum_{\{b\}} e^{K \sum_\ell (-1)^{a_\ell} + i\pi \sum_{p=\langle \ell_1 \ell_2 \ell_3 \ell_4 \rangle} b_p (a_{\ell_1} + a_{\ell_2} + a_{\ell_3} + a_{\ell_4})} \,. \tag{3.41}$$

Here the a_ℓ are $\mathbb{Z}_2$-variables labelled by links ℓ, where each link ℓ is defined by a pair of nearest neighbor points $\langle ij \rangle$. On the other hand, the b_p are $\mathbb{Z}_2$-variables labelled by plaquettes, each of which is specified by a set of four links $\langle \ell_1 \ell_2 \ell_3 \ell_4 \rangle$ forming a closed loop. What we will now show is that one way of evaluating $\mathcal{Z}(K)$ gives the right-hand side of (3.39), while another way of evaluating it gives the left-hand side of (3.39), and hence the two must be equal.

Let us start by reproducing the right-hand side. To do so, we first perform the sum over a_ℓ in (3.41). Since every link ℓ is shared by two plaquettes, say p_1 and p_2, we may rewrite,

$$\begin{aligned} \mathcal{Z}(K) &= \frac{1}{2^N} \sum_{\{b\}} \prod_\ell \sum_{a_\ell = 0,1} e^{K(-1)^{a_\ell} + i\pi a_\ell (b_{p_1} + b_{p_2})} \\ &= \frac{1}{2^N} \sum_{\{b\}} \prod_\ell \left(e^K + (-1)^{b_{p_1} + b_{p_2}} e^{-K} \right) \\ &= \frac{1}{2^N} \sum_{\{b\}} \prod_\ell 2 \cosh K \, (\tanh K)^{b_{p_1} + b_{p_2} \bmod 2} \,. \end{aligned} \tag{3.42}$$

Some additional trigonometric manipulations then allow one to rewrite this as

$$\mathcal{Z}(K) = \frac{1}{2^N} \sum_{\{b\}} \prod_\ell \sqrt{2 \sinh 2K} \, e^{\widetilde{K}(-1)^{b_{p_1} + b_{p_2}}} \,, \tag{3.43}$$

where $\widetilde{K}$ is such that

$$\tanh K = e^{-2\widetilde{K}} \qquad \Rightarrow \qquad \sinh 2K \sinh 2\widetilde{K} = 1 \,. \tag{3.44}$$

Exercise: Check these trigonometric manipulations.

Finally, noting that we may equivalently label each link by the pair of plaquettes that share it (instead of the pair of vertices it connects), we have

$$\begin{aligned} \mathcal{Z}(K) &= (\sinh 2K)^N \sum_{\{b\}} \prod_{\ell = \langle p_1 p_2 \rangle} e^{\widetilde{K}(-1)^{b_{p_1} + b_{p_2}}} \\ &= (\sinh 2K)^N Z_{\text{Ising}}(\widetilde{K}) \,. \end{aligned} \tag{3.45}$$

In other words, the role of vertices and plaquettes were exchanged, but the result is still the Ising model partition function.

Next we reproduce the left-hand side of (3.39). In this case, we first evaluate the sum over b_p. Then noting that

$$\frac{1}{2}\sum_{b_p=0,1} e^{i\pi a_\ell b_p} = \delta_{a_\ell,0}\ , \tag{3.46}$$

where the delta function is defined modulo 2, we have

$$\mathcal{Z}(K) = \sum_{\{a\}} e^{K\sum_\ell (-1)^{a_\ell}} \prod_{p=\langle \ell_1\ell_2\ell_3\ell_4\rangle} \delta_{a_{\ell_1}+a_{\ell_2}+a_{\ell_3}+a_{\ell_4},0}\ . \tag{3.47}$$

Note that if we impose $a_{\ell_1}+a_{\ell_2}+a_{\ell_3}+a_{\ell_4}=0 \mod 2$ on any plaquette, then the sets $\{a_\ell\}$ and $\{s_i\}$ are actually in 1-to-2 correspondence. Indeed,

- For every set of $\{s_i\}$, we can get the set of $\{a_\ell\}$ by defining

  $$a_{\langle ij\rangle} = s_i + s_j \mod 2\ . \tag{3.48}$$

  This choice satisfies the condition $a_{\ell_1}+a_{\ell_2}+a_{\ell_3}+a_{\ell_4}=0 \mod 2$.

- Conversely, for every set of $\{a_\ell\}$, we may define a set of $\{s_i\}$ by starting at some arbitrary lattice site, assigning it spin s_0, and then defining all other spins via

  $$s_i := s_0 + a_{\ell_1} + \cdots + a_{\ell_n}\ , \tag{3.49}$$

  where $\ell_1,\ldots,\ell_n$ form a path from site 0 to site i. Modulo a subtlety that we will discuss in a moment, the choice of path does not matter as long as $a_{\ell_1}+a_{\ell_2}+a_{\ell_3}+a_{\ell_4}=0 \mod 2$, but note that there are two choices of the initial s_0, so from the set $\{a_\ell\}$ there are two sets $\{s_i\}$.

Based on this 1-to-2 correspondence, we might then expect that

$$\sum_{\{a\}}\prod_{p=\langle \ell_1\ell_2\ell_3\ell_4\rangle} \delta_{a_{\ell_1}+a_{\ell_2}+a_{\ell_3}+a_{\ell_4},0} \longrightarrow \frac{1}{2}\sum_{\{s\}}\ . \tag{3.50}$$

This is almost correct, but on lattices with non-trivial topology (such as our current case with periodic boundary conditions) there is an additional subtlety. In particular, we assumed in the second bullet point that any choice of path $\ell_1,\ldots,\ell_n$ from site 0 to site i gave the same result. However, this can be false on non-contractible cycles. In particular, set $i=0$ and consider a path that goes from 0 to itself around a non-contractible cycle. If $a_{\ell_1}+\cdots+a_{\ell_n}=0 \mod 2$ along this cycle, then $s_0 \to s_0$, and we have periodic boundary conditions, but if $a_{\ell_1}+\cdots+a_{\ell_n}=1 \mod 2$ then we obtain anti-periodic boundary conditions. So what we really have is

$$\sum_{\{a\}}\prod_{p=\langle \ell_1\ell_2\ell_3\ell_4\rangle} \delta_{a_{\ell_1}+a_{\ell_2}+a_{\ell_3}+a_{\ell_4},0} \longrightarrow \frac{1}{2}\sum_{\text{b.c.}}\sum_{\{s\}}\ . \tag{3.51}$$

From this, we conclude that

$$\mathcal{Z}(K) = Z_{\text{Ising}/\mathbb{Z}_2}(K) \,, \tag{3.52}$$

thus proving the Kramers-Wannier duality relation.

3.4.2 Operators

Let us now briefly discuss the operators in the Ising model, and their behavior under duality. The most basic operators are the spin operators—in the presentation of the Ising model in terms of plaquettes in (3.45), these are written as

$$\sigma_p = (-1)^{b_p} \,, \tag{3.53}$$

which can be thought of as local operators living on the dual lattice. There is also an extended operator living on closed loops γ,

$$\eta(\gamma) = (-1)^{\sum_{\ell\in\gamma} a_\ell} \,. \tag{3.54}$$

From the expression of $\mathcal{Z}(K)$ in (3.41), it is easy to see that inserting this into the path-integral leads to a shift $b_p \to b_p + 1$ for plaquettes enclosed in the loop γ. In other words, this is the generator of the spin-flip $\mathbb{Z}_2$ symmetry that we had discussed before. It acts on the operators σ_p with a minus sign, pictorially



(3.55)

We could also try to define η on curves which are *not* closed, i.e. which have an endpoint. This gives a so-called *twist defect*, i.e. an operator inserted at a lattice site i which is then attached to a line of η. With its tail running to a boundary or infinity, this operator is denoted by[24]

$$\mu_i = (-1)^{a_{\ell_1} + a_{\ell_2} + \cdots} \,. \tag{3.56}$$

Here $\ell_1, \ell_2, \ldots$ form the tail starting at i; on a compact lattice the same open-line insertion has a μ at each endpoint.

Note that when we gauge the $\mathbb{Z}_2$ generated by η, on the one hand $\mathbb{Z}_2$ odd operators are projected out, which means that σ_p ceases to be a local operator, but on the other hand we make the η line transparent, so that $\mathbb{Z}_2$ twisted operators can become local; this sort of interchange was discussed in detail in Section 2.9. We thus see that, upon Kramers-Wannier duality, the

[24] In flat space the η tail could run off to infinity, but on a compact lattice such as the one we are considering here, the tail must end on a second μ operator. In other words, μ always comes in pairs on a compact manifold.

operators σ and μ are exchanged, and we also exchange the lattice and dual lattices, as well as the temperatures K and $\widetilde{K}$.

Finally, let us introduce the energy operator

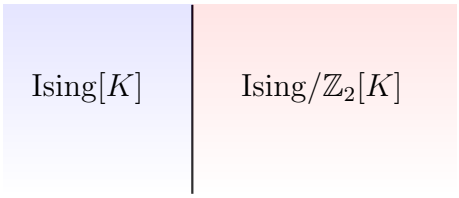
$$\epsilon_\ell = (-1)^{b_{p_1}+b_{p_2}} \ , \tag{3.57}$$

where ℓ is the link shared by p_1 and p_2. In the continuum limit, after subtracting its identity component, this flows to an operator whose role is to shift the temperature. In particular, if we start at the critical temperature $K_c = \widetilde{K}_c$ and modify the Hamiltonian to $H \to H - \delta K \sum_\ell \epsilon_\ell$, then we will change the coupling to $K_c \to K = K_c + \delta K$. Such a shift in K corresponds to an opposite shift in $\widetilde{K}$, namely $\widetilde{K} = K_c - \delta K$ to first order in δK, and this implies that when we pass to the Kramers-Wannier dual description, we must have $\epsilon \to -\epsilon$.

3.4.3 Ising fusion rules

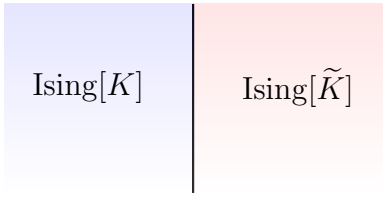
Let us summarize the above discussion. The Ising model has a spin operator σ and a $\mathbb{Z}_2$ symmetry generated by η such that $\eta\sigma = -\sigma$. There also exists a twist operator μ which is attached to an η line. Upon gauging η, the operators σ and μ exchange roles. The ungauged theory at temperature K and the gauged theory at temperature $\widetilde{K}$ are actually equivalent, i.e. $Z_{\text{Ising}/\mathbb{Z}_2}(K) \cong Z_{\text{Ising}}(\widetilde{K})$.

We would now like to connect this discussion to the Ising fusion rules given in (3.34). To do so, let us consider gauging the $\mathbb{Z}_2$ symmetry in half of the space, e.g. the right-hand side. When we perform the gauging, we impose Dirichlet boundary conditions at the boundary, which are topological; see Section 2.8.2. This should give us a topological interface between the Ising model at temperature K and the $\mathbb{Z}_2$-gauged Ising model at the same temperature,



. (3.58)

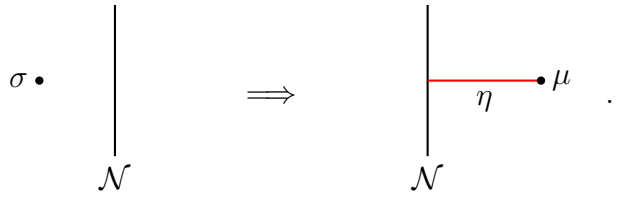
However, making use of Kramers-Wannier duality, we note that the right-hand side is also equivalent to the Ising model at temperature $\widetilde{K}$,



. (3.59)

Thus we see that at the critical temperature $K_c = \widetilde{K}_c$, i.e. $K_c = \frac{1}{2}\log(1+\sqrt{2})$, the *interface* between two theories becomes a *defect* in a single theory—in particular, it gives rise to a new symmetry generator $\mathcal{N}$. The conclusion is that the critical Ising model has an additional symmetry $\mathcal{N}$, which captures its self-duality under gauging of the $\mathbb{Z}_2$ symmetry.

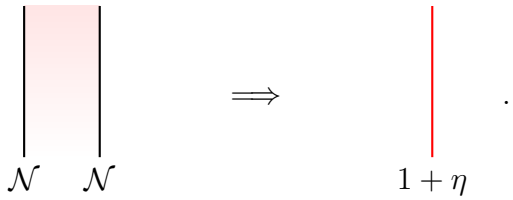
We may understand the fusion rules of $\mathcal{N}$ as follows. First, recall that upon $\mathbb{Z}_2$ gauging, the roles of σ and μ are interchanged. Thus moving σ past the defect $\mathcal{N}$ must turn it into μ, and hence the defect $\mathcal{N}$ must be able to absorb the line coming off of μ,[25]



(3.60)

From this we conclude that $\mathcal{N} \times \eta = \mathcal{N}$.

Next, imagine inserting two copies of $\mathcal{N}$ on e.g. the $\mathfrak{B}$-cycle of the torus. The first copy of $\mathcal{N}$ gauges the $\mathbb{Z}_2$ symmetry, while the second $\mathcal{N}$ ungauges it. This configuration thus corresponds to gauging the $\mathbb{Z}_2$ symmetry in the annulus between the two copies of $\mathcal{N}$. Recall from Section 2.9 that gauging a $\mathbb{Z}_2$ symmetry amounts to inserting a copy of $1+\eta$ on all cycles (this will be discussed further in Section 3.6). If we are inserting the defects $\mathcal{N}$ along the $\mathfrak{B}$-cycle, then the only cycle in the strip is again the $\mathfrak{B}$-cycle, and hence this configuration amounts to simply inserting $1+\eta$ along the $\mathfrak{B}$-cycle, i.e.



(3.61)

From this, we conclude that we have the fusion rules $\mathcal{N} \times \mathcal{N} = 1 + \eta$.

In conclusion, the fusion rules of the topological defects of the critical Ising model are given by

$$\eta^2 = 1\,, \qquad \eta \times \mathcal{N} = \mathcal{N}\,, \qquad \mathcal{N} \times \mathcal{N} = 1 + \eta\,. \tag{3.62}$$

One may further study the F-symbols of this model, upon which one finds that they take the form of (3.35), with the choice of overall sign of $F^{\mathcal{N}}_{\mathcal{N}\mathcal{N}\mathcal{N}}$ being $+1$. Let us finally note that the action of η and $\mathcal{N}$ on the operators $1, \epsilon$, and σ of the theory are as follows,

$$\begin{aligned} \eta\,|1\rangle &= |1\rangle\,, & \eta\,|\epsilon\rangle &= |\epsilon\rangle\,, & \eta\,|\sigma\rangle &= -|\sigma\rangle\,, \\ \mathcal{N}\,|1\rangle &= \sqrt{2}\,|1\rangle\,, & \mathcal{N}\,|\epsilon\rangle &= -\sqrt{2}|\epsilon\rangle\,, & \mathcal{N}\,|\sigma\rangle &= 0\,. \end{aligned} \tag{3.63}$$

The first line follows straightforwardly from our previous discussions. In the last equation of the second line, we are thinking of $\mathcal{N}$ as a map from the untwisted Hilbert space to itself, and $\mathcal{N}\eta = \mathcal{N}$ together with $\eta|\sigma\rangle = -|\sigma\rangle$ implies $\mathcal{N}|\sigma\rangle = -\mathcal{N}|\sigma\rangle = 0$. The action of $\mathcal{N}$ on the identity is simply its quantum dimension, which from the fusion algebra must be $\sqrt{2}$. Finally, the minus sign appearing in the action of $\mathcal{N}$ on ϵ comes from the fact that ϵ changes sign upon Kramers-Wannier duality, as explained in the previous subsection.

[25] The line coming off of μ should be anchored on $\mathcal{N}$ and not run off to infinity, since local moves should not change the asymptotic state.

3.5 Tambara-Yamagami categories and self-duality

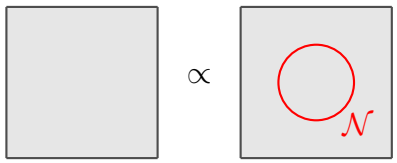
The Ising category studied above is one of the simplest examples of a more general class of categories known as *Tambara-Yamagami (TY) categories* [87]. These are categories which have invertible elements U_g labelled by elements g of a discrete Abelian group G, together with a single non-invertible element $\mathcal{N}$ with the following fusion rules,

$$U_{g_1} \times U_{g_2} = U_{g_1 g_2} \ , \qquad U_g \times \mathcal{N} = \mathcal{N} \ , \qquad \mathcal{N} \times \mathcal{N} = \sum_{g \in G} U_g \ . \tag{3.64}$$

The Ising category studied above corresponds to the case of $G = \mathbb{Z}_2$ (and a particular choice of F-symbols).

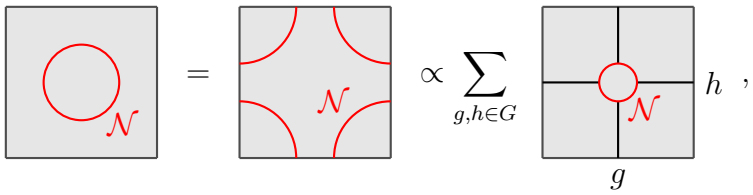
Previously, we saw that the non-invertible defect $\mathcal{N}$ captured self-duality under gauging the $\mathbb{Z}_2$ symmetry of the Ising model. More generally, it turns out that a theory is self-dual under gauging G if and only if the symmetry G admits a TY extension. One direction of this follows via the half-space gauging construction. Let us now discuss the other direction, at the level of the torus partition function. We begin with the schematic derivation, giving the more rigorous computation afterwards.

First assume that we have a defect $\mathcal{N}$, and insert a loop of it on a contractible cycle of the torus, which only changes the partition function by an overall factor that we divide out by,



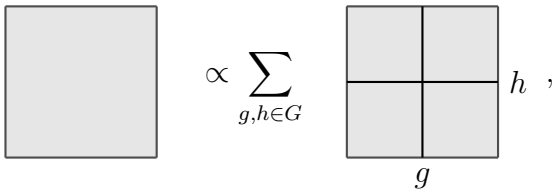
(3.65)

Then deforming $\mathcal{N}$ topologically so that it wraps the sum of all cycles using the fusion rules gives



(3.66)

where from now on, for simplicity, we label the lines corresponding to group-like symmetries $g \in G$ by the group element g itself, instead of by U_g. Finally, shrinking the bubble gives

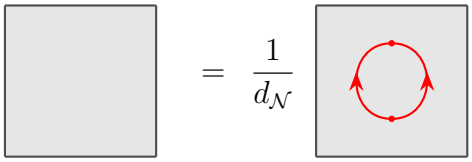


(3.67)

where the right-hand side is basically the G-gauged partition function. This is the statement of self-duality. Of course, in the computation above we have been very cavalier about issues

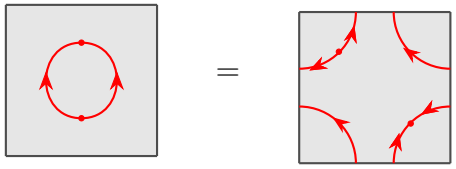
such as the orientation of $\mathcal{N}$ and overall normalization. A more precise treatment should keep track of the orientation of $\mathcal{N}$ and make use of the F-symbols to go from the second to the third expression in (3.66), as we do now.

The more precise derivation proceeds as follows. We begin by using our convention in (3.19) to write



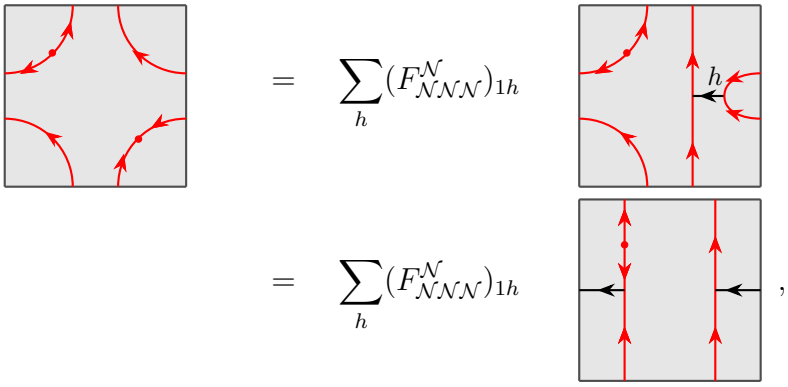
(3.68)

where all red lines represent $\mathcal{N}$—note that we are using that $\overline{\mathcal{N}} = \mathcal{N}$ for TY categories. Now we deform this topologically to obtain



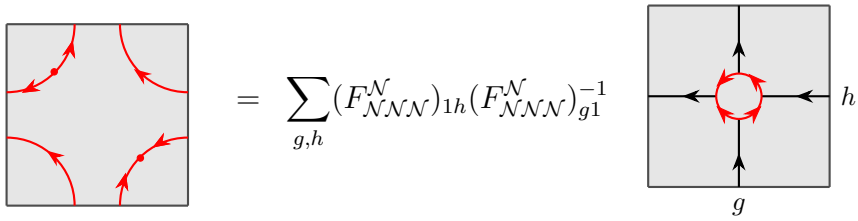
(3.69)

If we imagine an identity line as being stretched from the vertex in the bottom right to the middle of the line on the top right, then we may use an F-symbol to obtain[26]



(3.70)

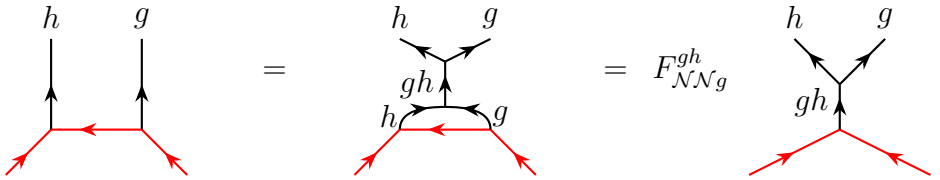
where in the second equality we used a topological move. Next we imagine an identity line stretched from the top left vertex to the middle of the top right line, do an F-move, and use a topological deformation to give



(3.71)

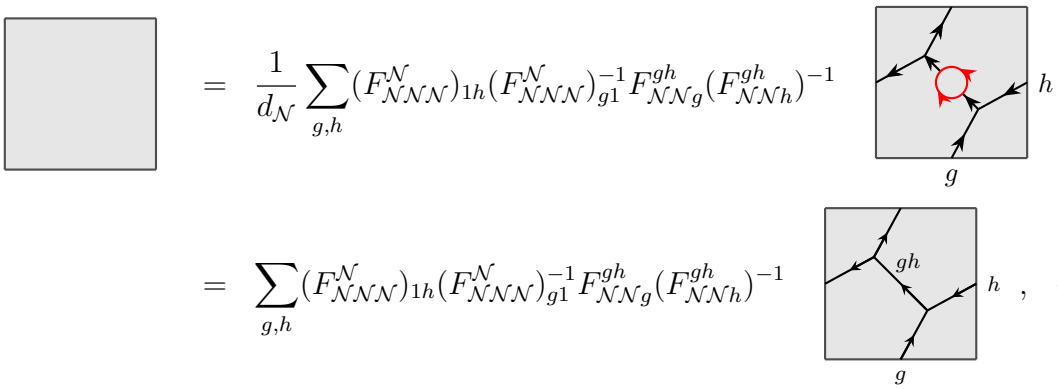
[26] For graphical simplicity, we drop the $\times$ that we were using to dress trivalent junctions in previous subsections. We can always reinstate them by just checking the number of lines going into/out of the vertex.

We must now shrink the internal bubble. In order to do so, let us consider the upper-left portion of the bubble. This may be manipulated as follows,



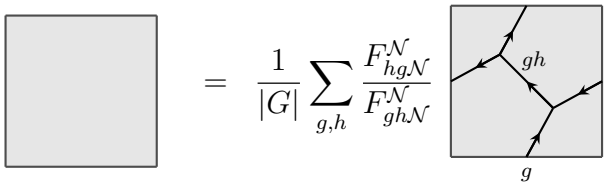
(3.72)

where in the first step we used the first equation of (3.7), and in the second step we used an F-move and the second equation of (3.18). Doing the same thing on the bottom right of (3.71) then gives



(3.73)

where we again made use of (3.18). In fact, it is possible to simplify the product of the F-symbols above by using the pentagon identity, giving the final result



(3.74)

This is precisely the result expected for self-duality under gauging of G with a discrete torsion specified by $F^\mathcal{N}_{hg\mathcal{N}}/F^\mathcal{N}_{gh\mathcal{N}}$.

Exercise: Verify that the pentagon identities actually give such a simplification. In particular, you will have to show that

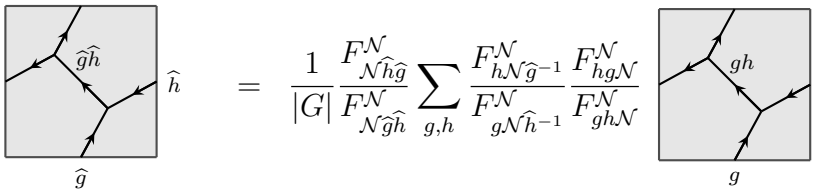
$$(F^\mathcal{N}_{\mathcal{N}\mathcal{N}\mathcal{N}})_{gh} = (F^\mathcal{N}_{\mathcal{N}\mathcal{N}\mathcal{N}})_{11}\frac{F^g_{g\mathcal{N}\mathcal{N}}F^\mathcal{N}_{g\mathcal{N}h^{-1}}}{F^\mathcal{N}_{\mathcal{N}hh^{-1}}F^1_{\mathcal{N}\mathcal{N}h^{-1}}}\,, \tag{3.75}$$

and

$$F^\mathcal{N}_{xy\mathcal{N}}F^z_{x\mathcal{N}\mathcal{N}}F^{x^{-1}z}_{y\mathcal{N}\mathcal{N}} = F^z_{xy,\mathcal{N},\mathcal{N}}\,, \qquad x,y,z\in G\,. \tag{3.76}$$

You will also need to use that $(F^\mathcal{N}_{\mathcal{N}\mathcal{N}\mathcal{N}})^2_{11} = 1/d^2_\mathcal{N} = 1/|G|$.

A similar result can be obtained when a background for the dual G symmetry is turned on [88],



$$= \frac{1}{|G|}\frac{F^{\mathcal{N}}_{\mathcal{N}\widehat{h}\widehat{g}}}{F^{\mathcal{N}}_{\mathcal{N}\widehat{g}\widehat{h}}}\sum_{g,h}\frac{F^{\mathcal{N}}_{h\mathcal{N}\widehat{g}^{-1}}}{F^{\mathcal{N}}_{g\mathcal{N}\widehat{h}^{-1}}}\frac{F^{\mathcal{N}}_{hg\mathcal{N}}}{F^{\mathcal{N}}_{gh\mathcal{N}}} \quad . \tag{3.77}$$

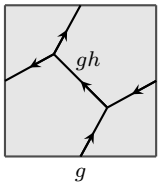
3.6 Gauging non-invertible symmetries

Just as invertible global symmetries can be promoted to gauge symmetries, it is natural to ask if there is a sense in which non-invertible symmetries can be gauged. Indeed, such a generalization exists (see e.g. [22, 89–91]).

3.6.1 Algebra objects

To begin, let us rephrase the gauging of an invertible symmetry as follows. As we have seen above, for discrete Abelian group G, the torus partition function of the gauged theory is given by,[27]

$$Z_{\mathcal{X}/G} = \frac{1}{|G|}\sum_{g,h\in G}\mu(g,h)\mu^{\vee}(h,g)$$

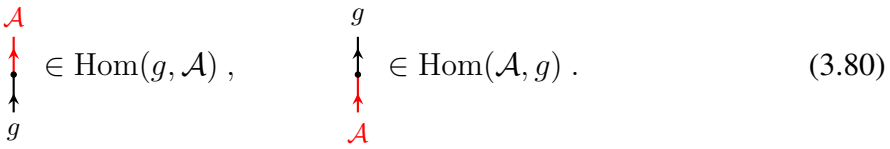


, (3.78)

where $\mu(g,h)\mu^{\vee}(h,g)$ captures the discrete torsion; this notation will be explained shortly. We may now rephrase this gauging prescription as follows. Begin by introducing the non-simple line

$$\mathcal{A} := \bigoplus_{g\in G} g \ . \tag{3.79}$$

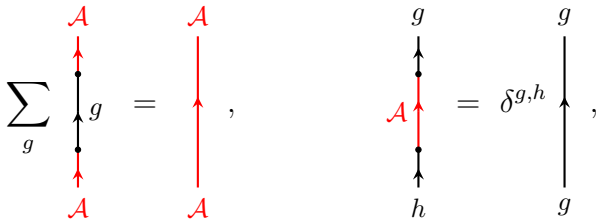
Because $\mathcal{A}$ is composed of $g \in G$, there exist non-zero, and in particular one-dimensional, hom-spaces $\mathrm{Hom}(g,\mathcal{A})$ and $\mathrm{Hom}(\mathcal{A},g)$, whose basis vectors can be chosen as inclusion and projection maps, respectively, for the constituents of $\mathcal{A}$. The basis vectors for these spaces are written graphically as follows,



(3.80)

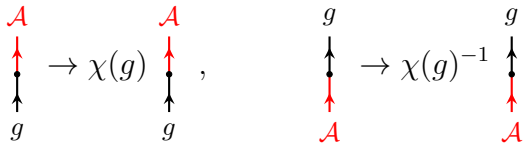
[27] A similar formula holds for non-Abelian G as long as we restrict the sum to commuting pairs.

In particular, for any G, we will always have $\dim \mathrm{Hom}(1, \mathcal{A}) = \dim \mathrm{Hom}(\mathcal{A}, 1) = 1$, capturing the presence of a topological endpoint for $\mathcal{A}$. We may choose a complete orthonormal basis such that the following conditions are satisfied,



(3.81)

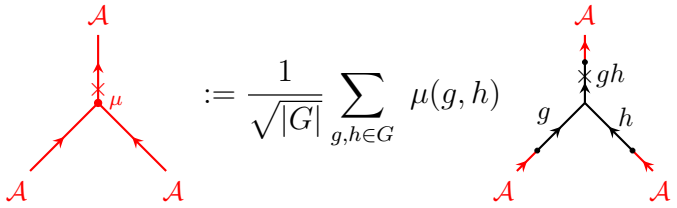
with the first condition capturing completeness and the second capturing orthonormality. We are always free to rescale these junction vectors by a phase,



(3.82)

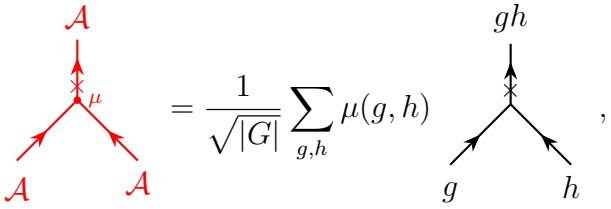
which preserves the two conditions. As in (3.24), this is referred to as a "gauge redundancy."

In addition to the line $\mathcal{A}$ itself, we must also specify rules for local fusion of the line, known as a *multiplication* junction μ. By using the hom spaces above, it is possible to represent the multiplication rule in terms of that of the constituent simple lines as,



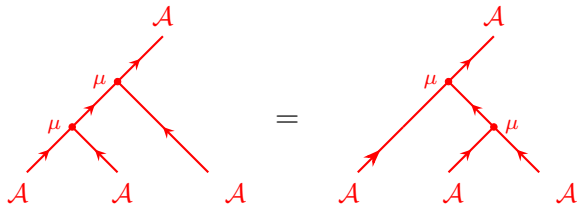
(3.83)

Note that for notational simplicity, one often drops the hom elements on the right-hand side and simply draws the simple internal lines, i.e.



(3.84)

but what is really meant by this is what is in (3.83). The multiplication must be chosen such that the following associativity condition is satisfied,



(3.85)

This condition may be rewritten in terms of the coefficients $\mu(g,h)$ by expanding the junctions using (3.83), shrinking an internal $\mathcal{A}$ line using (3.81), using an F-symbol of the constituent simple lines on the left, and comparing to the result on the right. This gives rise to the constraint

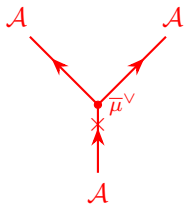
$$\mu(g,h)\mu(gh,k)\omega(g,h,k) = \mu(g,hk)\mu(h,k)\ , \tag{3.86}$$

or, in terms of the operation δ defined in (3.13),[28]

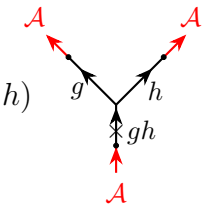
$$\omega(g,h,k) = \delta\mu(g,h,k)\ . \tag{3.88}$$

This tells us that the associator cocycle $\omega(g,h,k)$ must be cohomologically trivial—this encodes the lack of anomaly for the group G we will be gauging. The pair of $\mathcal{A}$ together with a multiplication operation satisfying the above is referred to as an *algebra*.

Similar to an algebra, one may also define a *coalgebra* by equipping $\mathcal{A}$ with a *comultiplication* operation, i.e. a splitting junction,

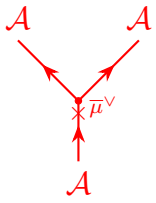


$$:= \frac{1}{\sqrt{|G|}} \sum_{g,h\in G} \mu^\vee(g,h)$$

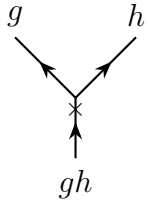


, (3.89)

or in simplified notation



$$= \frac{1}{\sqrt{|G|}} \sum_{g,h\in G} \mu^\vee(g,h)$$

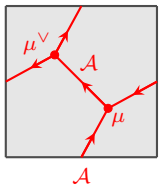


. (3.90)

The coefficients $\mu^\vee(g,h)$ are constrained to satisfy the upside-down analog of (3.85).

The algebra objects that we will be concerned with here will be equipped with *both* multiplication and comultiplication operations. Given such an algebra object, we may rewrite the gauged torus partition function in terms of it as follows,

$$Z_{\mathcal{X}/G} \quad =$$



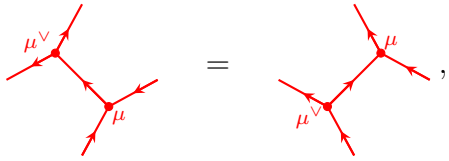
, (3.91)

[28] Note that the gauge redundancy in (3.82), which shifts the multiplication by

$$\mu(g,h) \sim \mu(g,h)\delta\chi(g,h)\ , \tag{3.87}$$

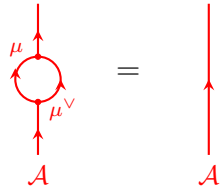
does not modify this relation.

where the choice of discrete torsion is now captured by the choice of (co-)multiplication. For the gauging process to be unambiguous (i.e. non-anomalous), it is clear that we must have the additional constraint



(3.92)

which is known as the *Frobenius* condition. Algebras which satisfy this constraint are referred to as *Frobenius algebras*. One also requires the consistency conditions shown below, known as the separability condition,



(3.93)

which ensures that configurations of $\mathcal{A}$ that are continuously deformable into one another encode the same gauging.[29] The upshot is then that discrete gaugings of G correspond to separable (symmetric) Frobenius algebras.

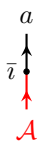
We may now generalize this to the non-invertible case. We proceed similarly to before by defining a certain object

$$\mathcal{A} = \bigoplus_{a\in\mathcal{C}} Z_a\, a\ , \tag{3.94}$$

with Z_a being non-negative integer coefficients; once again, we require $Z_1 = 1$. The hom-spaces $\mathrm{Hom}(a,\mathcal{A})$ and $\mathrm{Hom}(\mathcal{A},a)$ are now Z_a-dimensional, and the basis vectors may be represented graphically as before. Now, however, we must add a label keeping track of which basis vector we are considering,

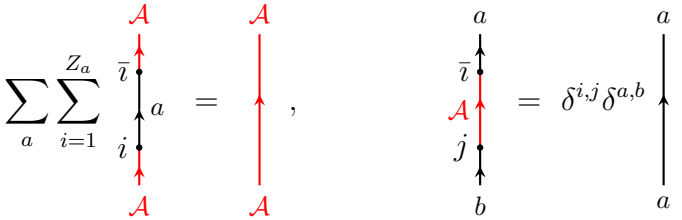


$\in \mathrm{Hom}(a,\mathcal{A})$,



$\in \mathrm{Hom}(\mathcal{A},a)$, $\quad i,\bar{\imath} = 1,\dots,Z_a$; (3.95)

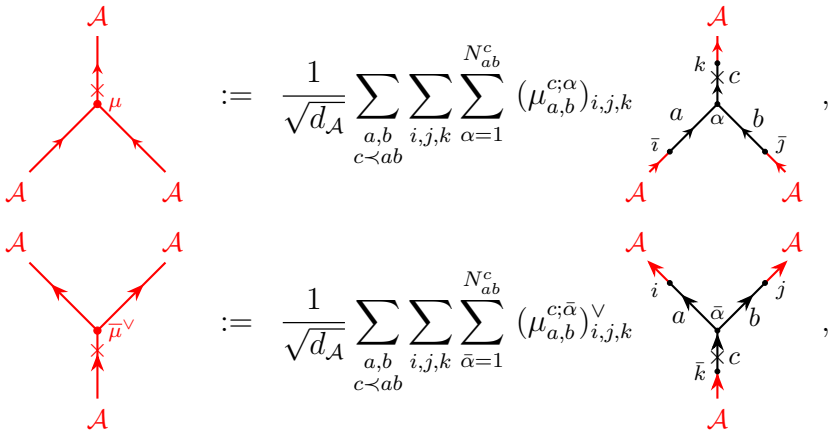
the completeness and orthonormality conditions are similarly modified,



(3.96)

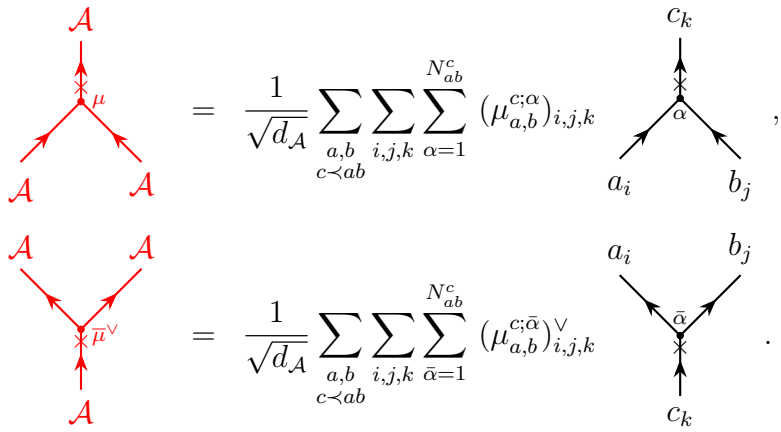
[29] In addition to the conditions discussed here, there is also a symmetricity condition; see e.g. Figure 4.5 of [74].

The multiplication and co-multiplication are defined in a similar manner to before,[30]



(3.97)

and it is once again common to use a short-hand notation for the right-hand side,



(3.98)

For consistency, we must again impose the associativity and separability constraints, as well as the Frobenius condition in (3.92). These constrain not only μ and $\mu^\vee$, but also the allowed coefficients Z_a. Of course, there need not be a unique solution—each different solution corresponds to a different algebra. One then gauges the algebra in exactly the same way as before by inserting a fine mesh of $\mathcal{A}$. For example, on the torus we have the configuration shown in (3.91).

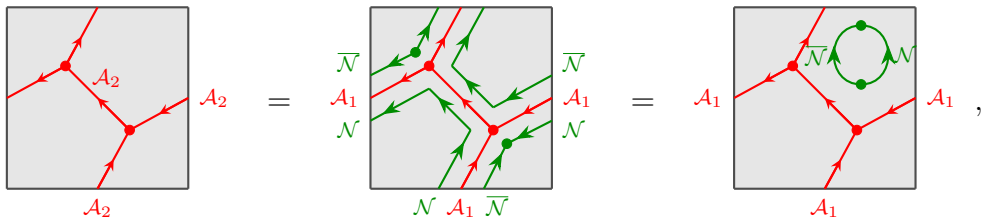
3.6.2 Morita equivalence

Note that it is useful to introduce a notion of equivalence classes of algebra objects, known as *Morita equivalence*. In particular, $\mathcal{A}_1$ is said to be Morita equivalent to $\mathcal{A}_2$, i.e. $\mathcal{A}_1 \sim \mathcal{A}_2$, when their categories of modules are equivalent as $\mathcal{C}$-module categories. A sufficient condition is that there exists a simple object $\mathcal{N}$ such that

$$\mathcal{A}_2 = \mathcal{N} \otimes \mathcal{A}_1 \otimes \overline{\mathcal{N}} \,. \tag{3.99}$$

[30] The notation $c \prec ab$ means that c is contained in the fusion of a and b.

Here the right-hand side carries the canonical algebra structure induced from $\mathcal{A}_1$. The reason for defining such equivalence classes is that gauging Morita equivalent objects gives the *same* theory. This is rather easy to understand pictorially, where the argument is schematically as follows,



(3.100)

whereby we may shrink the bubble to produce a factor of $d_{\mathcal{N}}$, which is then absorbed in the definition of the (co-)multiplication operations of $\mathcal{A}_1$.

A particular case of this is that of Tambara-Yamagami categories, encountered in Section 3.5. In that case, defining $\mathcal{A} = \bigoplus_g g$, we had

$$\mathcal{A} = \mathcal{N} \otimes \overline{\mathcal{N}} = \mathcal{N} \otimes \mathbb{1} \otimes \overline{\mathcal{N}} \,, \tag{3.101}$$

and hence $\mathcal{A}$ was Morita equivalent to the trivial algebra. In other words, gauging $\mathcal{A}$ should give the same theory as gauging $\mathbb{1}$, i.e. the original theory, and this is precisely the statement of Kramers-Wannier duality encountered above. As we saw in Section 3.5, the rigorous calculation is a bit more subtle than just (3.100).

3.6.3 Example: gauging Fibonacci

As a simple example, let us consider the Fibonacci category mentioned in Section 3.3.1, with F-symbols given in (3.33). We now try to gauge the object $\mathcal{A} = 1 \oplus W$. The first step in doing so is choosing (co-)multiplications satisfying the necessary consistency conditions. Since no non-trivial multiplicities appear in $\mathcal{A}$ or the fusion rules, we may drop the indices α, i, j, k, and we obtain e.g. the following conditions,

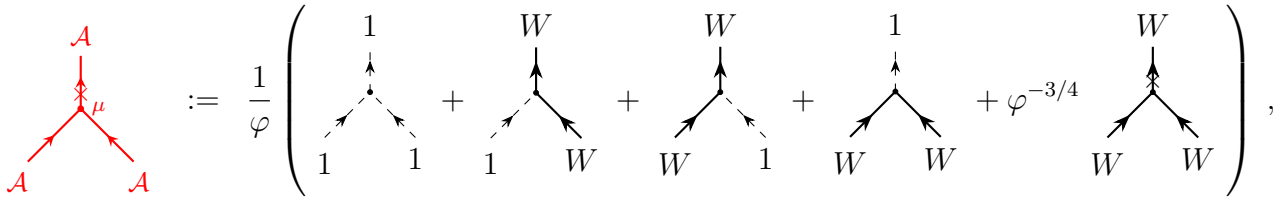
$$\begin{aligned} &\text{Separability}: && \sum_{a,b} N_{ab}^c \sqrt{\frac{d_a d_b}{d_c}} \mu_{ab}^c (\mu_{ab}^c)^\vee = d_{\mathcal{A}} && \forall\, c \in \mathcal{C} \\ &\text{Associativity}: && \sum_e \mu_{ab}^e \mu_{ec}^d (F_{abc}^d)_{ef} = \mu_{bc}^f \mu_{af}^d && \forall a,b,c,d,f \in \mathcal{C} \end{aligned} \tag{3.102}$$

where N_{ab}^c are the fusion coefficients appearing in (3.16), and μ_{ab}^c is non-zero only if N_{ab}^c is. The second of these is the non-invertible analog of the relation in (3.88).

One now attempts to find a solution to these equations. In general, this should be done with computer assistance, but in the current case things are simple enough to do by hand. For example, setting $(a,b,c,d) = (W,1,W,1)$ in the associativity constraint gives $\mu_{W1}^W = \mu_{1W}^W$, and likewise setting $(a,b,c,d) = (1,W,W,1)$ gives $\mu_{1W}^W = \mu_{11}^1$. Proceeding in this way, we ultimately find that

$$\mu_{11}^1 = \mu_{1W}^W = \mu_{W1}^W = \mu_{WW}^1 = \varphi^{3/4} \mu_{WW}^W = 1 \,, \tag{3.103}$$

and similarly for the co-multiplications. In other words, as long as we define trivalent junctions as



(3.104)

then we can consistently insert a mesh of $\mathcal{A}$ as in (3.91), which implements gauging of the Fibonacci symmetry. Note that since $\mathcal{A} = 1 + W = W \otimes \overline{W}$, this algebra is Morita trivial, and hence we expect that gauging this algebra will give back the original theory. It is a good exercise to check this explicitly.

Exercise: For those who are familiar with CFT, use the action of the Verlinde line W in $(G_2)_1$ and $(F_4)_1$ WZW models, together with the F-symbols above, to compute the various twisted partition functions, and then take the appropriate sum to explicitly show invariance under gauging.

3.7 Representation theory and the SymTFT

From the point of view of Physics, one of the most important aspects of symmetry groups is the fact that fields/operators in the theory are labelled by representations of the group. It is natural to ask what the analog of representation theory is for non-invertible symmetry. The answer is that irreducible representations of the tube algebra associated to a fusion category $\mathcal{C}$ are given by simple objects of the so-called *Drinfeld center* $\mathcal{Z}(\mathcal{C})$ [92–95]. More physically, irreducible representations of categorical symmetry in $(1+1)$d are in one-to-one correspondence with simple line operators in a certain $(2+1)$d topological theory known as the *Symmetry TFT (SymTFT)*. In this subsection we first review the basic idea of the SymTFT (closely following [96]), after which we illustrate the technique in the cases of $\mathcal{C} = \mathrm{Vec}^{\omega}(\mathbb{Z}_2)$ and $\mathcal{C} = \mathrm{Ising}$.

3.7.1 SymTFT primer

The SymTFT of a theory $\mathcal{X}$ in d-dimensions with global symmetry $\mathcal{C}$ is a topological theory SymTFT($\mathcal{C}$) in $(d+1)$-dimensions which, when compactified on an interval with appropriate boundary conditions, gives back the theory $\mathcal{X}$ of interest. One of the many nice properties of the SymTFT is that it decouples the dynamics of $\mathcal{X}$ from the symmetries. This makes the action of various topological manipulations ϕ (such as gauging a discrete symmetry or adding local counterterms in background fields) more transparent: indeed, by choosing appropriate

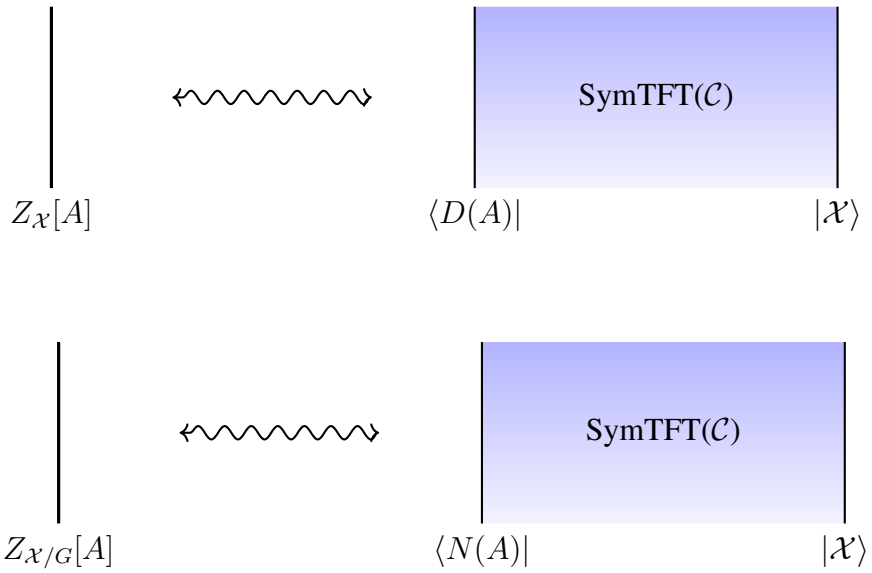


Figure 2: Schematic picture of the SymTFT. By imposing Dirichlet (resp. Neumann) boundary conditions on the left and the non-topological boundary condition (3.105) on the right, we may compactify to obtain $\mathcal{X}$ (resp. $\mathcal{X}/G$). In general, there exist other topological boundary conditions as well.

topological boundary conditions for the SymTFT before compactification to d-dimensions, we can obtain not only the original $\mathcal{X}$, but any theory of the form $\phi(\mathcal{X})$.

The basic idea is illustrated in Figure 2. The $(d+1)$-dimensional SymTFT is placed on an interval, where the right boundary is endowed with non-topological boundary conditions capturing the dynamics of $\mathcal{X}$, whereas the left boundary condition is topological. As described in Section 2.7, a theory in $(d+1)$ dimensions assigns a Hilbert space to d-dimensional subspaces, and hence the boundaries of SymTFT($\mathcal{C}$) can be labeled by appropriate elements of its Hilbert space. The dynamical boundary of SymTFT($\mathcal{C}$) is taken to be

$$|\mathcal{X}\rangle = \sum_a Z_{\mathcal{X}}[a]|a\rangle \ , \tag{3.105}$$

where $|a\rangle$ is an appropriate basis of the state space. When $\mathcal{C}$ is a finite group G, the a here represents the collective set of flat connections of G, and $Z_{\mathcal{X}}[a]$ denotes the partition function of $\mathcal{X}$ coupled to gauge fields a. When $\mathcal{C}$ is a fusion category rather than a group, the label a is an appropriate collective label for the topological defects in the theory.

For the following formulas, we specialize to a finite Abelian, non-anomalous zero-form symmetry G. The topological boundary of SymTFT($\mathcal{C}$) can take a number of forms, with common options being Dirichlet or Neumann boundary conditions. Dirichlet boundary conditions fix the fields a to certain values A, whereas Neumann boundary conditions allow a to fluctuate

freely. In the state notation, these can be written as

$$\begin{aligned} \text{Dirichlet}: \qquad & |D(A)\rangle = \sum_a \delta(a-A)|a\rangle \,, \\ \text{Neumann}: \qquad & |N(A)\rangle = \sum_a \exp\left(-i\int a\cup A\right)|a\rangle \,. \end{aligned} \tag{3.106}$$

The normalization factor in $\int a \cup A$ depends on the symmetry and is suppressed here. For Dirichlet states, A is a background gauge field for G; for Neumann states, it is a background gauge field for the quantum dual. Cup products use the natural pairing, and the sums and state normalizations include the suppressed gauge-volume factors.

Because the SymTFT is topological, the length of the interval is unimportant and can be shrunk to zero size, upon which one obtains a d-dimensional theory whose partition function can be computed by taking the inner product between the states on the left and right boundaries. In the case of Dirichlet boundary conditions, one obtains

$$\langle D(A)|\mathcal{X}\rangle = \sum_{a,a'} Z_{\mathcal{X}}[a]\,\delta(a'-A)\langle a'|a\rangle = \sum_{a,a'} Z_{\mathcal{X}}[a]\,\delta(a'-A)\delta(a-a') = Z_{\mathcal{X}}[A] \tag{3.107}$$

and hence reproduces the original d-dimensional theory $\mathcal{X}$, coupled to background fields A. On the other hand, by putting Neumann boundary conditions on the left, we obtain

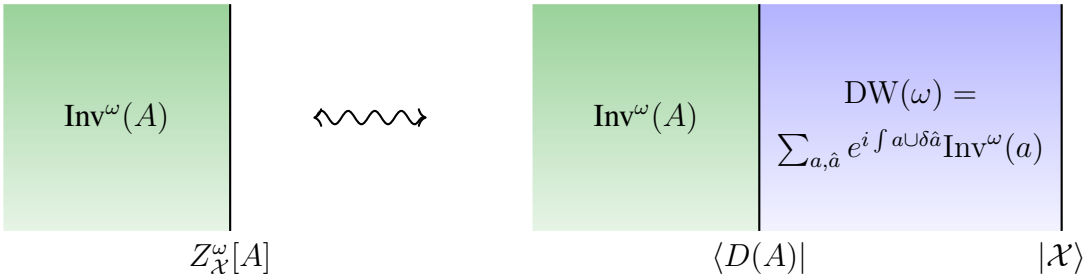
$$\langle N(A)|\mathcal{X}\rangle = \sum_{a,a'} Z_{\mathcal{X}}[a]\,\exp\left(i\int a'\cup A\right)\langle a'|a\rangle = \sum_{a} Z_{\mathcal{X}}[a]\,\exp\left(i\int a\cup A\right) = Z_{\mathcal{X}/G}[A]\,. \tag{3.108}$$

In other words, in this case we obtain a d-dimensional theory $\phi(\mathcal{X})$, where ϕ represents gauging the discrete symmetry G (which is indeed a topological operation). Choosing mixed Dirichlet-Neumann boundary conditions also allows us to obtain $\mathcal{X}$ with a subgroup of G gauged.

More generally, one can dress either boundary condition in (3.106) by a d-cocycle ν_d. The corresponding theories obtained by shrinking the slab are $\mathcal{X}$ stacked with a counterterm, and $\mathcal{X}/G$ with the gauging done with discrete torsion associated with the d-cocycle,

$$\begin{aligned} |D(A)_{\nu_d}\rangle &= \sum_a \delta(a-A)\exp\left(-i\int \nu_d(a)\right)|a\rangle \longleftrightarrow \langle D(A)_{\nu_d}|\mathcal{X}\rangle = Z_{\mathcal{X}}[A]\exp\left(i\int \nu_d(A)\right)\,, \\ |N(A)_{\nu_d}\rangle &= \sum_a \exp\left(-i\int a\cup A - i\int \nu_d(a)\right)|a\rangle \longleftrightarrow \langle N(A)_{\nu_d}|\mathcal{X}\rangle = Z_{(\mathcal{X}\times\nu_d)/G}[A]\,. \end{aligned} \tag{3.109}$$

One therefore expects that any two theories $\mathcal{X}$ and $\mathcal{X}'$ that are related by a topological manipulation ϕ, i.e. $\mathcal{X}' = \phi(\mathcal{X})$, can be obtained by the slab construction with the *same* SymTFT but with *different* topological boundary conditions.



(a) (b)

Figure 3: A theory with anomaly ω can be realized on the boundary of an invertible theory $\mathrm{Inv}^\omega(A)$. This is possible if and only if the Symmetry TFT is a (generalized) DW theory.

3.7.2 Invertible symmetry and Dijkgraaf-Witten theories

The SymTFT takes a particularly simple form when the symmetry in question is group-like. To see this, begin by considering a theory $\mathcal{X}$ with background gauge fields A and anomaly ω. As discussed in Section 2.7, the anomaly inflow paradigm, in modern language, says that such a theory may be realized on the boundary of an appropriate invertible theory $\mathrm{Inv}^\omega(A)$; see Figure 3(a).

Given Figure 3(a), one can promote the background gauge field A to a dynamical gauge field a both on the right boundary and in a slab in the bulk. The bulk theory within this slab is then a nontrivial TQFT, namely a *(generalized) Dijkgraaf-Witten (DW) theory*[31]

$$\mathrm{DW}(\omega) = \sum_{a,\hat{a}} \exp\left(i\int a\cup\delta\hat{a}\right)\mathrm{Inv}^\omega(a)\,, \tag{3.110}$$

where the displayed BF form assumes finite Abelian G, with a a G-valued 1-cochain and $\hat{a}$ a $(d-1)$-cochain for the dual group, and as commented above we suppress the normalization in the BF term. On the left of the slab, one further imposes Dirichlet boundary conditions to pin the dynamical field a to the background A. The theory $\mathcal{X}$ is recovered by taking the thin slab limit where the non-topological boundary condition $|\mathcal{X}\rangle$ and the Dirichlet boundary condition $|D(A)\rangle$ collide. We conclude that the SymTFT for an invertible symmetry with an anomaly is given by a (generalized) DW theory (3.110), i.e. a discrete gauge theory, potentially with a non-trivial twist.

3.7.3 Example: $\mathcal{C} = \mathrm{Vec}(\mathbb{Z}_2)$

We now illustrate the SymTFT technique for the simplest example of a $(1+1)$d theory with non-anomalous $\mathbb{Z}_2$ symmetry. By the above discussion, the SymTFT for such a theory is just

[31] The qualifier "generalized" here refers to the fact that the relevant invertible theories are not restricted to the group cohomology elements studied in the original work of Dijkgraaf and Witten [97].

$(2+1)$d $\mathbb{Z}_2$ gauge theory,

$$S = \pi \int_{M_3} a \cup \delta \widehat{a} \, . \tag{3.111}$$

This theory has four simple line operators, which can be written as

$$L_{(e,m)}(\gamma) := e^{i\pi e \oint_\gamma a} e^{i\pi m \oint_\gamma \widehat{a}} \, , \qquad e, m = 0, 1 \, . \tag{3.112}$$

The fusions of these line operators are straightforward,

$$L_{(e,m)}(\gamma) \times L_{(e',m')}(\gamma) = L_{(e+e',m+m')}(\gamma) \, , \tag{3.113}$$

but the braidings between them are non-trivial—in particular, inserting $L_{(e,m)}$ on a loop γ and $L_{(e',m')}$ on another loop γ', one finds that[32]

$$\langle L_{(e,m)}(\gamma) L_{(e',m')}(\gamma') \dots \rangle = (-1)^{(em'+me')\mathrm{Link}(\gamma,\gamma')} \langle \dots \rangle \, . \tag{3.114}$$

This may be derived from the path integral as follows. Inserting the line operators $L_{(e,m)}$ and $L_{(e',m')}$ amounts to modifying the action to

$$\begin{aligned} S &= \pi \int_{M_3} a \cup \delta \widehat{a} + \pi \int_\gamma (e a + m \widehat{a}) + \pi \int_{\gamma'} (e' a + m' \widehat{a}) \\ &= \pi \int_{M_3} a \cup \delta \widehat{a} + \pi \int (e \omega_\gamma + e' \omega_{\gamma'}) \cup a + \pi \int (m \omega_\gamma + m' \omega_{\gamma'}) \cup \widehat{a} \, , \end{aligned} \tag{3.115}$$

where ω_γ is the Poincaré dual of γ. Integrating out $\widehat{a}$ enforces the following,

$$\delta a = -(m \omega_\gamma + m' \omega_{\gamma'}) \, . \tag{3.116}$$

This means that if we introduce a two-manifold Σ such that $\partial \Sigma = m\gamma + m'\gamma'$, known as the *Seifert surface* associated to the curve, then we have $a = -\omega_\Sigma$, i.e. a is the Poincaré dual of Σ. Plugging this back into the action gives

$$\begin{aligned} -\pi \int (e \omega_\gamma + e' \omega_{\gamma'}) \cup \omega_\Sigma &= -\pi \int \omega_{(e\gamma + e'\gamma') \cap \Sigma} \\ &= -\pi \int \left[e \, \omega_{\gamma \cap \Sigma} + e' \, \omega_{\gamma' \cap \Sigma} \right] \, . \end{aligned} \tag{3.117}$$

Finally, we note that $\int \omega_{\gamma \cap \Sigma} = m' \, \mathrm{Link}(\gamma, \gamma')$ and likewise $\int \omega_{\gamma' \cap \Sigma} = m \, \mathrm{Link}(\gamma, \gamma')$, so that in total we get

$$-\pi \, (em' + e'm) \, \mathrm{Link}(\gamma, \gamma') \, , \tag{3.118}$$

and hence the path integral gets a factor of $e^{iS} = (-1)^{(em'+e'm)\mathrm{Link}(\gamma,\gamma')}$, which is the factor that appears in (3.114).

[32] We work in S^3 with zero framing and assume that these loops are not linked with other insertions.

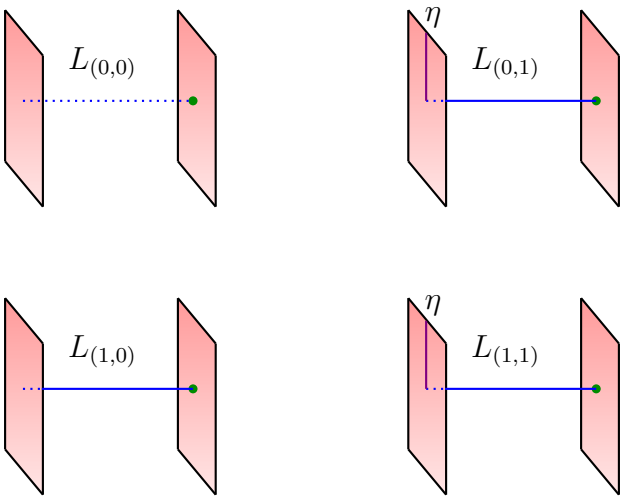


Figure 4: The four sectors of a theory with $\mathbb{Z}_2$ global symmetry. The red planes represent the two boundaries of the SymTFT. The green dot represents a non-topological operator on the dynamical boundary.

Now in order to get the original 2d theory from the $\mathbb{Z}_2$ gauge theory described above, we need to choose a dynamical boundary condition for the right boundary, and a topological boundary condition for the left boundary. For the topological boundary, let us first choose Dirichlet boundary conditions for the gauge field a. This means that $L_{(0,0)}$ and $L_{(1,0)}$ are trivialized on the boundary, whereas $L_{(0,1)}$ and $L_{(1,1)}$ both reduce to the same non-trivial defect on the boundary, which corresponds to the generator η of the boundary $\mathbb{Z}_2$ symmetry.

We may now consider the four configurations shown in Figure 4, in which various bulk lines stretch between the two boundaries. When ending on the non-topological boundary, the lines end on a potentially non-topological operator. Upon shrinking the bulk, these configurations define four sectors of operators in the original $(1+1)$d theory. Indeed, the two sectors in the first column define local operators, whereas the two sectors in the second column define operators in the η-twisted sector. We may measure the $\mathbb{Z}_2$ charge of each of these operators as follows. We begin by encircling the operators by the boundary line η. In the SymTFT picture, this encircling happens on the topological boundary. We may then pull the topological operator η into the bulk, upon which it becomes $L_{(0,1)}$. We then use the braiding of $L_{(0,1)}$ and the bulk line running perpendicular to the boundary to obtain the charge. This is shown for the sector labelled by $L_{(1,0)}$ in Figure 5. In this way, we find that the sectors labelled by $L_{(0,0)}$ and $L_{(0,1)}$ are $\mathbb{Z}_2$ even, while the sectors labelled by $L_{(1,0)}$ and $L_{(1,1)}$ are $\mathbb{Z}_2$ odd.

In total then, the four topological lines of the SymTFT correspond to the following four

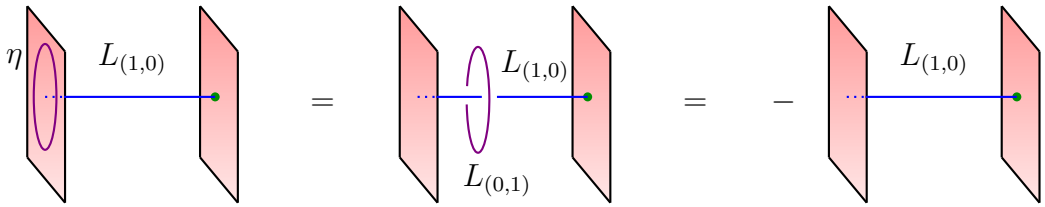


Figure 5: Measuring the $\mathbb{Z}_2$ charge of the operators in the sector labelled by $L_{(1,0)}$. By making use of the bulk braidings (3.114), we find that such operators are $\mathbb{Z}_2$ odd.

sectors of a theory with $\mathbb{Z}_2$ global symmetry,

$$\begin{aligned} L_{(0,0)} &\quad\leftrightarrow\quad \mathbb{Z}_2 \text{ even, untwisted}\,, \\ L_{(1,0)} &\quad\leftrightarrow\quad \mathbb{Z}_2 \text{ odd, untwisted}\,, \\ L_{(0,1)} &\quad\leftrightarrow\quad \mathbb{Z}_2 \text{ even, twisted}\,, \\ L_{(1,1)} &\quad\leftrightarrow\quad \mathbb{Z}_2 \text{ odd, twisted}\,. \end{aligned} \tag{3.119}$$

We see that the SymTFT knows about more than just usual representation theory (i.e. $\mathbb{Z}_2$ odd vs. even) of local operators—it also contains the data of twisted sectors. This will be crucial in the case of non-invertible symmetries, since non-invertible symmetries exchange untwisted and twisted sector operators (e.g. $\mathcal{N}$ exchanging σ and μ in the Ising model), and hence the representation theory of the non-invertible symmetry must accommodate both.

Let us note here that we could have also chosen the topological boundary condition to be Dirichlet boundary conditions for $\widehat{a}$, instead of a. This would simply exchange a and $\widehat{a}$, and consequently $L_{(1,0)}$ and $L_{(0,1)}$, throughout the above discussion. In particular, from the correspondences in (3.119), this exchanges the $\mathbb{Z}_2$ odd untwisted sector with the $\mathbb{Z}_2$ even twisted sector, while leaving the other sectors unchanged. Comparing this to the discussion around (2.126) and (2.127), we see that this is exactly the effect expected when gauging a $\mathbb{Z}_2$ symmetry in the boundary two-dimensional theory. We thus confirm that different discrete gaugings (which are examples of the topological manipulation ϕ discussed above) of a single theory correspond to different topological boundary conditions of a single SymTFT.[33] The boundary gauging is implemented by a bulk zero-form symmetry, namely the electromagnetic duality symmetry $\mathbb{Z}_2^{\text{EM}}$ exchanging $a \leftrightarrow \widehat{a}$. This will be important when we study the SymTFT for the Ising category below.

3.7.4 Example: $\mathcal{C} = \text{Vec}^{\omega}(\mathbb{Z}_2)$

We next consider the case of a $\mathbb{Z}_2$ symmetry with non-trivial 't Hooft anomaly. By the general discussion above, the SymTFT is a $\mathbb{Z}_2$ gauge theory with non-trivial Dijkgraaf-Witten twist,

[33] Note that there is no bosonic topological boundary condition for which $L_{(1,1)}$ is absorbed, since $L_{(1,1)}$ has non-trivial spin, and hence cannot be condensed.

which may be written in the following form

$$S = \pi \int_{M_3} a \cup \delta \widehat{a} + \widehat{a} \cup \widehat{a} \cup \widehat{a} \, . \tag{3.120}$$

Alternatively, introducing the vector $\mathbf{a} = (a, \widehat{a})$ and switching to continuum notation, we may write this as

$$S = \frac{1}{4\pi} \int_{M_3} \mathbf{a} \wedge K d\mathbf{a} \, , \qquad K = \begin{pmatrix} 0 & 2 \\ 2 & -2 \end{pmatrix} \, , \tag{3.121}$$

which is the so-called *K-matrix* presentation. As usual, we are allowed to do field redefinitions of a and $\widehat{a}$, so long as we are careful to preserve their quantization conditions. In particular, we can change $a \to a$, $\widehat{a} \to \widehat{a} + a$, in which case we have

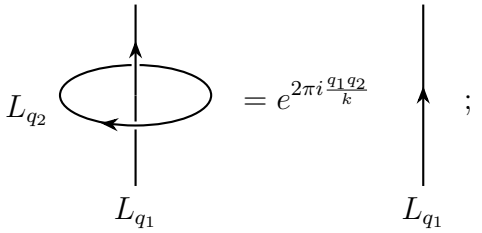
$$S = \frac{1}{4\pi} \int_{M_3} \mathbf{a} \wedge K d\mathbf{a} \, , \qquad K = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} \, . \tag{3.122}$$

In other words, the SymTFT for anomalous $\mathbb{Z}_2$ symmetry is simply $U(1)_2 \times U(1)_{-2}$ Chern-Simons theory.

In general, $U(1)_k$ Chern-Simons theory has $|k|$ lines L_q labelled by the charge $q = 0, \ldots, |k| - 1$, with grouplike fusion rules

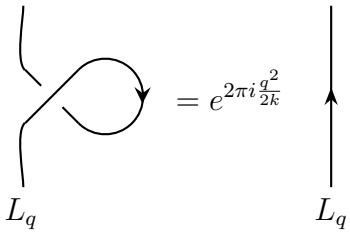
$$L_{q_1} \times L_{q_2} = L_{q_1 + q_2} \, , \tag{3.123}$$

with charges added modulo $|k|$, and braidings given by



(3.124)

see e.g. [98]. In addition, the lines have the following spins



(3.125)

In the special case of $k = \pm 2$, this means that there is exactly one nontrivial line, with spin $\pm i$, referred to as a "semion." Because of this, the theory $U(1)_2 \times U(1)_{-2}$ is known as the *double semion* theory [99, 100]. We denote the two semions by s and $\overline{s}$, each of which has full braiding

phase with another copy of itself $e^{\pm 2\pi i \frac{1}{2}} = -1$, and which have trivial linking with one another. The theory $U(1)_2 \times U(1)_{-2}$ thus contains four topological lines, $1, s, \overline{s}$, and $b := s\overline{s}$, with b having trivial spin.

We now consider topological boundary conditions for this theory. It is impossible to have boundary conditions which absorb s and $\overline{s}$, since these have non-trivial spin and cannot be condensed. Instead, we can only condense b, and from this we conclude the following correspondence between sectors,

$$\begin{aligned} 1 &\quad\leftrightarrow\quad \mathbb{Z}_2 \text{ even, untwisted}\,, \\ b &\quad\leftrightarrow\quad \mathbb{Z}_2 \text{ odd, untwisted}\,, \\ s &\quad\leftrightarrow\quad \mathbb{Z}_2 \text{ even, twisted}\,, \\ \overline{s} &\quad\leftrightarrow\quad \mathbb{Z}_2 \text{ odd, twisted}\,. \end{aligned} \tag{3.126}$$

In the twisted sectors, "even/odd" are bookkeeping labels after rephasing the symmetry action; the canonical projective action has eigenvalues $\pm i$ and squares to -1. In total then, we again have four sectors, just like for the non-anomalous $\mathbb{Z}_2$ symmetry studied before. However, unlike the previous case, note that there is no longer a topological boundary condition that exchanges the $\mathbb{Z}_2$ odd untwisted sector (b) with the $\mathbb{Z}_2$ even twisted sector (s). This corresponds to the fact that it is no longer possible to gauge the boundary $\mathbb{Z}_2$ symmetry, as expected from the fact that it was anomalous. More generally, non-trivial spin or self-braiding of lines in the SymTFT is closely related to the presence of anomalies on the boundary, as discussed in e.g. [101].

3.7.5 Example: $\mathcal{C} = \text{Ising}$

We next consider the case of a theory with Ising symmetry; the Drinfeld center in this case was obtained in [102, 103] in the mathematics literature and [96, 104, 105] in the physics literature. As described in [96], the SymTFT is obtained by gauging the $\mathbb{Z}_2$ EM duality symmetry of $(2+1)$d $\mathbb{Z}_2$ gauge theory.

Indeed, in order to obtain a duality defect $\mathcal{N}$ for $\mathbb{Z}_2$ on the boundary, one begins by considering a duality *interface* separating a theory $\mathcal{X}$ from the gauged theory $\mathcal{X}/\mathbb{Z}_2$. From the bulk perspective, the setup of interest is the insertion of a twist defect in the bulk, with the attached codimension-1 defect running parallel to the topological boundary; see Figure 6. This defect should have the property that it exchanges the Dirichlet and Neumann boundary conditions for a, which is equivalent to exchanging the Dirichlet boundary condition for a with the Dirichlet boundary condition for $\widehat{a}$. In other words, the defect in question is the generator of the $\mathbb{Z}_2^{EM}$ electromagnetic duality symmetry exchanging $a \leftrightarrow \widehat{a}$. In order to turn $\mathcal{N}$ from an interface into a defect, we must require that the theories $\mathcal{X}$ and $\mathcal{X}/\mathbb{Z}_2$ are equivalent, with a compatible action on the physical boundary. We then gauge the bulk $\mathbb{Z}_2^{EM}$ symmetry, so that the corresponding condensation defect becomes transparent, thereby making the twist defect into a genuine line operator in the bulk.

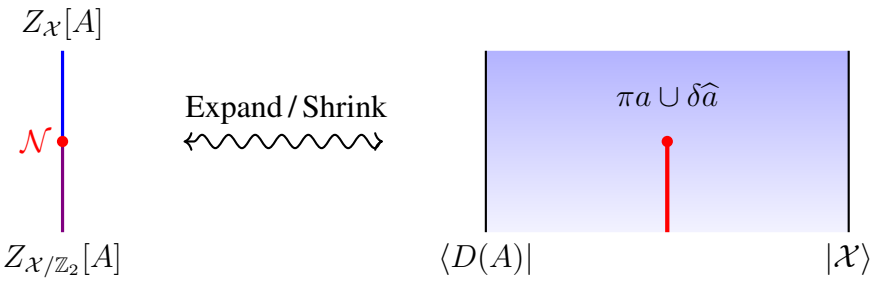


Figure 6: A $(1+1)$d QFT $\mathcal{X}$ with $\mathbb{Z}_2$ symmetry and another $(1+1)$d QFT $\mathcal{X}/\mathbb{Z}_2$ are separated by a topological interface $\mathcal{N}$. This setup can be expanded into a $(2+1)$d slab, where the $(2+1)$d $\mathbb{Z}_2$ SymTFT has an insertion of a twist defect parallel to the Dirichlet boundary.

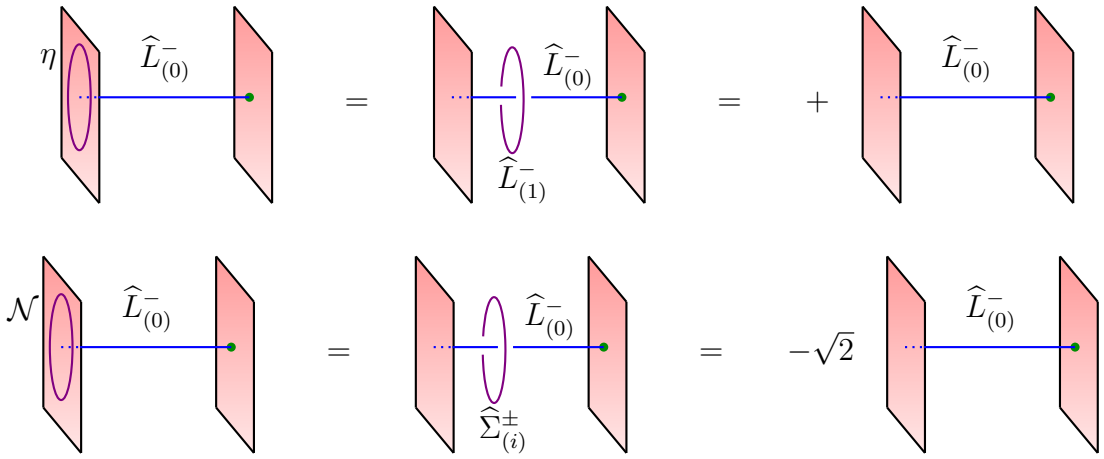


Figure 7: Measuring the charge of the operators in the sector labelled by $\widehat{L}^-_{(0)}$. By making use of the bulk braidings given in [96], we find that these states are $\mathbb{Z}_2$ even, but pick up a factor of $-\sqrt{2}$ when encircled by $\mathcal{N}$.

It is not hard to show that gauging the $\mathbb{Z}_2^{EM}$ symmetry of $\mathbb{Z}_2$ gauge theory gives a theory with a total of nine line operators, which we write as

$$\widehat{L}^{\pm}_{(0)}\ ,\ \widehat{L}^{\pm}_{(1)}\ ,\ \widehat{L}_{[01]}\ ,\ \widehat{\Sigma}^{\pm}_{(0)}\ ,\ \widehat{\Sigma}^{\pm}_{(1)}\ , \tag{3.127}$$

in the notation of [96]; we refer the readers to that reference for more details. If we start as before with the Dirichlet-Neumann orbit of boundary conditions on the topological boundary, then after gauging $\mathbb{Z}_2^{EM}$, we find that the operators $\widehat{L}^{\pm}_{(0)}$ become trivial on the boundary, whereas $\widehat{L}^{\pm}_{(1)}$ become the non-trivial operator η generating the $\mathbb{Z}_2$ symmetry of the boundary Ising category, and the operators $\widehat{\Sigma}^{\pm}_{(i)}$ all reduce to the non-invertible defect $\mathcal{N}$. Finally, for $\widehat{L}_{[01]}$, we find that it reduces to a non-simple line $1+\eta$ on the boundary.

What the above analysis tells us is that we have nine irreducible representations with a total of ten components in the boundary theory—three of which contain local operators (namely $\widehat{L}^{\pm}_{(0)}$, $\widehat{L}_{[01]}$), three of which contain operators in the η-twisted sector (namely $\widehat{L}^{\pm}_{(1)}$, $\widehat{L}_{[01]}$), and



Figure 8: When treating $\mathcal{N}$ as a map from untwisted operators to untwisted operators, it annihilates σ, but when treated as a map from untwisted operators to the η-twisted sector, it produces $\sqrt{2}\,\mu$.

four of which contain operators in the $\mathcal{N}$-twisted sector (namely $\widehat{\Sigma}^{\pm}_{(0)}$, $\widehat{\Sigma}^{\pm}_{(1)}$). We may now measure the charges of operators in each sector under elements of the Ising category by encircling them by the appropriate defects, as is done for operators in sector $\widehat{L}^{-}_{(0)}$ in Figure 7. We see that operators in this sector are $\mathbb{Z}_2$ even, but pick up a factor of $-\sqrt{2}$ under the action of $\mathcal{N}$. The local operator ϵ of the critical Ising model thus belongs to this sector. We may likewise label the other sectors by the operators that they contain in the critical Ising model,[34]

$$\begin{aligned} \widehat{L}^{+}_{(0)} &\leftrightarrow 1\,, & \widehat{L}^{-}_{(0)} &\leftrightarrow \epsilon\,, \\ \widehat{L}^{+}_{(1)} &\leftrightarrow \psi\,, & \widehat{L}^{-}_{(1)} &\leftrightarrow \overline{\psi}\,, \\ \widehat{L}_{[01]} &\leftrightarrow (\sigma,\mu)\,. \end{aligned} \tag{3.128}$$

We see that the sector labelled by $\widehat{L}_{[01]}$ contains both σ and μ. This is because the corresponding representation is two-dimensional—indeed, the non-invertible symmetry $\mathcal{N}$ maps between σ and μ.[35]

This may be seen in the SymTFT picture as follows. We begin by thinking of $\mathcal{N}$ as a map from the untwisted Hilbert space to the untwisted Hilbert space, which corresponds to wrapping σ by a loop of $\mathcal{N}$, with no outgoing lines (see Figure 8). This is expected to give a vanishing result, and can be understood in the bulk as follows. We begin by working in the ungauged picture, in which $\mathcal{N}$ lifts to a twist defect $\Sigma_{(i)}$ which is the boundary of the EM duality surface D_{EM}, and the representation that σ is in is labelled by the non-simple bulk line $L_{(0,1)} + L_{(1,0)}$. By shrinking the disk of D_{EM} in the bulk, we see that a non-zero value for this configuration requires the existence of a junction between $L_{(0,1)}$ and $L_{(1,0)}$, which by charge conservation cannot exist; see Figure 9. On the other hand, if we treat $\mathcal{N}$ as a map from the untwisted Hilbert space to the η-twisted Hilbert space, which corresponds to wrapping the operator in a loop of $\mathcal{N}$ with an outgoing η line (see Figure 8), then a non-zero result is indeed allowed by charge conservation (since η is given by $L_{(1,1)}$ in the bulk), and gives $\sqrt{2}\,\mu$.

[34] We have not discussed ψ and $\bar{\psi}$ in these lectures, but they are well-known operators in the η-twisted sector of the critical Ising model, and we include them here for completeness.

[35] More generally, the dimension of the representation is given, in the SymTFT description, by the dimension of the junction space of the bulk line on the topological boundary.

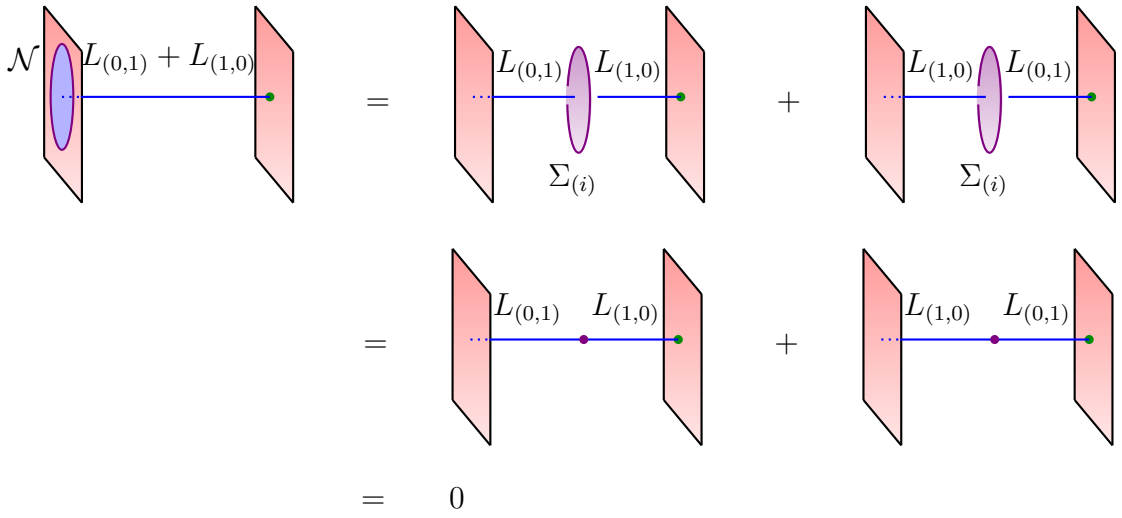


Figure 9: The bulk explanation for the fact that, when treated as a map between untwisted sector operators, $\mathcal{N}$ maps σ to 0.

Exercise: Fill in the details above. In particular, use the bulk braidings and quantum dimensions given in [96] to reproduce the actions in (3.63).

Further references

The first formal descriptions of categorical symmetries in $(1+1)$-dimensional systems were given in [74, 75], and by now there have been a vast number of developments including [92, 106–147]. By far the most well-understood examples of categorical symmetries are line operators in RCFTs. The study of line operators in RCFT was initiated in [148, 149]—far predating the modern study of categorical symmetry—and these results have since been expanded on in a vast number of directions [22, 74, 75, 92, 107, 111, 115, 117, 118, 122, 150–157].

As we discussed above, the representation theory of categorical symmetries is captured by the SymTFT, which has now been studied in various contexts, and in various dimensions [88, 158–173]. Recent results have also shed light on anomalies of non-invertible symmetries [101, 129, 161, 174–177], as well as their spontaneous breaking [178, 179].

An interesting topic which we have not had time to discuss here is the understanding of non-invertible symmetries on unorientable manifolds, and the relation to various structures in mathematics such as Jandl algebras. Some works in this direction include [114, 180–183].

4 Four dimensions and physical applications

Having discussed various aspects of non-invertible symmetries in $(1+1)$-dimensions, we now proceed to a similar discussion in $(3+1)$-dimensions. Fortunately, many of the conceptual tools developed previously, including the notion of half-space gauging, are immediately applicable. That being said, the space of non-invertible symmetries in higher dimensions remains less understood, and in fact there were no concrete examples of non-invertible symmetries in $(3+1)$-dimensions until the last few years [184–187]. As such, below we will not be able to give a comprehensive survey of non-invertible symmetries in $(3+1)$-dimensions, instead just offering some vignettes in particular models.

4.1 Duality defects in $(3+1)$-dimensions

Let us begin by discussing the particular case of duality defects in $(3+1)$-dimensions, which are the generalization of the TY defects in $(1+1)$-dimensions studied in Section 3.5. There are two main constructions of such defects in $(3+1)$-dimensions,

1. Start with a theory with a 0-form symmetry and 1-form symmetry with a certain anomaly, and gauge the 1-form symmetry. The 0-form symmetry will be broken due to the anomaly, but gives rise to a different, non-invertible symmetry.

2. Gauge a 1-form symmetry in half of the space, possibly with discrete torsion.

We now briefly review both of these constructions.

4.1.1 Gauging a theory with an anomaly

Say that we have a $(3+1)$-dimensional theory on a spin manifold M_4 with a zero-form symmetry $\mathbb{Z}_2^{(0)}$ and a one-form symmetry $\mathbb{Z}_2^{(1)}$. Let us denote the background gauge fields for these two symmetries by A_1 and B_2, and furthermore assume that they have a mixed 't Hooft anomaly given by the inflow action

$$\widehat{\mathcal{A}}[A_1, B_2] = \frac{\pi}{2}\int_{M_5} A_1 \cup \mathcal{P}(B_2)\ , \qquad \partial M_5 = M_4\ . \tag{4.1}$$

Here $\mathcal{P}$ is the Pontrjagin square, and the inflow expression is understood with its spin refinement: a zero-form symmetry transformation multiplies the partition function by $e^{\frac{i\pi}{2}\int_{M_4}\mathcal{P}(B_2)}$. This is the same as the set-up in Section 2.10. Recall that in that case we gauged $\mathbb{Z}_2^{(0)}$, and as a result of the anomaly the symmetry $\mathbb{Z}_2^{(1)}$ was extended by the dual $\widehat{\mathbb{Z}}_2^{(2)}$, in such a way that $\delta\widehat{A}_3 = -B_2\cup B_2$. This gave a non-trivial 3-group structure.

On the other hand, let us now consider what happens when we gauge $\mathbb{Z}_2^{(1)}$. In the current case, because the anomaly is quadratic in b_2, we can no longer absorb the M_5 dependence by changing the cocycle condition on the dual background field $\widehat{B}_2$. Thus we cannot salvage $\mathbb{Z}_2^{(0)}$ as part of a higher-group.

However, what we can do is the following. Although the generator of the $\mathbb{Z}_2^{(0)}$ symmetry is anomalous under $\mathbb{Z}_2^{(1)}$ transformations, we may stack it with an appropriate 3d TQFT such that the total anomaly is zero. This composite defect then becomes the generator of a non-invertible zero-form symmetry. Concretely, in the current case the inflow action for the defect is given by $\frac{\pi}{2}\int_{M_4}\mathcal{P}(b_2)$, and it is known that the simplest 3d TQFT that can match this anomaly is $U(1)_2$ Chern-Simons theory [188]. Denoting the original defect by $D(M_3)$, we conclude that the composite defect

$$\mathcal{N}(M_3) \propto \int \mathcal{D}c\, D(M_3)\, e^{\frac{i}{2\pi}\int_{M_3} c\wedge dc + \frac{i}{\pi}\int c\wedge b_2} \tag{4.2}$$

is a non-anomalous topological defect. Note that we are writing b_2 as a $U(1)$ gauge field here—we may always add a BF term to the action that makes it into a $\mathbb{Z}_2$ gauge field later.

The fusion rules may be obtained via direct computation. First of all, denoting the one-form symmetry generator by $L(\Sigma_2) = e^{i\oint b_2}$, it is clear that we have,

$$\begin{aligned} L(\Sigma_2)\times\mathcal{N}(M_3) &\propto \int \mathcal{D}c\, D(M_3)\, e^{\frac{i}{2\pi}\int_{M_3} c\wedge dc + \frac{i}{\pi}\int c\wedge b_2 + i\int_{\Sigma_2} b_2} \\ &= \int \mathcal{D}c\, D(M_3)\, e^{\frac{i}{2\pi}\int_{M_3} c\wedge dc + \frac{i}{\pi}\int (c+\pi\omega_{\Sigma_2})\wedge b_2}\,. \end{aligned} \tag{4.3}$$

For the discrete phases, let ω_{Σ_2} denote an integer lift of the mod-2 Poincaré dual of Σ_2; the following manipulations are understood in the cochain formulation. Then changing variables via $c \to c - \pi\omega_{\Sigma_2}$, we have

$$L(\Sigma_2)\times\mathcal{N}(M_3) = e^{\frac{i\pi}{2}\int \omega_{\Sigma_2}\cup\delta\omega_{\Sigma_2}}\,\mathcal{N}(M_3)\,. \tag{4.4}$$

The overall factor can be rewritten in terms of the *triple intersection number*,

$$Q(\Sigma_2) := \frac{1}{2}\int \omega_{\Sigma_2}\cup\delta\omega_{\Sigma_2}\,, \tag{4.5}$$

which finally gives

$$L(\Sigma_2)\times\mathcal{N}(M_3) = (-1)^{Q(\Sigma_2)}\mathcal{N}(M_3)\,. \tag{4.6}$$

As for the fusion of $\mathcal{N}(M_3)$ with itself (or its orientation reversal), we have,

$$\begin{aligned} \mathcal{N}(M_3)\times\mathcal{N}(M_3)^\dagger &\propto \int \mathcal{D}c\mathcal{D}\bar{c}\, e^{\frac{i}{2\pi}\int_{M_3}(c\wedge dc - \bar{c}\wedge d\bar{c}) + \frac{i}{\pi}\int_{M_3}(c-\bar{c})\wedge b_2} \\ &= \int \mathcal{D}c\mathcal{D}\widehat{c}\, e^{\frac{i}{\pi}\int c\wedge d\widehat{c} - \frac{i}{2\pi}\int \widehat{c}\wedge d\widehat{c} + \frac{i}{\pi}\int \widehat{c}\wedge b_2}\,, \end{aligned} \tag{4.7}$$

where in the second equality we have changed variables to $\widehat{c} := c - \bar{c}$. Note that the first term is simply a BF term, and we can easily integrate c out. Returning to $\mathbb{Z}_2$-valued fields (so that the preceding $U(1)$ two-form is πb_2), we are then left with

$$\mathcal{N}(M_3)\times\mathcal{N}(M_3)^\dagger \propto \sum_{\widehat{c}\in H^1(M_3,\mathbb{Z}_2)} e^{-\frac{i\pi}{2}\int \widehat{c}\cup\delta\widehat{c} + i\pi\int\widehat{c}\cup b_2}\,, \tag{4.8}$$

or in homology notation,

$$\mathcal{N}(M_3) \times \mathcal{N}(M_3)^\dagger \propto \sum_{\Sigma_2 \in H_2(M_3,\mathbb{Z}_2)} (-1)^{Q(\Sigma_2)} L(\Sigma_2) \ . \tag{4.9}$$

Note that we have been somewhat cavalier with the overall normalization—as shown in e.g. Appendix B of [186], the correct normalization is

$$\mathcal{N}(M_3) \times \mathcal{N}(M_3)^\dagger = \frac{1}{|H^0(M_3,\mathbb{Z}_2)|} \sum_{\Sigma_2 \in H_2(M_3,\mathbb{Z}_2)} (-1)^{Q(\Sigma_2)} L(\Sigma_2) \ . \tag{4.10}$$

This can be understood by noting that, from the point of view of M_3, the submanifold Σ_2 is codimension-1, so the right-hand side is analogous to the gauging of a zero-form symmetry in $(2+1)$-dimensions, whose coefficient is given in (2.104). Using the equivalence of $\mathcal{N}^\dagger$ and $\mathcal{N}$ in the spin theory, up to purely gravitational invertible factors, we obtain the following fusion rules,

$$\begin{aligned} &L(\Sigma_2)^2 = 1 \ , \qquad L(\Sigma_2) \times \mathcal{N}(M_3) = (-1)^{Q(\Sigma_2)} \mathcal{N}(M_3) \ , \\ &\mathcal{N}(M_3) \times \mathcal{N}(M_3) = \frac{1}{|H^0(M_3,\mathbb{Z}_2)|} \sum_{\Sigma_2 \in H_2(M_3,\mathbb{Z}_2)} (-1)^{Q(\Sigma_2)} L(\Sigma_2) \ . \end{aligned} \tag{4.11}$$

Let us note, incidentally, that the right-hand side of the final equation is often referred to as a *condensation defect*,

$$\mathcal{C}(M_3) := \frac{1}{|H^0(M_3,\mathbb{Z}_2)|} \sum_{\Sigma_2 \in H_2(M_3,\mathbb{Z}_2)} (-1)^{Q(\Sigma_2)} L(\Sigma_2) \ . \tag{4.12}$$

Let us now compare these fusion rules to the Ising fusion rules in (3.34). We see that they are extremely similar, with two main differences. First, in two dimensions all topological defects appearing were lines, but in the current case the duality defect is codimension-1, while the generators of the one-form symmetry are codimension-2. In particular, the fusion of two codimension-1 surfaces gives rise to a sum of codimension-2 surfaces. Because of this interplay between objects of various dimensions, the mathematical structure describing these symmetries is no longer a category, but rather a *higher-category*, though we will not give details of this mathematical structure.[36] Second, there are factors of $(-1)^{Q(\Sigma_2)}$ appearing above which do not appear in the Ising fusion rules—indeed, in the Ising case, all coefficients were required to be non-negative integers. More generally, the coefficients for d-dimensional defects can be d-dimensional TQFTs [190, 191]. In the case of lines, these were 1-dimensional topological quantum mechanical theories, i.e. theories of free n-dimensional qunits, with the resulting coefficient being the non-negative integer n itself.

As in $(1+1)$-dimensions, the presence of a duality defect tells us that the theory has a self-duality under gauging, with the role of the algebra object described in Section 3.6 now

[36] See e.g. [189] for some details in the fusion 2-category case.

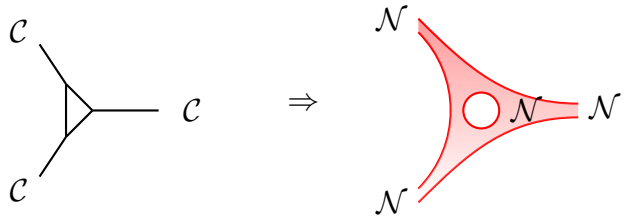


Figure 10: A theory with a Kramers-Wannier defect has a self-duality since the mesh of $\mathcal{C}$ can be replaced by a set of topologically trivial loops of $\mathcal{N}$.

being played by the condensation defect. This may be argued in a manner similar to Section 3.5. In particular, we begin by inserting a fine mesh of the condensate $\mathcal{C}(M_3)$ (which is itself a fine mesh of 2-surfaces), and then replacing it with pairs of $\mathcal{N}(M_3)$ as shown schematically in Figure 10. But assuming that we started with a fine enough mesh, each loop of $\mathcal{N}(M_3)$ is now contractible, and may be evaluated to a number. Thus with appropriate normalization we reobtain the original theory. Note that the factor of $(-1)^{Q(\Sigma_2)}$ in the expression for $\mathcal{C}(M_3)$ tells us that the gauging of $\mathbb{Z}_2^{(1)}$ must be performed with a certain discrete torsion in order to get a self-duality. In Section 4.3 we will see this in a bit more detail in a concrete example.

4.1.2 Half-space gauging

We may also generate non-invertible symmetries in $(3+1)$-dimensions by performing half-space gauging of non-anomalous symmetries. In particular, say that we gauge a $\mathbb{Z}_2^{(1)}$ one-form symmetry in half of the space, with Dirichlet boundary conditions at the interface,

$$\begin{array}{c|c} x<0 & x>0 \\ \mathcal{L}[B_2] & \mathcal{L}[b_2]+\pi b_2\cup B_2 \\ & \\ & b_2|_0=0 \end{array} \quad . \tag{4.13}$$

If the theories on the left and the right are the same, then this topological interface gives a topological defect in a single theory, and it is in fact non-invertible. The non-invertible nature can be verified by directly checking the fusion rules. Indeed, consider fusing the defect with its orientation reversal by first inserting $\mathcal{N}$ at $x=0$ and $\mathcal{N}^\dagger$ at $x=\epsilon$,

$$\begin{array}{c|c|c} x<0 & 0<x<\epsilon & x>\epsilon \\ \mathcal{L}[B_2] & \mathcal{L}[b_2]+\pi b_2\cup B_2 & \mathcal{L}[b_2]+\pi(b_2\cup\widehat{b}_2-\widehat{b}_2\cup B_2) \\ & & \\ & b_2|_0=0 & \widehat{b}_2|_\epsilon=0 \end{array} \tag{4.14}$$

and then taking the $\epsilon \to 0$ limit,

$x<0$	$x>0$
$\mathcal{L}[B_2]$	$\mathcal{L}[b_2] + \pi(b_2 \cup \widehat{b}_2 - \widehat{b}_2 \cup B_2)$
$b_2\|_0, \widehat{b}_2\|_0 = 0$	

. (4.15)

Note that away from the interface, we may integrate out $\widehat{b}_2$, which imposes $b_2 = B_2$. Thus away from the interface, we obtain $\mathcal{L}[B_2]$ everywhere. On the other hand, at the location of the interface, the Dirichlet boundary condition $\widehat{b}_2 = 0$ means that we no longer have $b_2 = B_2$. Instead, we are left with a copy of $\mathbb{Z}_2^{(1)}$ gauge theory in the tubular neighborhood of the interface at M_3, which gives precisely a condensate $\mathcal{C}(M_3)$ of the type in (4.12); see [190] for details.[37] The upshot is that we reobtain fusion rules

$$\mathcal{N}(M_3) \times \mathcal{N}(M_3)^\dagger = \mathcal{C}(M_3) \tag{4.16}$$

of the familiar type. We likewise have $L(\Sigma_2) \times \mathcal{N}(M_3) = \mathcal{N}(M_3)$, as follows immediately from the Dirichlet boundary conditions at the location of the interface. Hence this is again a defect of duality type.

We should note that the two constructions of non-invertible symmetries discussed above, i.e. via gauging in a theory with an anomaly or via half-space gauging, may give the same defects in some cases, but are not in general interchangeable. For example, there exist non-invertible defects which can be constructed via half-space gauging, but which do not arise from any theory with an anomaly.[38] There could also be non-invertible defects which cannot be constructed via either of these pictures.

4.2 Maxwell theory

We will now switch gears and discuss non-invertible symmetries in $(3+1)$d in a variety of concrete models. Our first example will be Maxwell theory [187], which we had discussed before in Sections 2.5 and 2.7. Unlike before, in the current case we will also allow for a theta angle,

$$\mathcal{L} = \frac{1}{2e^2} f \wedge *f + \frac{\theta}{8\pi^2} f \wedge f \ . \tag{4.17}$$

The theta term is a total derivative and hence at the classical level seems to do nothing, but at the quantum level it has important effects. The resulting theory is labelled by two parameters

[37] In the current case, the condensate does not contain $(-1)^{Q(\Sigma_2)}$ though. This is because the current gauging does not involve discrete torsion.

[38] Such non-invertible symmetries are sometimes referred to as *intrinsically non-invertible* [192].

e and θ, which we group into a single complex parameter

$$\tau = \frac{\theta}{2\pi} + \frac{2\pi i}{e^2} \,. \tag{4.18}$$

Let us denote the theory labelled by the complex coupling τ by $\mathcal{X}[\tau]$. An important property of Maxwell theory (on spin manifolds) is that it has a Montonen-Olive duality symmetry $\mathrm{SL}(2,\mathbb{Z})$ [193], which says (suppressing the standard gravitational factors) that[39]

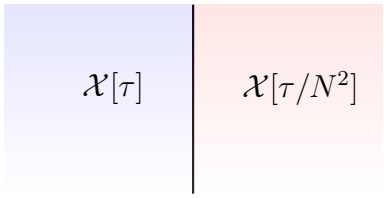
$$\mathcal{X}[\tau] \cong \mathcal{X}\left[\frac{a\tau+b}{c\tau+d}\right] \,, \qquad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{Z}) \,. \tag{4.19}$$

For example, taking $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, we find that $\mathcal{X}[\tau] \cong \mathcal{X}\left[-1/\tau\right]$. This is an extremely non-trivial identity, which relates the theory at strong coupling to the theory at weak coupling.

The presence of the theta angle modifies the currents in (2.46) to

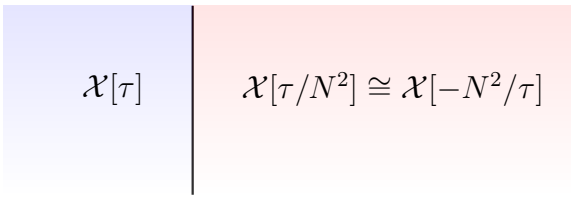
$$j^E = \frac{1}{e^2} f - \frac{\theta}{4\pi^2} * f \,, \qquad j^M = \frac{1}{2\pi} * f \,, \tag{4.20}$$

but as before they are both conserved, and they correspond to two one-form symmetries $U(1)^{(1)}_E$ and $U(1)^{(1)}_M$. In order to construct non-invertible defects, we now consider half-space gauging subgroups of these two symmetry groups. For example, consider gauging a $\mathbb{Z}^{(1)}_N \subset U(1)^{(1)}_E$ with $N > 1$ in half-space with Dirichlet boundary conditions on the boundary. Gauging $\mathbb{Z}^{(1)}_N$ amounts to shifting $a \to \frac{1}{N} a$,[40] and hence maps $\mathcal{X}[\tau]$ to $\mathcal{X}[\tau/N^2]$. We thus get an interface,



. (4.21)

In general, the theories on the left and right of the interface are distinct. However, recall that we have the duality $\mathcal{X}[\tau] \cong \mathcal{X}[-1/\tau]$, so this interface can alternatively be written as an interface between $\mathcal{X}[\tau]$ and $\mathcal{X}[-N^2/\tau]$,



. (4.22)

In particular, when $\tau = iN$ we see that the theories on the left and the right are the same, and hence we get a defect in a single theory. By the same arguments as in Section 4.1, this defect is topological, and gives a duality defect $\mathcal{N}$.

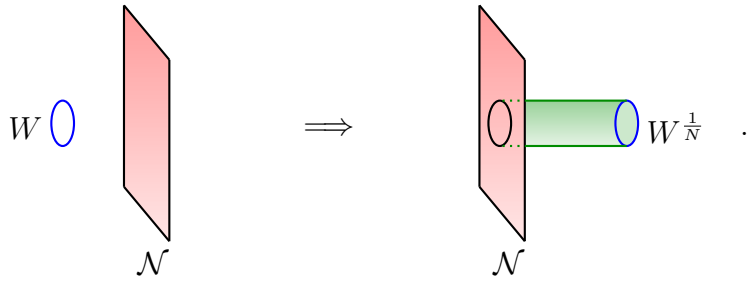
[39] Recall that $\mathrm{SL}(2,\mathbb{Z})$ is the group of 2×2 integer matrices with unit determinant.

[40] This follows since gauging $\mathbb{Z}^{(1)}_N$ keeps only Wilson lines of charge $q \in N\mathbb{Z}$. Thus the minimal charge in the gauged theory is N times the minimal charge of the original theory, and this can alternatively be understood as rescaling $a \to \frac{1}{N} a$.

In the case of the Ising model, moving the local operator σ past the non-invertible defect $\mathcal{N}$ gave rise to a non-local operator μ, as shown in (3.60)—this was one of the hallmarks of non-invertibility. In the current case, a similar phenomenon occurs when we try to move a Wilson line through $\mathcal{N}(M_3)$.[41] In a common electric frame, the outgoing line is instead a fractional 't Hooft line. Indeed, since the gauging step maps $a \to \frac{1}{N}a$, the charge-1 Wilson line becomes a charge $\frac{1}{N}$ Wilson line upon passing it, but such a Wilson line is not well-defined as a line—it must be thought of as attached to a surface,[42]

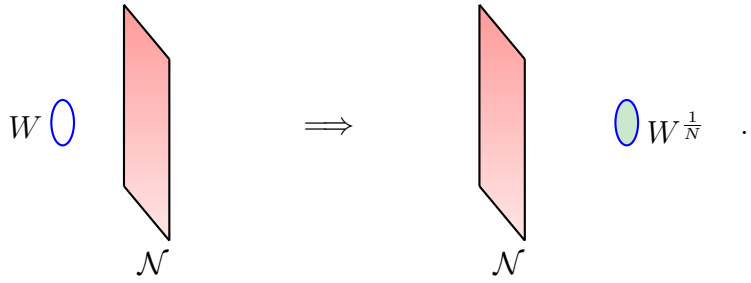
$$e^{i\oint a} \to \text{“ } e^{\frac{i}{N}\oint a} \text{ ”} = e^{\frac{i}{N}\int f} \,, \tag{4.23}$$

or pictorially,



(4.24)

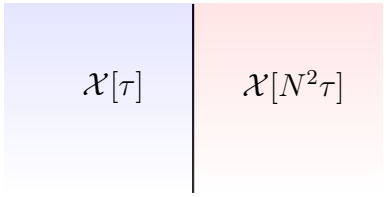
If the cycle on which W was defined is contractible, we can further deform the right-hand side to obtain a droplet,



(4.25)

Either way, we see that the Wilson line is mapped to a non-genuine line operator, i.e. a line attached to a surface.

Thus far we have discussed the gauging of $\mathbb{Z}_N^{(1)} \subset U(1)_E^{(1)}$. But it is also possible to gauge a subset $\mathbb{Z}_N^{(1)} \subset U(1)_M^{(1)}$ of the magnetic one-form symmetry. In this case one can show that $a \to Na$, so that $\mathcal{X}[\tau]$ is mapped to $\mathcal{X}[N^2\tau]$,



(4.26)

[41] In the following formulas and diagrams, the right-hand operators are written in the gauged electric frame, before the S-duality identification.

[42] It is not well-defined since it is not gauge invariant.

Using Montonen-Olive duality to rewrite $\mathcal{X}[N^2\tau] \cong \mathcal{X}[-\frac{1}{N^2\tau}]$, we see that for $\tau = i/N$ we obtain a new defect, which is topological and non-invertible by the usual arguments. Hence at both $\tau = iN$ and $\tau = i/N$, we have non-invertible duality defects, which are constructed by gauging $\mathbb{Z}_N^{(1)}$ subgroups of the electric and magnetic one-form symmetries, respectively.

Even more generally, we could consider simultaneously gauging $\mathbb{Z}_p^{(1)} \times \mathbb{Z}_q^{(1)} \subset U(1)_E^{(1)} \times U(1)_M^{(1)}$. However, in this case we must be careful about the values p and q. As discussed in Section 2.7, the electric and magnetic one-form symmetries have a mixed anomaly and hence cannot be gauged simultaneously, but it is still possible for there to be non-anomalous subgroups. To analyze this, let us restrict B_E and B_M in (2.79) to be $\mathbb{Z}_p^{(1)}$ and $\mathbb{Z}_q^{(1)}$ gauge fields by including appropriate BF terms, in which case the anomaly inflow action becomes

$$\widehat{\mathcal{A}}[B_E, B_M] = \int_{M_5} \frac{1}{2\pi} B_E \wedge dB_M + \frac{p}{2\pi} B_E \wedge da_E + \frac{q}{2\pi} B_M \wedge da_M \ . \tag{4.27}$$

Here a_E and a_M are two-form gauge fields on M_5. Integrating out a_E and a_M allows us to replace $B_E \to \frac{2\pi}{p} B_E$, $B_M \to \frac{2\pi}{q} B_M$, where B_E (resp. B_M) now has periods in $\mathbb{Z}_p$ (resp. $\mathbb{Z}_q$),

$$\widehat{\mathcal{A}}[B_E, B_M] = \frac{2\pi}{pq} \int_{M_5} B_E \cup \delta B_M \ . \tag{4.28}$$

Since B_M can be shifted by q times any cochain, we may replace δB_M by $(1 - qy)\delta B_M$ for any integer y. We now make use of Bézout's identity, which claims that there exist integers x, y such that $px + qy = 1$ if and only if p and q are coprime.[43] What this means is that when $\gcd(p, q) = 1$, we may rewrite the inflow action as

$$\widehat{\mathcal{A}}[B_E, B_M] = \frac{2\pi}{pq} \int_{M_5} (1 - qy) B_E \cup \delta B_M = \frac{2\pi x}{q} \int_{M_5} B_E \cup \delta B_M \ . \tag{4.29}$$

Imposing that B_M is flat mod q, we then conclude that $\widehat{\mathcal{A}}[B_E, B_M] \in 2\pi\mathbb{Z}$, and hence that the anomaly is trivialized.

The conclusion is thus that we are allowed to gauge $\mathbb{Z}_p^{(1)} \times \mathbb{Z}_q^{(1)} \subset U(1)_E^{(1)} \times U(1)_M^{(1)}$ as long as $\gcd(p, q) = 1$. Doing so sends $\tau \to (q/p)^2 \tau$, and coupling this with Montonen-Olive duality, we conclude that there is a non-invertible duality symmetry at every point $\tau = ip/q$ with positive coprime p, q and $pq > 1$. For $p = q = 1$, this construction gives the ordinary invertible S-duality defect.

[43] More generally, Bézout's identity says that for integers p and q with greatest common divisor $d = \gcd(p, q)$, there exist integers x and y such that $px + qy = d$.

Exercise: Consider the $(1+1)$-dimensional theory of a single compact boson at radius R. This theory has two $U(1)$ zero-form symmetries, namely the momentum and winding symmetries $U(1)_{\text{mom}}$ and $U(1)_{\text{wind}}$. Argue that gauging a $\mathbb{Z}_p \times \mathbb{Z}_q$ subgroup of $U(1)_{\text{mom}} \times U(1)_{\text{wind}}$ leads to a change in the radius by $R \to \frac{q}{p}R$. (Note that we must take $\gcd(p,q) = 1$ in order to avoid a mixed anomaly). Then use this to argue that there are non-invertible defects at every radius $R^2 \in \mathbb{Q}$ with $R^2 > 0$ and $R^2 \neq 1$ (*Hint: follow the same steps as for Maxwell theory, replacing S-duality with T-duality*). At $R^2 = 1$, this construction reduces to ordinary invertible T-duality.

4.3 Massless QED

Let us next analyze the example of massless QED [194, 195], i.e. Maxwell theory coupled to a unit charge Dirac fermion,

$$\mathcal{L} = \frac{1}{4e^2} f_{\mu\nu} f^{\mu\nu} + i\overline{\Psi}(\partial_\mu - i a_\mu)\gamma^\mu \Psi \ . \tag{4.30}$$

We work on spin manifolds, without dynamical magnetic monopoles, and suppress gravitational background terms. In the above action we have not included a θ angle, but this does not lead to a loss of generality—indeed, in massless QED the theta angle is not physical. The reason for this is the ABJ anomaly discussed in Section 2.7. In particular, doing an axial rotation of Ψ of the form $\Psi \to e^{i\alpha\gamma_5/2}\Psi$ produces a theta angle α, and conversely any theta angle can be removed by simply doing a field redefinition of Ψ.

In the presence of the fermion Ψ, the current j^E defined in (2.46) is no longer conserved, since the conservation of j^E followed from the equation of motion $d * f = 0$, which now picks up a source term.[44] As a result, the defects that we obtained by gauging subgroups of $U(1)^{(1)}_E$ are no longer present. On the other hand, the $U(1)^{(1)}_M$ symmetry is not affected by the charged fermion, and hence we can still consider half-space gauging of $\mathbb{Z}^{(1)}_N \subset U(1)^{(1)}_M$. However, since we can no longer use Montonen-Olive duality in QED (electromagnetic duality is broken by the presence of the electrically charged fermion), in order to obtain a non-invertible defect we will have to perform our gauging in a somewhat clever way.

Our starting point is the action in (2.75) without the background field B_E, i.e.

$$S[B_M] \ = \ \int \frac{1}{2e^2} f \wedge * f + \frac{1}{2\pi} f \wedge B_M \ . \tag{4.31}$$

Restricting B_M to a $\mathbb{Z}^{(1)}_N$ background, we are free to add local counterterms in B_M, and we choose to do this as follows,

$$S[B_M] \ = \ \int \frac{1}{2e^2} f \wedge * f + \frac{1}{2\pi} f \wedge B_M + \frac{Nk}{4\pi} B_M \wedge B_M \ , \tag{4.32}$$

[44] If the fermion had non-unit charge $q = N$, we would not break $U(1)^{(1)}_E$ completely, but rather to a $\mathbb{Z}^{(1)}_N$ subgroup.

where we assume that $\gcd(k,N) = 1$. We now gauge $\mathbb{Z}_N^{(1)} \subset U(1)_M^{(1)}$, which is done by including a BF term and making B_M dynamical, i.e. $B_M \to b_M$,

$$S[b_M] \;=\; \int \frac{1}{2e^2} f \wedge *f + \frac{1}{2\pi} f \wedge b_M + \frac{N}{2\pi} b_M \wedge da_M + \frac{Nk}{4\pi} b_M \wedge b_M \,. \tag{4.33}$$

Including local counterterms before gauging is known as including discrete torsion; it is sometimes also referred to as *stacking with an SPT* before gauging.

What is the effect of such a gauging? Note that because $\gcd(k,N) = 1$, there exists an integer p such that $pk = 1 \bmod N$. We may then rewrite the action $S[b_M]$ as follows,

$$S[b_M] \;=\; \int \frac{1}{2e^2} f \wedge *f + \frac{1}{8\pi^2} \left(-\frac{2\pi p}{N}\right) f \wedge f + \frac{Nk}{4\pi} \left(b_M + \frac{p}{N} f\right)^2 \,, \tag{4.34}$$

where we have dropped the BF term and terms that evaluate to $2\pi\mathbb{Z}$. Integrating out b_M then gives, up to an overall gravitational counterterm, the following,

$$S[b_M] \;=\; \int \frac{1}{2e^2} f \wedge *f + \frac{1}{8\pi^2} \left(-\frac{2\pi p}{N}\right) f \wedge f \,. \tag{4.35}$$

Hence we see that the gauging of $\mathbb{Z}_N^{(1)} \subset U(1)_M^{(1)}$ with discrete torsion labelled by k shifts $\theta \to \theta - \frac{2\pi p}{N}$, where p is the mod N inverse of k. On the other hand, this gauging does not affect the coupling e at all.

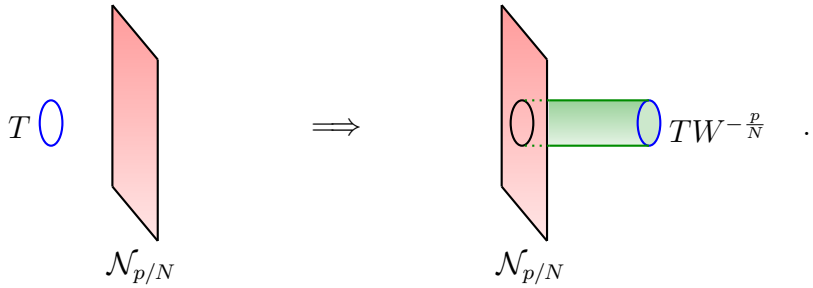
The importance of this is that, as we discussed above, the θ angle in QED can be removed by a redefinition of the fermion Ψ. As such, the theory obtained by half-space gauging of $\mathbb{Z}_N^{(1)} \subset U(1)_M^{(1)}$ with discrete torsion can be brought back to the original theory by chiral rotation,

$$\mathcal{X}[\tau] \;\Big|\; \mathcal{X}[\tau - \tfrac{p}{N}] \cong \mathcal{X}[\tau] \,, \tag{4.36}$$

and hence half-space gauging gives rise to a topological defect $\mathcal{N}_{p/N}$ in a single theory.

How does this defect act on operators of the theory? On the elementary field Ψ, it acts as a standard axial rotation with angle $\alpha = \frac{2\pi p}{N}$. In this sense, we see that the axial symmetry survives as a non-anomalous symmetry for angles $\alpha \in 2\pi\mathbb{Q}$, at the cost of becoming non-invertible. The non-invertible nature can be seen by analyzing the action of the symmetry on extended operators of the theory—in particular, when acting on 't Hooft lines it gives them fractional electric charge via the Witten effect, and hence 't Hooft lines are mapped to non-

genuine lines attached to surfaces,



(4.37)

Note that the existence of these defects is independent of the coupling e—i.e. at any coupling, we have an infinite number of non-invertible defects labelled by $\mathbb{Q}$.[45]

One important implication of this defect is that massless QED is *natural*, i.e. we do not expect $m\overline{\Psi}\Psi$ to be produced radiatively. Indeed, if we add a mass $m\overline{\Psi}\Psi$, then we explicitly break the non-invertible symmetry.[46] Another application of the non-invertible symmetry is that it can be used to prove helicity conservation in electron-positron scattering—for more on this, see [195].

Before moving on to more realistic models, let us discuss a slightly more explicit construction of the above non-invertible defect. As we have already said, the defect involves an axial rotation by $\frac{2\pi p}{N}$, so we might imagine that it is constructed out of the axial current $\mathcal{N}_{\frac{p}{N}} \overset{?}{\sim} e^{\frac{2\pi i p}{N}\frac{1}{2}\int *j_A}$.[47] But because of the ABJ anomaly,

$$d * j_A = \frac{1}{4\pi^2} f \wedge f \ , \tag{4.38}$$

we see that j_A is not conserved, and hence that $\mathcal{N}$ defined in this manner would not be topological. On the other hand, moving the right-hand side to the left, we see that

$$d\left(*j_A - \frac{1}{4\pi^2} a \wedge da\right) = 0 \ , \tag{4.39}$$

and hence the operator

$$\mathcal{N}_{\frac{p}{N}}(M_3) \overset{?}{\sim} e^{\frac{2\pi i p}{N}\frac{1}{2}\int_{M_3}\left(*j_A - \frac{1}{4\pi^2} a\wedge da\right)} \tag{4.40}$$

is topological. In flat space, this operator is also gauge-invariant, and hence it is a well-defined generator for the invertible axial symmetry. But for more general spacetimes (or in the presence of magnetic monopoles), it is not gauge-invariant, due to the fractional Chern-Simons level $k = \frac{p}{N}$.

[45] Contrast this with the case of Maxwell theory, where in each theory labelled by $\tau \in i\mathbb{Q}$ with $\mathrm{Im}\,\tau > 0$ and $\tau \neq i$, we had constructed one non-invertible defect.

[46] Previous arguments for this relied on the invertible $U(1)_A$, which is itself anomalous, and no longer a symmetry in topologically non-trivial spacetimes.

[47] The extra factor of $\frac{1}{2}$ follows from the comments in footnote 12.

However, there is a well-known way to make this term gauge-invariant. In particular, for $p = 1$, we replace it with a *fractional quantum Hall system*,

$$\frac{N}{4\pi} c dc + \frac{1}{2\pi} c da \ , \tag{4.41}$$

where c and a are properly quantized gauge fields, which in particular means that $\oint dc, \oint da \in 2\pi\mathbb{Z}$. Integrating out c would naively give $c \stackrel{?}{=} -\frac{a}{N}$, and plugging this back into (4.41) would then reproduce $-\frac{1/N}{4\pi} a da$. But in reality the equation $c \stackrel{?}{=} -\frac{a}{N}$ is not quite correct, since it would imply that c is not properly quantized. The correct, gauge invariant expression is the one in (4.41).

Thus, at the cost of introducing a new degree of freedom c on M_3, we are able to construct a topological, gauge-invariant defect,

$$\mathcal{N}_{\frac{1}{N}}(M_3) := \int \mathcal{D}c \ e^{i \int_{M_3} \left(\frac{\pi}{N} * j_A + \frac{N}{4\pi} c dc + \frac{1}{2\pi} c da \right)} \ . \tag{4.42}$$

For $N > 1$, the new degree of freedom c is the origin of the non-invertible nature of $\mathcal{N}_{\frac{1}{N}}(M_3)$.

To see that this is non-invertible, we may compute the fusion rules of this defect with its orientation reversal, following the steps in Section 4.1.1. Starting with,

$$\mathcal{N}_{\frac{1}{N}}(M_3) \times \mathcal{N}_{\frac{1}{N}}(M_3)^\dagger = \int \mathcal{D}c \mathcal{D}\bar{c} \ e^{i \int_{M_3} \frac{N}{4\pi} c dc - \frac{N}{4\pi} \bar{c} d\bar{c} + \frac{1}{2\pi} (c - \bar{c}) da} \tag{4.43}$$

we change variables via $\bar{c} \to c - \widehat{c}$, integrate out c, switch to homology notation, and note that the generators of the magnetic one-form symmetry take the form $L(\Sigma_2) = e^{\frac{i}{N} \oint_{\Sigma_2} da}$ to obtain for even N

$$\mathcal{N}_{\frac{1}{N}}(M_3) \times \mathcal{N}_{\frac{1}{N}}(M_3)^\dagger = \frac{1}{|H^0(M_3, \mathbb{Z}_N)|} \sum_{\Sigma_2 \in H_2(M_3, \mathbb{Z}_N)} e^{\frac{2\pi i}{N} Q(\Sigma_2)} L(\Sigma_2) \ , \tag{4.44}$$

For even N, $Q(\Sigma_2)$ is defined by the integer-lift expression in (4.5), now using ω_{Σ_2} as an integer lift of the mod-N Poincaré dual. For odd N, the spin-theory condensate is untwisted and the exponential phase in this equation is omitted. The even-N phase comes from the discrete torsion in gauging $\mathbb{Z}_N^{(1)} \subset U(1)_M^{(1)}$.

For $p = 1$, we used a standard fractional quantum Hall system to define the defect $\mathcal{N}_{1/N}$. A similar procedure may be carried out for more general p such that $\gcd(p, N) = 1$. In that case, we couple to the so-called *minimal* $\mathbb{Z}_N^{(1)}$ *TQFT* $\mathcal{A}^{N,p}$, defined in [188].

4.4 Massless QCD below the electroweak scale

We now make our example slightly more realistic, and consider massless QCD below the electroweak scale, where $SU(2) \times U(1)_Y$ is broken to $U(1)_{EM}$ [194, 195].[48] The IR chiral

[48] We take the direct-product color and electromagnetic gauge group; in electron-charge units, $a/3$ is the properly quantized $U(1)$ connection. Thus the defects below need not be minimal.

Lagrangian takes the following form,

$$\mathcal{L}_{IR} = \frac{1}{2} d\pi^0 \wedge *d\pi^0 + g\,\pi^0 f \wedge f + \dots \,, \tag{4.45}$$

where we have focused on the part that involves the neutral pion. What we will now show is that, using a non-invertible symmetry, we may predict the value of the $\pi^0 \to \gamma\gamma$ coupling g, and in particular prove that it must be non-zero.

To see this, note that for massless quarks, the QCD Lagrangian is known to have the following symmetry,

$$U(1)_{A_3}: \qquad \begin{pmatrix} u \\ d \end{pmatrix} \to e^{i\alpha\gamma_5\sigma_3/2} \begin{pmatrix} u \\ d \end{pmatrix} , \tag{4.46}$$

whose corresponding classically conserved current is

$$j_\mu^{A_3} = \frac{1}{2}\bar{u}\gamma_5\gamma_\mu u - \frac{1}{2}\bar{d}\gamma_5\gamma_\mu d \,. \tag{4.47}$$

In fact, this symmetry has an ABJ anomaly of exactly the same type as before, and hence we may use precisely the same construction to obtain a non-invertible symmetry for every element of $U(1)_{A_3}$ labelled by $\alpha = 2\pi p/N$. We now take $p = 1$ and consider an insertion of this non-invertible defect at $x = 0$, giving

$$\begin{aligned} &\int_{x<0} \left(\frac{1}{2} d\pi^0 \wedge *d\pi^0 + g\pi^0 f \wedge f \right) + \int_{x>0} \left(\frac{1}{2} d\pi^0 \wedge *d\pi^0 + g\pi^0 f \wedge f \right) \\ &\qquad + \int_{x=0} \left(\frac{2\pi}{N} * j^{A_3} + \frac{N}{4\pi} c dc + \frac{1}{2\pi} c da \right) . \end{aligned} \tag{4.48}$$

Note that in the chiral Lagrangian, the axial current becomes simply

$$j^{A_3} = -f_\pi d\pi^0 + \dots \,, \tag{4.49}$$

where $f_\pi \sim 92.4$ MeV is the pion decay constant, and corresponds to the symmetry shifting the neutral pion by $\pi^0 \to \pi^0 - \alpha f_\pi$. We may then compute the following equations of motion,

$$\begin{aligned} \pi^0: &\qquad \pi^0|_{x=0^+} - \pi^0|_{x=0^-} = -\frac{2\pi}{N} f_\pi \,, \\ c: &\qquad N dc + f = 0 \,, \\ a: &\qquad 2g \left(\pi^0|_{x=0^+} - \pi^0|_{x=0^-} \right) f = \frac{1}{2\pi} dc \,. \end{aligned} \tag{4.50}$$

Plugging the first equation into the third gives

$$-\frac{4\pi g f_\pi}{N} f = \frac{1}{2\pi} dc \,, \tag{4.51}$$

and then further using the second equation gives

$$4\pi g f_\pi = \frac{1}{2\pi} \qquad \Rightarrow \qquad g = \frac{1}{8\pi^2 f_\pi} \,. \tag{4.52}$$

Thus, as promised, the presence of the non-invertible symmetry $\mathcal{N}_{\frac{1}{N}}$ in QCD tells us the coupling of the $\pi^0 f \wedge f$ term in the chiral Lagrangian, and hence the leading chiral prediction for the rate of $\pi^0 \to \gamma\gamma$, once small quark masses are restored.

4.5 Axion models

Let us now move on to some beyond the Standard Model (BSM) models. In particular, we consider axion-Maxwell theory [196–198],

$$\mathcal{L} = \frac{v^2}{2} d\phi \wedge *d\phi + \frac{1}{2e^2} f \wedge *f - \frac{k}{8\pi^2} \phi f \wedge f \ , \tag{4.53}$$

where $\phi \sim \phi + 2\pi$ is the axion field. We take $k \in \mathbb{Z}$ and work on spin manifolds. The theory of a compact scalar was studied previously in Section 2.6, where it was shown that it has a 0-form and 2-form symmetry. In total then, we can try to define the following four currents

$$j^{(1)}_{\text{shift}} = v^2 d\phi \ , \quad j^{(3)}_{\text{winding}} = \frac{1}{2\pi} * d\phi \ , \quad j^{(2)}_E = \frac{1}{e^2} f \ , \quad j^{(2)}_M = \frac{1}{2\pi} * f \ , \tag{4.54}$$

which satisfy the following

$$d * j^{(1)}_{\text{shift}} = -\frac{k}{8\pi^2} f \wedge f \ , \quad d * j^{(3)}_{\text{winding}} = 0 \ , \quad d * j^{(2)}_E = \frac{k}{4\pi^2} d\phi \wedge f \ , \quad d * j^{(2)}_M = 0 \tag{4.55}$$

using the equations of motion and the Bianchi identities. When $k = 0$, we see that there are four conserved currents, and hence four symmetries,

$$U(1)^{(0)}_{\text{shift}} \ , \quad U(1)^{(2)}_{\text{wind.}} \ , \quad U(1)^{(1)}_E \ , \quad U(1)^{(1)}_M \ , \tag{4.56}$$

which are the symmetries of the decoupled axion and Maxwell theories, but for non-zero k the shift symmetry and electric one-form symmetries $U(1)^{(0)}_{\text{shift}}$ and $U(1)^{(1)}_E$ are broken.

For the shift symmetry $U(1)^{(0)}_{\text{shift}}$, we see that the symmetry is broken by an ABJ-like term of the same form as the $U(1)_A$ symmetry in QED. As such, we can resurrect a $\mathbb{Q}$ subset as non-invertible symmetries by stacking the naive defect constructed out of $*j^{(1)}_{\text{shift}}$ with an appropriate fractional quantum Hall system. Alternatively, we can gauge a subgroup of $U(1)^{(1)}_M$ in half-space. Either way, we get a non-invertible zero-form symmetry whose defect takes the form analogous to (4.42), with the opposite Chern-Simons sign for $k > 0$.

Next we consider $U(1)^{(1)}_E$. As in the previous case, non-conservation of $*j^{(2)}_E$ means that $\mathcal{N}^{(1)}_\alpha(M_2) \overset{?}{\sim} e^{i\alpha \oint_{M_2} *j^{(2)}_E}$ is no longer topological. We may attempt to obtain a topological defect by instead using the modified current,

$$\mathcal{N}^{(1)}_\alpha(M_2) \overset{?}{\sim} e^{i\alpha \oint_{M_2} \left(*j^{(2)}_E - \frac{k}{4\pi^2} \phi \wedge f \right)} \ , \tag{4.57}$$

but this term does not, in general, respect the identification $\phi \sim \phi + 2\pi$. The solution, as before, is to couple the defect to new degrees of freedom which, when naively integrated out, reproduce the above non-gauge-invariant expression. The final result is that, for $\alpha = \frac{2\pi p}{N}$ with p and N coprime, we may write

$$\mathcal{N}^{(1)}_{p/N}(M_2) = \int \mathcal{D}\lambda \mathcal{D}c \, e^{i \oint_{M_2} \left(\frac{2\pi p}{Ne^2} * f + \frac{N}{2\pi} \lambda dc + \frac{pk}{2\pi} \phi dc + \frac{1}{2\pi} \lambda f \right)} \ . \tag{4.58}$$

Here $\lambda \sim \lambda + 2\pi$ is a scalar and c is a $U(1)$ gauge field on M_2.

Exercise: Check that naively integrating out λ and c reproduces the ill-defined defect in (4.57).

This is our first example of a non-invertible one-form symmetry in $(3+1)$d,[49] combining two generalizations of the notion of symmetry. Note that for $k = 1$, this defect may also be obtained by *higher half-space gauging* of a $\mathbb{Z}_N^{(1)} \times \mathbb{Z}_N^{(2)} \subset U(1)_M^{(1)} \times U(1)_{\text{wind}}^{(2)}$ on a codimension-1 slice of spacetime, as described in [196].

We will now use this non-invertible symmetry to make some non-trivial physical claims. As we have just said, the non-invertible symmetry $\mathcal{N}_{p/N}^{(1)}(M_2)$ is constructed by gauging a subgroup of $U(1)_M^{(1)} \times U(1)_{\text{wind}}^{(2)}$, and as such it forms an algebra with these symmetries. Thus if we break the $U(1)_M^{(1)} \times U(1)_{\text{wind}}^{(2)}$ symmetry, we must also break the full non-invertible one-form symmetry. Likewise, breaking $U(1)_M^{(1)}$ breaks the full non-invertible zero-form symmetry.

Now say that we try to UV complete axion-Maxwell theory, and upon doing so break some of the symmetries. If we call the scales of symmetry breaking $E_{\text{shift}}, E_{\text{wind}}, E_E$, and E_M (i.e. these are the scales below which the symmetries are emergent), then from the condensation structure explained above, we must have

$$E_{\text{shift}} \lesssim E_M \ , \qquad E_E \lesssim \min\{E_{\text{wind}}, E_M\} \ . \tag{4.59}$$

In particular, one typically expects that $E_E \approx m_E$ and $E_M \approx m_M$ with $m_{E,M}$ the masses of the lightest electrically/magnetically charged particles and $E_{\text{wind}} \lesssim \sqrt{T}$ with T the tension of a dynamical axion string, from which we conclude that

$$m_E \lesssim m_M \ , \qquad m_E \lesssim \sqrt{T} \ . \tag{4.60}$$

Under these assumptions, the lightest electrically charged excitation cannot be parametrically heavier than the lightest monopole.

For a semiclassical illustration, consider the $SU(2)$ Georgi-Glashow model coupled to an axion. In this model, we have gauge bosons g_μ^a, an adjoint Higgs field H, and an axion ϕ coupled to the $SU(2)$ instanton density, with the Higgs being used to break down to the axion-Maxwell theory described above. In this case we have

$$m_E \sim m_W \sim gv \ , \tag{4.61}$$

where $v^2 = \langle \text{Tr} H^2 \rangle$ and g is the $SU(2)$ coupling at the scale v. On the other hand, since the winding symmetry is preserved in this model, we have $E_{\text{wind}} = \infty$. Finally, $m_M \sim \frac{v}{g}$. Thus the above inequalities imply that

$$gv \lesssim \frac{v}{g} \quad \Rightarrow \quad g \lesssim 1 \ , \tag{4.62}$$

[49] In $(2+1)$-dimensions, non-Abelian anyons give a wealth of examples.

i.e. the semiclassical Higgs description is consistent with this hierarchy in the weak-coupling regime. This is sensible: when the theory is strongly coupled at the Higgsing scale, it is believed to confine instead of flowing to axion-Maxwell [67].

4.6 Neutrino mass models

One sharp signal of BSM physics is non-zero neutrino masses,

$$|m_2^2 - m_1^2| \sim (10\,\mathrm{meV})^2\,, \qquad |m_3^2 - m_2^2| \sim (50\,\mathrm{meV})^2\,, \qquad \sum_\nu m_\nu \leq 150\,\mathrm{meV}\,. \tag{4.63}$$

The bound on the sum is cosmology dependent. There are two standard ways to account for these masses,

- **Majorana mass:** In this case we use an effective dimension-5 operator (the *Weinberg operator*),

$$\mathcal{L} \supset \frac{1}{\Lambda} Y^N_{ij} (HL_i)(HL_j) \qquad \overset{\langle H\rangle \sim v}{\longrightarrow} \qquad \frac{v^2}{\Lambda} Y^N_{ij} \nu_i \nu_j\,. \tag{4.64}$$

- **Dirac mass:** In this case we add a right-handed neutrino (whose left-handed conjugate we denote by N) and use the renormalizable interaction,

$$\mathcal{L} \supset Y^N_{ij} H L_i N_j \qquad \overset{\langle H\rangle \sim v}{\longrightarrow} \qquad v\, Y^N_{ij} \nu_i N_j\,. \tag{4.65}$$

We will focus mainly on the latter case here. A troublesome feature is the tiny Yukawa couplings needed to reproduce experiment, i.e. $Y_\nu / Y_\tau \lesssim 10^{-11}$. In the current section, we briefly explain how non-invertible symmetries can be used to make this result natural [199].

The key fact is that, in order to break a higher-form symmetry in an EFT, we cannot just modify the action by adding higher-derivative terms in the original fields, since local operators are not charged under the higher-form symmetry. Instead, the breaking of higher-form symmetries must involve the introduction of new degrees of freedom.

Now say that we have a $U(1)^{(1)}_M$ symmetry in our theory. Breaking this symmetry requires finite action configurations carrying magnetic charge. Let us assume that this happens via monopoles. The mass and cutoff in this case are given by

$$m_M \sim \frac{v}{g}\,, \qquad \Lambda \sim g v\,, \tag{4.66}$$

with v some e.g. Higgsing scale that determines the monopole mass. The quantum vacuum contains such monopoles which propagate for proper time $\delta t \sim 1/\Lambda$, and hence give rise to terms in the quantum effective action of the scale

$$\delta\mathcal{L} \sim e^{-S_{\mathrm{mon}}} \sim e^{-m_M \delta t} \sim e^{-\#/g^2}\,. \tag{4.67}$$

	$SU(2)_L$	$U(1)_Y$	$U(1)_{\mu-\tau}$	$U(1)_{\widetilde{L}}$
L_e	2	-1	0	1
L_μ	2	-1	1	1
L_τ	2	-1	-1	1
$\bar{e}_e$	1	2	0	-1
$\bar{e}_\mu$	1	2	-1	-1
$\bar{e}_\tau$	1	2	$+1$	-1
N_e	1	0	0	0
N_μ	1	0	-1	0
N_τ	1	0	$+1$	0
H	2	$+1$	0	0

Table 2: Charges of lepton and Higgs sectors under the gauge groups in this model, as well as the global $U(1)_{\widetilde{L}}$ symmetry corresponding to Standard Model lepton number, which is not conserved quantum mechanically.

We thus see that there is a natural way to violate magnetic one-form symmetries non-perturbatively, i.e. by exponentially small amounts.

In situations in which there is a non-invertible zero-form symmetry coming from half-space gauging of the magnetic one-form symmetry, there is consequently also a natural way to break the non-invertible zero-form symmetry by exponentially small amounts, since breaking the one-form symmetry also breaks the non-invertible symmetry. The goal is then to build models in which the neutrino masses explicitly break such a non-invertible symmetry. This would mean that having $m_\nu = 0$ would be natural as long as the non-invertible symmetry is preserved, but if the non-invertible symmetry is violated by $e^{-\#/g^2}$, as we have described above, then we would expect $m_\nu \sim e^{-\#/g^2}$ as well. This gives a mechanism for having exponentially small neutrino masses.

We now describe one particular model with this behavior, in the context of the Dirac mass case; other models can be found in [199] and follow-up works. We begin by coupling the Standard Model to right-handed neutrinos and gauging $U(1)_{\mu-\tau}$. The charges of the lepton and Higgs sectors under the gauge groups (ignoring color) are shown in Table 2. Note that we have also included the charge under the global symmetry $U(1)_{\widetilde{L}}$, corresponding to Standard Model lepton number.

In fact, it is known that in the Standard Model, an anomaly with $SU(2)_L$ breaks $U(1)_{\widetilde{L}}$ to a non-anomalous $\mathbb{Z}_3^{\widetilde{L}}$ subgroup. This subgroup prevents generation of the Dirac mass term HLN, and hence the presence of $\mathbb{Z}_3^{\widetilde{L}}$ ensures that having massless neutrinos is natural.

However, in the current model we are also gauging $U(1)_{\mu-\tau}$. It turns out that $\mathbb{Z}_3^{\widetilde{L}}$ has yet another ABJ anomaly with $U(1)_{\mu-\tau}$, which this time breaks it altogether. However, as described in previous subsections, it is still possible to salvage $\mathbb{Z}_3^{\widetilde{L}}$ as a *non-invertible symmetry*, and once again, it turns out that this non-invertible symmetry can be obtained by half-space

gauging of a magnetic one-form symmetry. In a UV completion where monopole effects break the magnetic one-form symmetry and the associated instanton effects are the only source of violation of the protecting non-invertible symmetry, exponentially suppressed Dirac masses can therefore be generated.

Further references

Early works on non-invertible symmetries in higher dimensions include [184–187, 200], and there have now been countless follow-ups, including but not limited to [93, 96, 124, 191, 192, 194, 195, 201–237].

Relations between higher-dimensional non-invertible symmetries and M-theory/string theory have been explored in [202, 214, 215, 223, 227, 228, 238, 239], while the role played by non-invertible symmetries on string worldsheets has been studied in [240, 241], with some recent phenomenological applications discussed in [242–252]. Other applications to phenomenology or amplitudes include [194–199, 253–257]. Finally, lattice realizations in various dimensions have been discussed in e.g. [185, 258–270].

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